A Note on Residual Networks: Skip Connections Unlock d^2_{model} Parameters per Layer

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Abstract

We investigate the geometric and representational consequences of skip connections. We find that at equal and even similar hidden widths, the function space they generate is generically different from the skipless one. Furthermore, we prove that in networks of attention and $\tanh$, GELU, SwiGLU or GeGLU blocks, skip connections increase the functional dimension over the skipless case by at least d^2_{model} per block at any depth without normalization (resp. $d_{\text{model}}(d_{\text{model}}-1)/2$ under pre-RMSNorm). Without normalization, the increase is exactly d^2_{model} per block in networks of $\tanh$ blocks or of GELU blocks of width at least $2d_{\text{model}}$. We show how to explicitly remove the $\#Layers \times O(d^2_{\text{model}})$ parameters in the skipless case. Finally, by explicitly removing $O(d^2_{\text{model}})$ parameters in residual networks, a counterexample to the statement of Wang and Wang (2025) that the symmetry group of a transformer is the product of the groups of its blocks is provided.

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Author contributions. Marko Karbevski (MK) conceived and led the study. Antonij Mijoski (AM) proved the absorption result for a single ReLU block (Theorem 3.5(i), Theorem A.11), the difficult case, paving the way forward. MK extended the tools to GELU (Theorem A.13), proved the cases of ReLU^2 and of the gated (GLU) variants (Theorem 3.4) and the results at depth (Theorems 3.6 and 3.7, Theorems 6.1 and 6.2), and worked on the symmetry breaking across layers and its counting, the parameter counts at model scale and the applications. Both authors worked on the experiments and on the writing. All of MK's work on the results, the proofs and the experiments was done with AI assistants.

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1 Introduction

Skip connections are a building block of almost every deep network trained today (He et al., 2016). The rationale behind the choice in deployed models is stated to be optimization and signal propagation: they make deep networks easier to train (Balduzzi et al., 2017; He et al., 2016; MacDonald et al., 2023; Orhan and Pitkow, 2018).

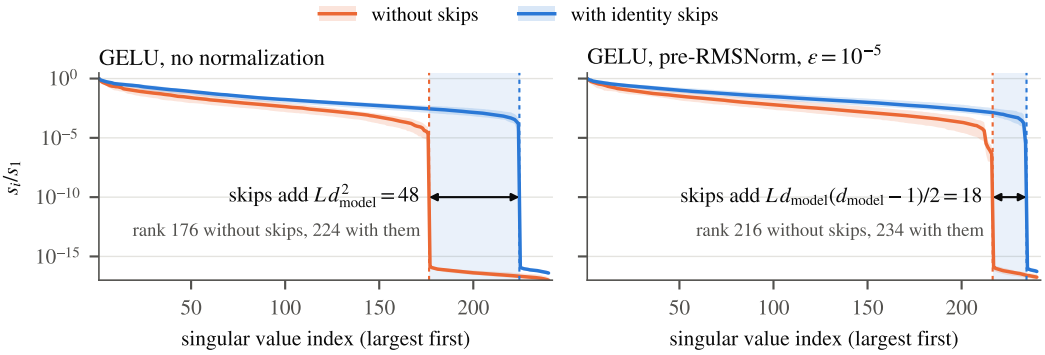


Figure 1: Singular values of the Jacobian of the parameter-to-output map, with and without skips at the same weights ($L = 3$ GELU blocks, $d_{\text{model}} = 4$; median and range over 20 draws of $\mathcal{N}(0, 1/\text{fan-in})$ weights and $\mathcal{N}(0, 1)$ inputs). The skips add Ld^2_{model} to its rank, the functional dimension, and $Ld_{\text{model}}(d_{\text{model}} - 1)/2$ under RMSNorm. Details in Figure 4.

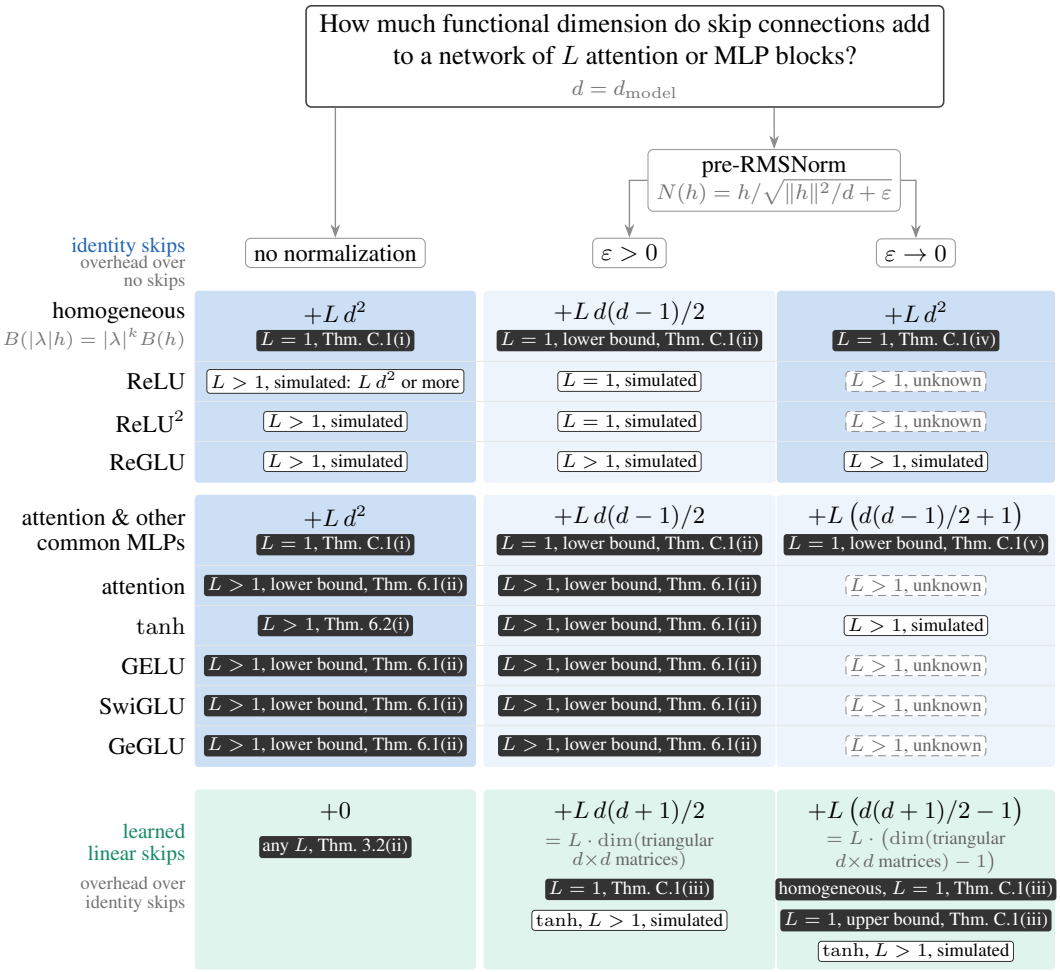


Figure 2: Values from the symmetry counts (Theorem 3.3(iii), Theorem A.24(iii)). Black tags are proven, by the result they name; simulations attain the lower bounds (Appendix F).

Recent skipless models challenge their practical necessity. Ji et al. (2026) remove every skip from a ViT-B, a vision transformer (Dosovitskiy et al., 2021), and train it from scratch on ImageNet-1k (Russakovsky et al., 2015), outperforming the original model in terms of top-1 accuracy. In language modeling, He and Hofmann (2024); He et al. (2023) manage to train transformer variants without skips and/or normalization.

On the representational side, Lin and Jegelka (2018) show that deep single-neuron residual networks with biases are L^1 universal approximators. The result is however limited to the narrow regime with respect to the hidden layers and requires biases, which are excluded in a large portion of both academic and deployed models (Karpathy, 2023; Liu et al., 2026; Mistral AI, 2026; Team OLMo et al., 2025; Yang et al., 2025) (see Appendix B for more details).

In this work we investigate what differences the skip connections induce in wide-hidden-layer MLP and attention layers. Do residual networks compute functions that skipless networks of the same widths cannot? And how many more directions in function space do their parameters reach?

Fix $d_{\text{model}} \geq 2$. Let $\mathcal{S}$ (skipless) and $\mathcal{R}$ (residual) be the sets of functions $x \mapsto h_L D$ computed by networks of L MLP or attention blocks of fixed kinds and widths, without and with skip connections, and without biases or normalization: $h_0 = xE$, then $h_\ell = B_\ell(h_{\ell-1})$ in $\mathcal{S}$ and $h_\ell = h_{\ell-1} + B_\ell(h_{\ell-1})$ in $\mathcal{R}$, over all embeddings E, blocks B_ℓ and readouts D (Section 2). For a single MLP block B with $N \geq d_{\text{model}}$ hidden units, write $\mathrm{Id} + B$ for the residual block $h \mapsto h + B(h)$, and let $\mathcal{S}_{N'}$ be the set

Proven	L **(blocks)**	**Attention** (≥ 2 tokens)	**GELU**	**tanh**	**SwiGLU, GeGLU**	**ReLU**	**ReLU^2, ReGLU**
$\mathcal{S} \subseteq \overline{\mathcal{R}}$	≥ 1	✓ Prop. 6.3(i)					
with pre-RMSNorm	≥ 1	✓ if $\varepsilon = 0$ unknown if $\varepsilon > 0$ Prop. 6.3(ii)					
$\mathcal{S} \cap \mathcal{R}$	$= 1$	$\emptyset$ Thm. 3.8(ii)	measure zero Thm. 3.5(i)	$\emptyset$ Thm. 3.4(iv)	$\emptyset$ Thm. 3.4(iii)	measure zero Thm. 3.5(i)	$\emptyset$ Thm. 3.4(iii)
extra units for a skipless block to equal a residual one	$= 1$	none suffices Thm. 3.8(i)	d_{model} for almost every block Thm. 3.5(ii)	none suffices Thm. 3.4(iv)	none suffices Thm. 3.4(i)	d_{model} for almost every block Thm. 3.5(ii)	none suffices Thm. 3.4(i)
$\mathcal{S} \cap \mathcal{R}$	≥ 2	measure zero Thm. 3.6(ii)		$\emptyset$ Thm. 3.4(iv)	$\emptyset$ Thm. 3.4(iii)	measure zero if first hidden dim $< 2d_{\text{model}}$ Thm. 3.7 otherwise unknown	$\emptyset$ Thm. 3.4(iii)
with pre-RMSNorm, $\varepsilon > 0$	≥ 1	measure zero Thm. 3.6(ii)			$\emptyset$ Thm. 3.4(iii)	unknown	$\emptyset$ Thm. 3.4(iii)
with pre-RMSNorm, $\varepsilon = 0$	≥ 1	measure zero, at any widths Prop. 4.2					
symmetries removed by the skips	≥ 1	Ld_{model}^2 Thm. 3.3(i)					
with pre-RMSNorm, $\varepsilon > 0$	≥ 1	$Ld_{\text{model}}(d_{\text{model}} - 1)/2$ Thm. 3.3(i)					
with pre-RMSNorm, $\varepsilon = 0$	≥ 1	$L\big(d_{\text{model}}(d_{\text{model}} - 1)/2 + 1\big)$ Thm. 3.3(i)				Ld_{model}^2 Lem. A.2(i)	
$\operatorname{fdim} \mathcal{R} - \operatorname{fdim} \mathcal{S}$	$= 1$	$= d_{\text{model}}^2$ Thm. C.1(i)					
with pre-RMSNorm, $\varepsilon > 0$	$= 1$	$\geq d_{\text{model}}(d_{\text{model}} - 1)/2$ Thm. C.1(ii)					
with pre-RMSNorm, $\varepsilon = 0$	$= 1$	$\geq d_{\text{model}}(d_{\text{model}} - 1)/2 + 1$ Thm. C.1(v)				$= d_{\text{model}}^2$ Thm. C.1(iv)	
$\operatorname{fdim} \mathcal{R} - \operatorname{fdim} \mathcal{S}$	≥ 2	$\geq Ld_{\text{model}}^2$ Thm. 6.1(ii)	$\geq Ld_{\text{model}}^2$ Thm. 6.1(ii) $= Ld_{\text{model}}^2$ if every hidden dim $\geq 2d_{\text{model}}$ Thm. 6.2(ii)	$= Ld_{\text{model}}^2$ Thm. 6.2(i)	$\geq Ld_{\text{model}}^2$ Thm. 6.1(ii)	$\geq Ld_{\text{model}}^2$ near small writes Rem. A.20	
with pre-RMSNorm, $\varepsilon > 0$	≥ 2	$\geq Ld_{\text{model}}(d_{\text{model}} - 1)/2$ Thm. 6.1(ii)				unknown	unknown
with pre-RMSNorm, $\varepsilon = 0$	≥ 2	unknown					

Table 1: Proven for skipless ($\mathcal{S}$) and residual ($\mathcal{R}$) networks of L blocks, without biases or normalization, and with pre-RMSNorm where marked; assumptions in Section 2. $\mathcal{S} \cap \mathcal{R}$: $\emptyset$ if $ED \neq 0$, at any widths; measure zero at equal widths. Hidden dim: number of hidden units of a block (heads for attention). Measured: Table 3.

of maps computed by blocks of the same kind with N' hidden units and no skip. Then

networks, any blocks:	$\mathcal{S} \subseteq \overline{\mathcal{R}}$	Prop. 6.3(i)
smooth blocks, any depth:	$\mathcal{S} \cap \mathcal{R}$ has measure zero in $\mathcal{R}$	Thm. 3.6(ii)
ReLU, first block under $2d_{\text{model}}$ units:	$\mathcal{S} \cap \mathcal{R}$ has measure zero in $\mathcal{R}$	Thm. 3.7
ReLU, GELU, almost every B:	$\text{Id} + B \in \mathcal{S}_{N+d_{\text{model}}} \setminus \mathcal{S}_{N+d_{\text{model}}-1}$	Thm. 3.5(ii)
ReLU^2, ReGLU, SwiGLU, GeGLU, every B:	$\text{Id} + B \notin \mathcal{S}_{N'}$ for every N'	Thm. 3.4(i)

Here smooth blocks are attention and $\tanh$, GELU, SwiGLU or GeGLU blocks, and $\overline{\mathcal{R}}$ is the closure of $\mathcal{R}$ under uniform convergence on bounded sets of inputs. $\mathcal{S} \cap \mathcal{R}$ has measure zero in $\mathcal{R}$ when the residual networks that compute a function of $\mathcal{S}$ have their weights in a set of Lebesgue measure zero, and almost every B means every B outside such a set.

The first relation holds because scaling each write up by a power of t, and the next read down by as much, leaves a skipless network unchanged and makes the skips of a residual network negligible, with an error of order $1/t$ (Proposition 6.3(i), Remark A.25). The second holds for $d_{\text{in}}, d_{\text{out}} \geq d_{\text{model}}$: a skipless network keeps a change of basis at every hidden state, so its functional dimension stays below that of the residual networks (Theorem 3.6(ii)). The third holds because near almost every input a generic residual ReLU network is linear of rank d_{model}, while a skipless one uses disjoint units of its first block at an input and at its negative (Theorem 3.7). The last two say that a skip is worth exactly d_{model} hidden ReLU or GELU units, and more than any number of the other units (Theorem 3.5(ii), Theorem 3.4(i)).

Section 2 provides a common framework for the standard MLP, the MLP with a skip connection (including one with learned weights) as well as the attention layer.

In Section 3 we realize skip connections as a gauge collapse of larger networks: networks whose blocks have d_{model} more units with the identity as activation, read with a matrix U and written with a matrix V (Figure 3a). Only UV enters the function, so $(U, V) \mapsto (UK, K^{-1}V)$ changes nothing for every invertible K. The same holds at each hidden state of a skipless network, between the write of one block and the read of the next (Figure 3b). Without normalization, spending these changes of basis from the readout back sets every U and V to the identity whenever every UV is invertible. The result is the residual network, with the same function and the same functional dimension, the number of independent directions in which the parameters move the function (Grigsby et al., 2025) (Theorem 3.2). A skip connection is thus a gauge fix, in the language in which Hashimoto et al. (2024) describe parameter redundancies. The fix ties the hidden states together: the residual network keeps one change of basis for all of them (Figure 3c), and the skipless network with the same parameters keeps one per hidden state. With L blocks the skips therefore remove Ld_{model}^2 dimensions of symmetry, and $Ld_{\text{model}}(d_{\text{model}} - 1)/2$ under RMSNorm (Theorem 3.3). The skipless ViT-B of Ji et al. (2026) thus does more with less: it outperforms the residual ViT-B with about 8 percent less functional dimension.

In Section 4 we extend the collapse to normalized networks. Only orthogonal changes of basis pass through RMSNorm, so invertible learned skips reduce to triangular or symmetric positive definite matrices, and the identity skip is both a gauge fix and a constraint (Theorem 4.1).

In Section 5 we spend the changes of basis that a trained network keeps to remove parameters without changing its function. Without skips, the change of basis at each hidden state sets one $d_{\text{model}} \times d_{\text{model}}$ block of a read or of a write to the identity, or to a triangular matrix under normalization (Proposition 5.1(i)). For the skipless ViT-B of Ji et al. (2026), assuming standard pre-LayerNorm blocks, this removes 8.5 percent of its parameters. With skips, one global change of basis is left. When the embedding and the readout count among the parameters, it is a counterexample to the statement of Wang and Wang (2025) that the symmetry group of a transformer is the product of the groups of its blocks (Section 5.2). With an untied readout it also removes 0.10, 0.05 and 0.03 percent of the parameters of Llama-3.1-8B, -70B and -405B (Grattafiori et al., 2024).

In Section 6 we ask when the removed symmetry is a gain in functional dimension. It is when the symmetries are the only redundancy of both networks, the equality condition. We prove this for networks with one block, attention included, with linear algebra and limits along rays (Theorem C.1). For networks of smooth blocks we prove it at any depth for the residual network, so the skips add

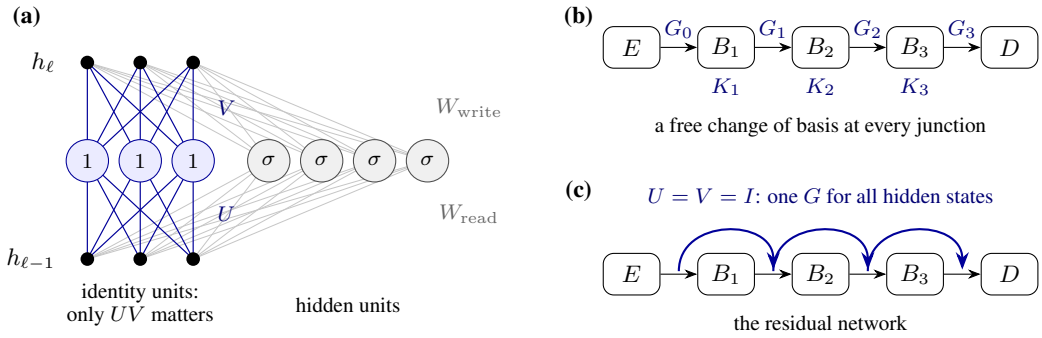


Figure 3: (a) A mixed vanilla MLP block with $d_{\text{model}} = 3$ identity units and 4 hidden units, drawn as one wide layer: its read places U and W_{read} side by side, its map is $[\,\text{Id}, \text{Id}, \text{Id}, \sigma, \sigma, \sigma, \sigma\,]$, and its write stacks V and W_{write}. Nothing nonlinear happens between the read U and the write V of the identity units, so only UV enters the function (Proposition 3.1). (b) A network of such blocks, or of blocks without identity units, between an embedding E and a readout D: each hidden state has a free change of basis G_ℓ, and the identity units of each block a free K_ℓ. (c) Spending them sets $U = V = I$ in every block, which is the residual network (Theorem 3.2); the frozen identity units carry each hidden state into the next in the same basis, so one change of basis is left for all hidden states.

at least Ld_{model}^2, and at least $Ld_{\text{model}}(d_{\text{model}} - 1)/2$ under RMSNorm (Theorem 6.1(ii)). Without normalization, in networks of $\tanh$ blocks or of GELU blocks of width at least $2d_{\text{model}}$, they add exactly Ld_{model}^2 (Theorem 6.2). Figure 1 shows the gain in the singular values of the Jacobian.

Table 1 and Figure 2 summarize what is proved for each kind of block, and Table 2 lists the results with their support. Section 7 discusses related work, and Section 8 separates what is proved, measured and conjectured.

2 Setup

A block is a read, a fixed map and a write. Every network here is a chain of blocks with d_{model} identity units each: deleting them gives the network without skips, and freezing their weights at the identity gives the residual network.

Blocks. The hidden state of a network is a matrix $h \in \mathbb{R}^{T \times d_{\text{model}}}$ with one row of width d_{model} per token: the samples of a batch, or the tokens of a sequence in a transformer. A *block* is a triplet $B = (W_{\text{read}}, f, W_{\text{write}})$ of a trainable *read* $W_{\text{read}} \in \mathbb{R}^{d_{\text{model}} \times p}$, a fixed map $f : \mathbb{R}^{T \times p} \to \mathbb{R}^{T \times q}$ and a trainable *write* $W_{\text{write}} \in \mathbb{R}^{q \times d_{\text{model}}}$, and it maps h to $B(h) = f(hW_{\text{read}})W_{\text{write}}$. In a *vanilla MLP* block, $p = q = m$ and $f = [\,\sigma, \ldots, \sigma\,]$ applies an activation σ to each of the m hidden units, row by row. In a *GLU* block (Shazeer, 2020), $p = 2m$ and $q = m$: f splits the coordinates of a row into gates z_{g} and values z_{v} and returns $g(z_{\text{g}}) \odot z_{\text{v}}$, with the gate function g the sigmoid in GLU, $\max(z, 0)$ in ReGLU, GELU in GeGLU and $z/(1 + e^{-z})$ in SwiGLU. In an *attention* block with H heads of width d_h, the read holds the query, key and value projections, f is the attention map, which mixes the rows, and the write is the output projection. A transformer layer consists of two blocks, so L counts sub-blocks.

Identity units. Give a block d_{model} more units whose activation is the identity, read with $U \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ and written with $V \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$. The result is the *mixed block*

$$\tilde{B}(h) \;=\; \underbrace{hUV}_{\text{identity units}} \;+\; \underbrace{f(hW_{\text{read}})\,W_{\text{write}}}_{\text{the block } B} \;=\; \tilde{f}\big(h\,[\,U \mid W_{\text{read}}\,]\big)\begin{bmatrix} V \\ W_{\text{write}} \end{bmatrix}, \tag{1}$$

with $\tilde{f}\big(\big[\,u \;\big|\; z\,\big]\big) = \big[\,u \;\big|\; f(z)\,\big]$ for the input u of the identity units and the input z of f. For a vanilla MLP block, $\tilde{f} = [\,\text{Id}, \ldots, \text{Id}, \sigma, \ldots, \sigma\,]$ is the *mixed activation*, with d_{model} identities followed by m copies of σ, and a residual block is a block of width $d_{\text{model}} + m$ in which d_{model} units have the

Table 2: The main results, informally; *Where* links to the precise statements. Networks have L blocks (two per transformer layer) and hidden states of width d_{model}; $\mathcal{G}$ is the group of changes of basis at a hidden state. *Support*: proof; measured, as the rank of a Jacobian on a batch of inputs (Appendix F); checked, by evaluating a construction in double precision; count, by arithmetic on released configurations.

Result	Where	What it shows	Support
Skips are gauge fixes	Proposition 3.1, Theorem 3.2	Without normalization, trainable identity units and invertible learned skips reduce to identity skips, with the same functions and functional dimension; one global change of basis is left.	Proof, checked
Symmetry removed	Theorem 3.3(i), Section 4	Against the skipless network with the same parameters, the skips remove $L \dim \mathcal{G}$ dimensions of symmetry: Ld_{model}^2 without normalization; with $\varepsilon > 0$, $Ld_{\text{model}}(d_{\text{model}} - 1)/2$ under RMSNorm, $L(d_{\text{model}} - 1)(d_{\text{model}} - 2)/2$ under LayerNorm.	Proof
Identity units are new	Thms. 3.4(i), 3.5(ii), 3.6(ii) and (iii), 3.7	Without biases, a block never absorbs its identity units for ReLU^2, ReGLU, SwiGLU and GeGLU, whatever its width; for ReLU and GELU, d_{model} more units do, and fewer only on a set of weights of measure zero. At any depth, with or without pre-RMSNorm, a residual network of smooth blocks equals a skipless one of the same widths only on a set of weights of measure zero, and, without normalization, for GELU blocks also with fewer than $Ld_{\text{model}}/2$ more hidden units in all. A residual ReLU network equals a skipless one whose first block has fewer than $2d_{\text{model}}$ units only on a set of weights of measure zero.	Proof
Learned skips under RMSNorm	Theorem 4.1	Learned skips reduce to triangular or symmetric matrices; the identity skip fixes $d_{\text{model}}(d_{\text{model}} - 1)/2$ dimensions of gauge and, under the equality condition, imposes $d_{\text{model}}(d_{\text{model}} + 1)/2$ constraints.	Proof, checked, measured
Reduction without skips	Proposition 5.1(i)	One $d_{\text{model}} \times d_{\text{model}}$ block per hidden state becomes the identity, or triangular under normalization: 8.5 percent of a skipless ViT-B.	Proof, count
One global change of basis	Section 5.2	A symmetry outside the product of the groups of the blocks; with an untied readout it also removes one block, for example 0.03 to 0.10 percent of Llama-3.1.	Proof, checked, count
Gain, one block	Theorem C.1(i)–(iv)	The skip adds exactly d_{model}^2 to the functional dimension (tanh, GELU, ReLU, ReLU^2, ReGLU, SwiGLU, GeGLU, attention), and under RMSNorm at least $d_{\text{model}}(d_{\text{model}} - 1)/2$ with $\varepsilon > 0$ and exactly d_{model}^2 for ReLU, ReLU^2 and ReGLU with $\varepsilon = 0$; a learned skip adds exactly $d_{\text{model}}(d_{\text{model}} + 1)/2$ more.	Proof, measured
Gain, any depth	Theorems 6.1 and 6.2	For smooth blocks the residual network satisfies the equality condition, so the skips add at least $L \dim \mathcal{G}$, and, without normalization, exactly Ld_{model}^2 in networks of tanh blocks or of GELU blocks of width at least $2d_{\text{model}}$.	Proof
Gain, deeper networks	Figure 4, Section 6.2	The gain equals the removed symmetry in tanh, GELU and gated MLPs, with and without RMSNorm, and in nanoGPT, in every draw, except for skipless ReGLU networks under RMSNorm with $\varepsilon = 10^{-5}$, where some directions fall below double precision.	Measured
Skipless networks as limits	Proposition 6.3, Remark A.25	Without biases or normalization, every skipless network is a limit of residual networks of the same widths, with an error of order $1/t$ when each write is scaled up by a power of t and the next read down by as much. Under pre-RMSNorm with $\varepsilon = 0$, so is every skipless network whose hidden states never vanish, with only the writes scaled.	Proof, checked

identity as activation and their read and write frozen at the identity matrix (Lin and Jegelka, 2018, Section 4).

Networks. A network of L mixed blocks has hidden states $h_0 = xE$ and $h_\ell = \tilde{B}_\ell(h_{\ell-1})$ for $\ell = 1, \dots, L$, and output $y = h_L D$, with an embedding $E \in \mathbb{R}^{d_{\text{in}} \times d_{\text{model}}}$ and a readout $D \in \mathbb{R}^{d_{\text{model}} \times d_{\text{out}}}$. Deleting the identity units gives the network *without skips*, or *skipless* network, $h_\ell = B_\ell(h_{\ell-1})$; freezing $U_\ell = V_\ell = I$ gives the *residual* network, $h_\ell = h_{\ell-1} + B_\ell(h_{\ell-1})$; and merging U_ℓ and V_ℓ into $A_\ell = U_\ell V_\ell$ gives the network with *learned skips*, $h_\ell = h_{\ell-1}A_\ell + B_\ell(h_{\ell-1})$. The skipless and the residual network have the same P parameters, and the skipless network is our baseline. Biases may be added inside and after f, and then the embedding and the readout have biases too.

Normalization. In a *pre-norm* network the read of every block, and the readout, see $N(h)$ instead of h, where N is RMSNorm, which divides each row $h^{(t)}$ by $\sqrt{\|h^{(t)}\|^2/d_{\text{model}} + \varepsilon}$; a learned gain after N is folded into the next read. The identity units, which carry the skip, read h itself. LayerNorm is RMSNorm after the projection that removes from each row its mean (Proposition A.23), and we treat it that way.

Scope. Unless we say otherwise, $d_{\text{model}} \geq 2$, networks have no biases, inputs and outputs of dimension at least d_{model}, and blocks of width at least d_{model}, where an attention block has heads of width $d_h \leq d_{\text{model}}$ and width Hd_h; normalization is absent or pre-RMSNorm with $\varepsilon > 0$. We call attention, $\tanh$, GELU, SwiGLU and GeGLU blocks *smooth*, and the MLP blocks of a network of smooth blocks are all of one kind.

Changes of basis. For an invertible G, the *change of basis* G at a hidden state h_ℓ replaces every matrix W that writes h_ℓ (V_ℓ and W^ℓ_{write}, or E if $\ell = 0$) by WG, and every matrix W that reads it ($U_{\ell+1}$ and $W^{\ell+1}_{\text{read}}$, or D if $\ell = L$) by $G^{-1}W$. Through RMSNorm a read sees $N(hG)$, and $N(hG) = N(h)Q$ for all h exactly when $G = cQ$ with Q orthogonal and $c > 0$, where $c = 1$ if $\varepsilon > 0$ (Lemma A.1); such a read is replaced by $Q^{-1}W$. We write $\mathcal{G}$ for the group of changes of basis that pass through the reads: $\mathrm{GL}(d_{\text{model}})$ without normalization, $\mathrm{O}(d_{\text{model}})$ under RMSNorm with $\varepsilon > 0$, and the scaled orthogonal matrices $\mathrm{CO}(d_{\text{model}})$ when $\varepsilon = 0$. Biases add a translation at each hidden state, which the skips do not tie (Lemma A.4). A block can also have *symmetries inside*, which leave the hidden states alone: the rescalings of hidden units for positively homogeneous activations such as ReLU, of the values of a GLU unit, and in ReGLU of its gates as well, and in attention an invertible map per head between queries and keys and another between values and outputs. We write s for their dimension, which is the same with and without skips.

Functional dimension. Let F_θ be the function computed at parameters $\theta \in \mathbb{R}^P$. Following Grigsby et al. (2025), the *functional dimension* at θ is the largest rank of the Jacobian of $\theta \mapsto (F_\theta(x_1), \ldots, F_\theta(x_n))$ over finite batches of inputs, and the *redundancy* is P minus it; *generic* values are those taken on an open set of full measure. A symmetry changes θ without changing F_θ. The changes of basis and the symmetries inside blocks act *freely* at θ if no continuous family of them, other than the identity, fixes θ; then the redundancy at θ is at least the dimension of these symmetries. They act freely at generic parameters when the embedding has rank d_{model} and every block has width at least d_{model} (Lemma A.5, Remark A.6). A network satisfies the *equality condition* if at generic parameters the redundancy is exactly this dimension. It is a hypothesis. We prove it for networks with one block, without normalization and, for the residual network, under RMSNorm (Theorem C.1), and for residual networks of smooth blocks at any depth (Theorem 6.1). Without normalization, we also prove it for skipless networks of $\tanh$ blocks or of GELU blocks of width at least $2d_{\text{model}}$ (Theorem 6.2). We measure it in deeper networks (Section 6.2).

Remark 2.1 (Large models). The measurements use small networks, because the exact functional dimension of a model of realistic size is out of reach, for three reasons. *Memory.* The batch Jacobian has one column per parameter and at least as many rows as the functional dimension; for ViT-B, with 86M parameters, it fills over 50 petabytes in double precision. *Compute.* Reading its rank takes a factorization of order P^3 operations, about 10^{24} for ViT-B, and iterative methods would have to resolve a kernel of millions of dimensions. *Sampling.* The rank is read at a gap in the singular values, and the floor of rounding error rises with the size of the matrix. A direction that the network uses only on a small fraction of inputs, such as a hidden unit that is rarely active, has singular values that grow only like the square root of the number of inputs that reach it. In high dimension such inputs can be rare, so resolving these directions takes far more than the minimum of about P/d_{out} inputs. In our small networks the gap spans many orders of magnitude (Figure 4).

3 Skip connections are gauge fixes

The identity units and the hidden states between blocks are linear junctions; spending their changes of basis turns trainable identity units into skip connections, and against the network without skips this removes the change of basis at every hidden state but one.

3.1 Junctions

Nothing nonlinear happens between the read of the identity units and their write. Without skips, nothing nonlinear happens between the write into a hidden state and its reads either. Both are linear junctions: only the products of the matrices on the two sides enter the function.

Proposition 3.1 (Junctions)**. (i) Only UV matters.** *In a mixed block, $(U, V) \mapsto (UK, K^{-1}V)$ preserves the function for every $K \in \mathrm{GL}(d_{\mathrm{model}})$.*
(ii) Hidden states are junctions. *In a skipless network, and in a network of mixed blocks, the map $W \mapsto WG$ on the matrices that write a hidden state and $W \mapsto G^{-1}W$ on those that read it preserves the function for every $G \in \mathcal{G}$. For $\varepsilon = 0$ and $G = cQ$, a read through RMSNorm maps to $Q^\top W$ instead.*
(iii) One group per junction. *The maps of (i) and (ii) commute when they act at different junctions. A skipless network therefore carries one copy of $\mathcal{G}$ per hidden state, and a network of mixed blocks one more copy of $\mathrm{GL}(d_{\mathrm{model}})$ per block.*

Proof in Appendix A.1.

Part (i) is the merge: a mixed block depends on U and V only through $A = UV$, so a network of mixed blocks is the network with learned skips A_ℓ. Under RMSNorm the identity units still read the raw stream, so their junction keeps all of $\mathrm{GL}(d_{\mathrm{model}})$, and only the hidden states are restricted to $\mathcal{G}$. For vanilla MLP blocks without normalization, both parts are cases of Proposition C.1 of Zhao et al. (2023), which turns every change of basis that passes through the activation of a hidden layer into a symmetry: part (i) with the mixed activation $\tilde{f}$ of (1), which passes the matrices $\mathrm{diag}(K, I)$, since $\tilde{f}\big(\tilde{z}\,\mathrm{diag}(K, I)\big) = \tilde{f}(\tilde{z})\,\mathrm{diag}(K, I)$, and part (ii) with the identity as the activation at the hidden states. Under RMSNorm with $\varepsilon > 0$, part (ii) for the skipless network is the case of the radial rescaling of their Example C.6. For one hidden layer, no wider than the wider of its input and its output, their Theorem 3.3 also shows that the change of basis moves the parameters in d_{model}^2 independent directions for $\mathrm{GL}(d_{\mathrm{model}})$ and in $d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2$ for $\mathrm{O}(d_{\mathrm{model}})$, and Lemma A.5 extends this to deep networks.

3.2 The gauge fix

The changes of basis at the junctions bring the read and the write of every identity unit to the identity. Let $\Lambda_\ell = A_{\ell+1}A_{\ell+2}\cdots A_L$, with $A_\ell = U_\ell V_\ell$ and $\Lambda_L = I$.

Theorem 3.2 (Skip connections are gauge fixes)**. (i) Same function.** *Without normalization, if every A_ℓ is invertible, the network of mixed blocks and the network with learned skips compute the same function as the residual network with the embedding $E\Lambda_0$, the reads $\Lambda_{\ell-1}^{-1}W_{\mathrm{read}}^{\ell}$, the writes $W_{\mathrm{write}}^{\ell}\Lambda_\ell$ and the readout D.*
(ii) *At these parameters the residual network has the same functional dimension as the network of mixed blocks and as the network with learned skips.*
(iii) One change of basis is left. *Without normalization, or under pre-RMSNorm with $\varepsilon > 0$, let every block be nonzero for some value of its parameters, and let $G_0, \ldots, G_L \in \mathcal{G}$. The map $E \mapsto EG_0$, $W_{\mathrm{read}}^{\ell} \mapsto G_{\ell-1}^{-1}W_{\mathrm{read}}^{\ell}$, $W_{\mathrm{write}}^{\ell} \mapsto W_{\mathrm{write}}^{\ell}G_\ell$, $D \mapsto G_L^{-1}D$ preserves the function of every residual network if and only if $G_0 = G_1 = \cdots = G_L$.*

A special case of Theorem A.3, proved in Appendix A.2.

The sweep. Multiplying each hidden state h_ℓ by an invertible G_ℓ, as in (iii), turns the skip A_ℓ into $G_{\ell-1}^{-1}A_\ell G_\ell$, so the G_ℓ must be chosen together. From the readout back, a skip becomes the identity when $G_{\ell-1} = A_\ell G_\ell$, and $G_L = I$ gives $G_\ell = \Lambda_\ell$, which yields the parameters of (i). In a mixed block, the merge $(U_\ell, V_\ell) \mapsto (U_\ell K, K^{-1}V_\ell)$ of Proposition 3.1(i) then sets U_ℓ and V_ℓ to I. The product of all skips ends up in the embedding, or, run from the embedding forward, in the readout (Appendix A.2). So without normalization, networks with invertible learned skips or identity units compute the same functions as residual networks, with the same functional dimension: the gauge fix loses nothing.

Why one change of basis is left. Part (iii) is the tie. After changes of basis at $h_{\ell-1}$ and h_ℓ the branch becomes $B_\ell(\cdot\, G_{\ell-1}^{-1})\, G_\ell$, and the residual update computes

$$h_{\ell-1}G_{\ell-1} + B_\ell(h_{\ell-1})\, G_\ell \;=\; h_\ell\, G_\ell \;+\; h_{\ell-1}\,(G_{\ell-1} - G_\ell). \tag{2}$$

The frozen identity units carry $h_{\ell-1}$ in the basis $G_{\ell-1}$, and the branch writes in the basis G_ℓ; unless the two agree, the defect $h_{\ell-1}(G_{\ell-1} - G_\ell)$ is passed on to every later block. In the language of gauge theory, the change of basis of a skipless network is local in depth, and the frozen identity units restrict it to a global one (Hashimoto et al., 2024).

When the block dominates the skip. The defect is carried by the skip: relative to $h_\ell G_\ell$ it is at most $\|h_{\ell-1}\|\|G_{\ell-1} - G_\ell\|/\|h_\ell G_\ell\|$, which is small where the branch dominates, $\|B_\ell(h_{\ell-1})\| \gg \|h_{\ell-1}\|$. There changes of basis that differ only across this block are approximate symmetries, and the residual block differs from its branch alone by a relative $\|h_{\ell-1}\|/\|B_\ell(h_{\ell-1})\|$, although in general no skipless block of the same kind and width equals it exactly (Section 3.4). The optimizer ARO shares one rotation through the skips, as the exact symmetry requires, and relaxes it to one rotation per transformer layer where the skips are weak (Gong et al., 2026, Section 6.4). In our check, a change of basis at every hidden state after one block, so that only the skip of that block joins two bases, changes the output by a relative 0.25 when the branch is about half the size of the skip, and by 5×10^{-3} when it is about fifty times larger (Appendix F).

3.3 What the skips remove

Theorem 3.3 (What the skips remove)**. (i)** *The changes of basis that preserve the function at all parameters form a group of dimension* $(L+1)\dim\mathcal{G}$ *in a skipless network and* $\dim\mathcal{G}$ *in a residual one, so the* L *skips remove* $L\dim\mathcal{G}$ *dimensions of symmetry.*
(ii) *If in both networks the symmetries act freely at generic parameters, the functional dimension at these parameters is at most* $P - (L+1)\dim\mathcal{G} - s$ *without skips and* $P - \dim\mathcal{G} - s$ *with them.*
(iii) *If in addition both networks satisfy the equality condition, the skips raise the functional dimension by exactly* $L\dim\mathcal{G}$*, which is* Ld_{model}^2 *without normalization.*

Proof in Appendix A.3.

Parameter counts understate residual networks. Table 4 in Appendix A.3 keeps the books. The identity units of a block add $2d_{\text{model}}^2$ parameters and d_{model}^2 dimensions of symmetry, so on balance d_{model}^2 of functional dimension, and the gauge fix removes their $2d_{\text{model}}^2$ parameters together with $2d_{\text{model}}^2$ dimensions of symmetry. With biases every hidden state also carries a translation, and the gain is again Ld_{model}^2 (Lemma A.4). Under the equality condition the residual network thus has functional dimension $P - \dim\mathcal{G} - s$ and the skipless one $P - (L+1)\dim\mathcal{G} - s$. At equal parameter count the residual network has $L\dim\mathcal{G}$ more independent directions in function space, Ld_{model}^2 without normalization and $Ld_{\text{model}}(d_{\text{model}}-1)/2$ under RMSNorm; in this sense a residual network with P parameters is worth a skipless one with $P + L\dim\mathcal{G}$, and the skipless network can shed its surplus after training (Section 5.1). Comparisons at equal parameter count therefore favor residual networks. The skipless ViT-B of Ji et al. (2026) outperforms the residual one with about 8 percent less functional dimension: it does more with less (Section 5.1). A comparison at equal functional dimension would give the skipless network $L\dim\mathcal{G}$ more parameters, Ld_{model}^2 without normalization and about $Ld_{\text{model}}^2/2$ with it. Added as hidden units of $\tanh$ or GELU blocks, these parameters reach the functional dimension of the residual network; gated units also bring rescalings and need a few more (Figure 4).

3.4 Are the identity units new?

The gain needs identity units that the other units of a block cannot absorb. A linear branch absorbs them: $h + hA = h(I + A)$, the deep linear residual network of Hardt and Ma (2017), whose identity parametrization adds no functional dimension. So do d_{model} more ReLU, GELU or SiLU units, since $\sigma(z) - \sigma(-z) = z$. With fewer units the common blocks cannot. A block is *nondegenerate* if the

columns of W_{read} are nonzero and pairwise non-collinear and the rows of W_{write} are nonzero. For a set S of hidden units, $W_{\text{read}}[:, S]$ and $W_{\text{write}}[S, :]$ keep the columns and the rows of the units in S.

We ask whether a block can absorb a linear term. Let $M \neq 0$ be a $d_{\text{model}} \times d_{\text{model}}$ matrix, and let B_θ be a vanilla or GLU MLP block with parameters $\theta = (W_{\text{read}}, W_{\text{write}})$ (Section 2). We look for parameters θ' of a block with the same activation or gate function, and of any width, such that

$$xM + B_\theta(x) = B_{\theta'}(x) \qquad \text{for all row vectors } x \in \mathbb{R}^{1 \times d_{\text{model}}}. \tag{3}$$

Theorem 3.4 (Homogeneous, gated and bounded blocks)**.** *Let the blocks be positively homogeneous of degree $k \neq 1$, as ReLU^2 and ReGLU blocks are, or gated with a gate that is differentiable at 0 and vanishes there, as SwiGLU and GeGLU blocks are.*
(i) *No block $B_{\theta'}$ of any width satisfies (3).*
(ii) *For gated blocks, and for homogeneous blocks with $k > 1$, no composition of one or more blocks without skips equals a composition of residual blocks, whatever their numbers and widths.*
(iii) *For the blocks of (ii), with an embedding E and a readout D, no skipless network equals a residual network with $ED \neq 0$, also under pre-RMSNorm with $\varepsilon > 0$.*
(iv) *Without normalization, (i), (ii) and (iii) also hold for* $\tanh$ *blocks.*

Proof in Appendix A.4.

Theorem 3.5 (ReLU and GELU blocks)**.** *Let σ be ReLU or GELU.*
(i) *If B_θ is nondegenerate, a block $B_{\theta'}$ of the same width satisfies (3) if and only if some set S of hidden units has $W_{\text{read}}[:, S]\, W_{\text{write}}[S, :] = -M$; for generic θ there is none.*
(ii) *If M is invertible and W_{read} has rank d_{model}, a block $B_{\theta'}$ with d_{model} more hidden units satisfies (3), and for generic θ none with fewer does.*

Proof in Appendix A.4.

Theorem 3.6 (GELU and smooth blocks at depth)**. (i)** *For GELU blocks and almost every choice of weights, a composition of $L \geq 2$ residual blocks equals no composition of L blocks without skips of the same widths.*
(ii) *With an embedding and a readout, the same holds for networks of smooth blocks of any depth $L \geq 1$, also under pre-RMSNorm with $\varepsilon > 0$.*
(iii) Too few extra units. *For GELU blocks with an embedding and a readout, and almost every choice of weights, a residual network of L blocks equals no skipless network of L blocks with fewer than $Ld_{\text{model}}/2$ more hidden units in all.*
(iv) Enough extra units. *For GELU blocks with an embedding and a readout, a residual network whose reads all have rank d_{model} equals a skipless network with d_{model} more hidden units in every block.*

Proof in Appendix A.4.

Theorem 3.7 (ReLU blocks at depth)**.** *Consider networks of ReLU blocks with an embedding and a readout. For almost every choice of weights, a residual network equals no skipless network whose first block has fewer than $2d_{\text{model}}$ hidden units, whatever its depth and its other widths.*

Proof in Appendix A.4.

Theorem 3.4 compares how the two sides scale along rays from the origin; for $\tanh$ blocks both sides are bounded, and xM is not. In Theorem 3.5(i) the even part of the activation matches the hidden units of the two blocks up to sign and scale, and its odd part, $z/2$, turns a sign flip into a linear term, as Petzka et al. (2020, Section 3) show for two-layer ReLU networks with one output. So a residual block is not a block of the same kind and width without a skip, and the identity units of a mixed block are determined by its function: if two mixed blocks of the same kind and width compute the same function, the difference M of their merged matrices satisfies (3), so $M = 0$ for the blocks of Theorem 3.4, and M is one of finitely many matrices for those of Theorem 3.5. The merge of Proposition 3.1(i) is thus the only continuous symmetry that the identity units add, and the gauge fix of Theorem 3.2 is well defined. In Theorem 3.5(ii), two units with opposite reads add a rank-one

linear term, and the matching of part (i) shows that every extra unit adds at most one. Applied to the reads EW_{read} and the writes $W_{\text{write}}D$, Theorem 3.5(i) holds for networks of one block with an embedding E and a readout D, with $M = ED$.

Theorem 3.6(i) and (ii) compare dimensions. A change of basis at each hidden state leaves a skipless network unchanged. Its functional dimension is therefore at most $P - 2d_{\text{model}}^2 - s$, from the changes of basis at h_0 and h_L alone, and at most $P - (L+1)d_{\text{model}}^2 - s$ when every block has width at least d_{model} (Theorem 3.3); here s counts the symmetries inside the blocks, and $s = 0$ for networks of GELU blocks. With small writes, a residual network is, to first order, the linear map xED plus one wide block that holds all its hidden units and heads, and its functional dimension there is $P - d_{\text{model}}^2 - s$ (Lemma A.19). The larger dimension keeps almost every residual network outside the skipless ones (Lemma A.18). In Theorem 3.6(iii) each extra hidden unit adds $2d_{\text{model}}$ parameters against a gap of Ld_{model}^2, so absorbing the skips takes between $Ld_{\text{model}}/2$ and Ld_{model} extra units, the latter by Theorem 3.6(iv). Theorem 3.7 compares ranks. Near almost every input, a generic residual ReLU network is linear of rank d_{model}. The units of the first block that are active at an input and at its negative are disjoint, so with fewer than $2d_{\text{model}}$ units a skipless network has fewer than d_{model} active units at one of the two, and some direction of inputs leaves it unchanged there.

Attention. Attention mixes tokens; a skip keeps each token where it is. Two inputs make this precise. If all tokens are equal, mixing changes nothing, so the heads must reproduce the skip. If one token is zero, its query is zero, so attention gives it an equal mix of all tokens. The skip gives it zero.

An *attention block* with parameters $\theta = (W_{Q,i}, W_{K,i}, W_{V,i}, W_{O,i})_{i=1}^{H}$ maps $h \in \mathbb{R}^{T \times d_{\text{model}}}$, whose rows are the $T \geq 2$ tokens, to

$$B_\theta(h) = \sum_{i=1}^{H} P_{\Gamma_i}(h)\, h\, W_{V,i} W_{O,i}, \qquad P_\Gamma(h) = \operatorname{softmax}\big(h\,\Gamma\, h^\top\big), \qquad \Gamma_i = W_{Q,i} W_{K,i}^\top / \sqrt{d_h}, \tag{4}$$

where $W_{Q,i}, W_{K,i}, W_{V,i} \in \mathbb{R}^{d_{\text{model}} \times d_h}$ and $W_{O,i} \in \mathbb{R}^{d_h \times d_{\text{model}}}$. The softmax acts on each row of the $T \times T$ matrix of scores, with or without a causal mask, so each row of $P_\Gamma(h)$ is a probability vector. Head i has the score matrix Γ_i and the value–output product $W_{V,i}W_{O,i}$.

Theorem 3.8 (No absorption in attention)**.** *Let θ and θ' be parameters of attention blocks with any numbers and widths of heads, and let $M \neq 0$ be a $d_{\text{model}} \times d_{\text{model}}$ matrix.*
(i) *For almost every $h \in \mathbb{R}^{T \times d_{\text{model}}}$, $hM + B_\theta(h) \neq B_{\theta'}(h)$.*
(ii) *With an embedding E and a readout D with $ED \neq 0$, the residual network $x \mapsto \big(xE + B_\theta(xE)\big)D$ differs from every skipless network $x \mapsto B_{\theta'}(xE')D'$.*

A special case of Theorem A.21, proved in Appendix A.5.

The proof gives a gap: some input whose rows have norm at most 1 has a row in which $hM + B_\theta(h)$ and $B_{\theta'}(h)$ differ by at least $\|M\|/3$, where $\|M\|$ is the spectral norm (Theorem A.21(ii)). It uses only that each row of attention weights sums to 1 and is uniform for a zero query, so it also holds with rotary position embeddings. Adding heads does not help, whereas d_{model} more ReLU or GELU units absorb the skip of an MLP block (Theorem 3.5(ii)).

The softmaxes must match. With three or more tokens there is a sharper reason why a skip cannot be absorbed: the heads of B_θ and $B_{\theta'}$ must cancel each other exactly, and then nothing is left to produce hM. What tells heads apart is where their softmaxes have poles in the complex plane.

Proposition 3.9 (The softmaxes must match)**.** *Let $T \geq 3$, let θ and θ' be parameters of attention blocks, and suppose that $hM + B_\theta(h) = B_{\theta'}(h)$ for all h in a nonempty open set. Then $M = 0$, and for every matrix Γ, the heads with score matrix Γ have the same sum of value–output products in B_θ as in $B_{\theta'}$.*

Follows from Lemma A.22, proved in Appendix A.5.

Lemma A.22 shows that, with three or more tokens, a linear term hM and heads with distinct score matrices cancel on an open set of inputs only if each of them vanishes. With two tokens they can cancel. The softmax of two scores is the logistic function $1/(1+e^{-z})$ of their difference z, and $1/(1+e^{-z}) + 1/(1+e^{z}) = 1$ gives $P_\Gamma(h)h + P_{-\Gamma}(h)h = 2P_0(h)h$: two heads with opposite score matrices add up to a uniform head. A skip still cannot be absorbed when $T = 2$ (Theorem 3.8(i)). With a causal mask, the same proof runs on the last row, which attends to all tokens.

4 Under normalization: triangular skips

Only orthogonal changes of basis pass through RMSNorm, so a skip matrix splits into an orthogonal part, which is gauge, and a triangular or symmetric part, which no change of basis removes: the identity skip fixes the first and constrains the second.

Under normalization the identity units, which carry the skip, read h, while the block reads $N(h)$. We therefore start from the network with learned skips

$$h_\ell = h_{\ell-1}A_\ell + f_\ell\big(N(h_{\ell-1})\,W^\ell_{\text{read}}\big)\,W^\ell_{\text{write}}, \qquad y = N(h_L)\,D. \tag{5}$$

A change of basis at h_ℓ must now pass through N, so it lies in $\mathcal{G}$, which is $\mathrm{O}(d_{\text{model}})$ for $\varepsilon > 0$ (Lemma A.1). The skips read the raw stream and absorb any change of basis: $A_\ell \mapsto G_{\ell-1}^{-1}A_\ell G_\ell$. Let $\Pi_\ell = A_1 \cdots A_\ell$, with $\Pi_0 = I$.

Theorem 4.1 (RMSNorm: a gauge fix and a constraint)**.** *Consider pre-RMSNorm networks with $\varepsilon > 0$, and the network with learned skips of (5).*
(i) Triangular skips. *If every A_ℓ is invertible, there are orthogonal $Q_1, \ldots, Q_L$, with $Q_0 = I$, such that every $Q_{\ell-1}^\top A_\ell Q_\ell$ is upper triangular with positive diagonal, and the network with the skips $Q_{\ell-1}^\top A_\ell Q_\ell$, the reads $Q_{\ell-1}^\top W^\ell_{\text{read}}$, the writes $W^\ell_{\text{write}}Q_\ell$ and the readout $Q_L^\top D$ computes the same function. Other orthogonal Q_ℓ make every skip symmetric positive definite instead.*
(ii) Orthogonal skips are gauge. *If every A_ℓ is orthogonal, the network computes the same function as the residual network with the embedding E, the reads $\Pi_{\ell-1}W^\ell_{\text{read}}$, the writes $W^\ell_{\text{write}}\Pi_\ell^\top$ and the readout $\Pi_L D$.*
(iii) What each skip adds. *Under the equality condition for the networks with learned skips, with identity skips and without skips, the functional dimension of the network with learned skips exceeds that of the residual network by $L\,d_{\text{model}}(d_{\text{model}}+1)/2$, and that of the residual network exceeds that of the skipless network by $L\,d_{\text{model}}(d_{\text{model}}-1)/2$.*

A special case of Theorem A.24, proved in Appendix A.8.

The identity skip is a gauge fix and a constraint. Of the d^2_{model} entries of a skip matrix $A = RQ$, the $d_{\text{model}}(d_{\text{model}}-1)/2$ of the orthogonal factor Q are gauge: changes of basis remove them exactly (part (i)). The $d_{\text{model}}(d_{\text{model}}+1)/2$ of the triangular factor R, or of the symmetric factor S in $A = SQ$, remain, and under the equality condition setting them to the identity lowers the functional dimension by as much (part (iii)), which we measure at $d_{\text{model}} = 4$ and $L = 3$ (Appendix F). So under RMSNorm an identity skip fixes the gauge at a hidden state and also imposes $d_{\text{model}}(d_{\text{model}}+1)/2$ constraints; without normalization it only fixes the gauge. Rotations in the skips are pure gauge: SliceGPT inserts them before it deletes rows and columns (Ashkboos et al., 2024a, Sections 3.3 and 3.4), and van Nierop (2024, Section 4.6) inserts them into LayerNorm transformers to recover one copy of the stream symmetry per skip, at the cost of as many parameters.

LayerNorm and DyT. LayerNorm is RMSNorm after a projection: $\mathrm{LN}(h) = \gamma \odot N(hZ) + \beta$, with Z the projection that removes the mean, so a LayerNorm network does not use the direction of the all-ones vector and normalizes the other $d_{\text{model}} - 1$ (Ashkboos et al., 2024a; Baroni et al., 2026; Theus et al., 2025). Folding the gains into the next reads and centering every write turns it into an RMSNorm network of width $d_{\text{model}} - 1$ (Proposition A.23), with $\mathrm{O}(d_{\text{model}} - 1)$ at each hidden state, so the skips remove $L(d_{\text{model}} - 1)(d_{\text{model}} - 2)/2$ dimensions, as we measure in nanoGPT (Section 6.2). An elementwise read such as DyT, $\gamma \odot \tanh(\alpha h) + \beta$ with a learnable scalar α (Zhu et al., 2025), lets through only signed permutations and one positive scale, which α absorbs, so the skips remove only L dimensions (Appendix F).

Low-rank skips. The low-rank version of LAuReL trains its skip matrix as $A = I + UV^\top$ with $U, V \in \mathbb{R}^{d_{\text{model}} \times r}$ (Menghani et al., 2025). Without normalization such a skip, when invertible, is a learned skip, and Theorem 3.2(i) removes all $2d_{\text{model}}r$ of its parameters. Under RMSNorm, making it triangular would give up the low rank, since a triangular matrix has $d_{\text{model}}(d_{\text{model}}+1)/2$ entries against the $2d_{\text{model}}r$ of the factors. The product itself offers a change of basis of width r: $(U, V) \mapsto (UK, VK^{-\top})$ leaves $UV^\top$ unchanged for every $K \in \mathrm{GL}(r)$, so r rows of U can be set to the identity, which removes r^2 parameters per skip. The redundancy we measure under RMSNorm

is larger, $r(2r-1)$ per skip when $2r \le d_{\text{model}}$: rotations of the two hidden states that keep the skip within rank r of the identity add $r(r-1)$ (Appendix F). Whether triangular or symmetric skips train better than full or low-rank ones is an empirical question left for future work; a trained full skip can always be reduced afterwards (Section 5.3).

The skip keeps the length of the input. With $\varepsilon = 0$, $N(ch) = N(h)$ for every $c > 0$, so the first normalization of a skipless network discards the length of each input token. The skip of a residual network passes it on.

Proposition 4.2 (Absorption under RMSNorm with $\varepsilon = 0$)**.** *Consider pre-RMSNorm networks with $\varepsilon = 0$, with blocks of any kind and width.*
(i) *Every skipless network satisfies $F(cx) = F(x)$ for all $c > 0$.*
(ii) *A residual network with this property computes $x \mapsto N(xE)D$.*
(iii) *The residual networks that equal a skipless network have parameters in a set of measure zero.*

Proof in Appendix A.9.

So with $\varepsilon = 0$ no skipless network of any width equals a generic residual network, although skipless networks are limits of residual ones (Proposition 6.3(ii)).

5 Exact reduction after training

Every change of basis that a trained network keeps can be spent on structure without changing its function: one block per hidden state without skips, one block in all with them, and the orthogonal part of every learned skip.

5.1 Without skips: one block per hidden state

At a hidden state h_ℓ, pick an invertible $d_{\text{model}} \times d_{\text{model}}$ block W_1, formed by d_{model} columns of a matrix that reads h_ℓ or by d_{model} rows of a matrix that writes it.

Proposition 5.1 (One free block per hidden state)**. (i)** *In a skipless network or a network of mixed blocks, the map $W \mapsto WG$ on the matrices that write h_ℓ and $W \mapsto G^{-1}W$ on those that read it turns W_1 into the identity for $G = W_1$ if W_1 is part of a read, and for $G = W_1^{-1}$ if it is part of a write. Under RMSNorm with $\varepsilon > 0$, $G = Q$ from a QR factorization $W_1 = QT$, or $W_1^\top = QT$ for a write, turns W_1 into a triangular matrix. Done at every hidden state, this removes d_{model}^2, respectively $d_{\text{model}}(d_{\text{model}}-1)/2$, parameters per hidden state without changing the function.*
(ii) *In a residual network, the same map with one G at all hidden states preserves the function and turns one such block, in any read or write of the network, into the identity or a triangular matrix.*

Proof in Appendix A.6.

On a skipless pre-RMSNorm network with $d_{\text{model}} = 16$ and $L = 4$ the reduction zeroes 8.1 percent of the parameters and changes the function by a relative 1.8×10^{-15}; on a residual network the same step changes it by 1.5, because there the hidden states share one change of basis.

Three patterns. A transformer layer without skips has two hidden states, before its attention block and before its MLP block, and each can fix one block. (i) *Queries and output projections*, when $Hd_h = d_{\text{model}}$: $G = W_Q$ sets the queries to the identity and moves W_Q into the write of the block before, and $G = W_O^{-1}$ sets the output projection to the identity and moves W_O into the read of the next MLP block. This is how Graef (2024) removes $2d_{\text{model}}^2$ parameters per layer in transformers without skips or normalization, fifteen percent of the weights of a skipless version of Mistral-7B by his count. Any d_{model} columns of a read that form an invertible matrix will do instead, for instance some queries, some keys and some values. (ii) *MLP blocks only*: d_{model} columns of the read of each MLP block and d_{model} rows of its write are set to the identity, absorbed by the attention blocks around it, which stay full; again $2d_{\text{model}}^2$ parameters per layer. (iii) *Under normalization* the change of basis must be orthogonal, and the chosen block becomes triangular instead: $d_{\text{model}}(d_{\text{model}}-1)/2$ parameters per hidden state.

A skipless Vision Transformer. The skipless ViT-B of Ji et al. (2026) does more with less. It outperforms the residual ViT-B, which has the same parameters, in top-1 accuracy on ImageNet-1k. Assuming standard pre-LayerNorm blocks, which Proposition A.23 turns into RMSNorm of width $d_{\text{model}} - 1 = 767$, each of its 25 hidden states carries a free $\mathrm{O}(767)$. That is 24 more copies than in the residual ViT-B (Appendix E), so where the symmetries are the only redundancy, it reaches its accuracy with about 8 percent less functional dimension. The surplus can also be removed after training: one triangular 767×767 read block at each hidden state zeroes $25 \cdot 767 \cdot 766/2 = 7.3$M parameters, 8.5 percent of the model, without changing its function. The saving is in memory if the triangular blocks are stored packed, and in compute only if triangular multiplies run at dense speed, for example in the rectangular full packed format of Gustavson et al. (2010); measuring this is left for future work.

5.2 With skips: one global change of basis

A residual network keeps one change of basis for all its hidden states (Theorem 3.2(iii)). With the embedding and the readout as parameters, it is a symmetry of the whole model, with two consequences.

A counterexample. Wang and Wang (2025) characterize the symmetries of transformer blocks and state that the symmetry group of a full transformer is the product of the groups of its blocks, since residual connections and LayerNorm prevent any coupling across layers (their Theorem 7). When the embedding and the readout are among the parameters, as in the models they name and in their end-to-end checks, the change of basis that the skips leave gives counterexamples. Applied at every hidden state from the embedding to the readout, a global permutation of the stream coordinates preserves the function for every placement of the normalization and with a tied or an untied readout, as Verma and Elbayad (2024, Section 3.3) use to merge BERT models, and in pre-norm models with an untied readout so does a global rotation, the construction behind Theorem 1 of Ashkboos et al. (2024a). Both change the token embedding, which no element of the product of per-block groups does, so both lie outside that product (Appendix D, Table 5). For a bare stack of blocks that acts on a fixed input and is read in fixed coordinates, both change the function, and the counterexample does not apply.

What it also removes. The same change of basis can fix one block in the whole model, in any one read or write and at any depth (Proposition 5.1(ii)). In a pre-RMSNorm model with an untied readout it makes one $d_{\text{model}} \times d_{\text{model}}$ block triangular and removes $d_{\text{model}}(d_{\text{model}} - 1)/2$ parameters. We take Llama-3.1 as an example, a family with grouped-query attention (Grattafiori et al., 2024): the reduction removes 8.4M parameters in Llama-3.1-8B, 34M in Llama-3.1-70B and 134M in Llama-3.1-405B, that is 0.10, 0.05 and 0.03 percent (Table 6). The same holds for other pre-RMSNorm language models with multi-head or grouped-query attention and an untied readout, while multi-head latent attention, which computes keys and values from a shared low-rank latent through trainable up-projections (DeepSeek-AI, 2024, Section 2.1.1), is beyond the scope of this paper. The reduction is exact, but small; the interest of this symmetry is the counterexample above.

Tied readouts. With a tied readout, the readout $E^\top$ reads h_L through the gain g of the final normalization, and a global rotation Q must keep one matrix E at both ends, which requires $Q\,\mathrm{diag}(g')\,Q^\top = \mathrm{diag}(g)$ for some gain g'; for distinct gains only signed permutations do so. One $d_{\text{model}} \times d_{\text{model}}$ matrix inserted between the final normalization and the readout restores the rotations and absorbs the gain, but it costs d_{model}^2 parameters and frees only $d_{\text{model}}(d_{\text{model}} + 1)/2$: for identity skips it does not pay (Appendix F).

5.3 Learned skips after training

Networks trained with learned skips reduce exactly after training. Without normalization every invertible skip matrix is removed, the product of all of them moves into the embedding or the readout (Theorem 3.2(i)), and the change of basis that is left fixes one more block, in any one read or write (Proposition 5.1(ii)). Under RMSNorm every skip becomes triangular or symmetric

positive definite (Theorem 4.1(i)), which removes $d_{\text{model}}(d_{\text{model}}-1)/2$ parameters per skip; started at any hidden state instead of h_0, the recursion of Theorem A.24(i) also makes one block of a read or a write of that state triangular (Appendix A.8). With a tied readout under RMSNorm, the embedding also reads h_L, which fixes the rotations at h_0 and h_L, for generic gains up to signs and permutations; those at $h_1, \dots, h_{L-1}$ stay free. They make any $L-1$ of the skips triangular, and the remaining one keeps its full matrix: $(L-1)d_{\text{model}}(d_{\text{model}}-1)/2$ parameters are removed. One more $d_{\text{model}} \times d_{\text{model}}$ matrix inserted before the readout frees the rotations at h_0 and h_L and absorbs the gain, $d_{\text{model}}(d_{\text{model}}-1)/2 + d_{\text{model}}(d_{\text{model}}-1)/2 + d_{\text{model}} = d_{\text{model}}^2$ parameters, exactly its cost: it does not pay. Without normalization the tie only requires $G_L = G_0^{-\top}$: any $L-1$ of the skips become identities, and the free G_0 sets one block of any one read or write to the identity, which removes Ld_{model}^2 parameters. At $d_{\text{model}} = 8$ and $L = 5$ the function changes by 5×10^{-16} for triangular skips, 4×10^{-15} for symmetric ones and 7×10^{-16} with a tied readout, where 112 parameters are removed (Appendix F). Low-rank skips are removed entirely without normalization, and give up at least the r^2 parameters inside the product under RMSNorm (Section 4).

6 When the count is functional dimension

The symmetry removed by the skips is a gain in functional dimension when the symmetries are the only redundancy of both networks; we prove this for one block, attention included, prove a gain of at least $L \dim \mathcal{G}$ at any depth for networks of smooth blocks, and measure it in deeper MLPs and small transformers. In exchange the skips lose no function in the limit: every skipless network without biases or normalization is a limit of residual networks of the same widths.

The gain of Theorem 3.3(iii) needs the equality condition, for the residual and for the skipless network. A *ReLU*^{2} block (So et al., 2021) uses $\sigma(z) = \max(z, 0)^2$, and a *ReGLU* block (Shazeer, 2020) is the GLU block with the gate $\max(z, 0)$. The symmetries inside a block rescale its m hidden units for ReLU and ReLU^2, so each block contributes m to s, and gates and values separately for ReGLU, so each block contributes $2m$; $\tanh$ and GELU blocks contribute nothing. An attention block of H heads of width d_h contributes $2Hd_h^2$: in each head, a change of basis between queries and keys and one between values and outputs.

6.1 One block

For networks with one block the equality condition is a theorem, except for the skipless network under RMSNorm, for which we prove only the bound of Theorem 3.3(ii) (Theorem C.1 in Appendix C). Let $d_{\text{model}} \geq 2$, and let the block have width at least d_{model}, with inputs and outputs of dimension at least d_{model}: a $\tanh$, GELU, ReLU, ReLU^2, ReGLU, SwiGLU or GeGLU block, or an attention block with heads of width at most d_{model}, causal or not, on sequences of at least two tokens. Without normalization, with or without biases, at generic parameters, the skipless and the residual network both satisfy the equality condition, so the skip raises the functional dimension by exactly d_{model}^2; for attention without biases this extends to a learned absolute position embedding. Under pre-RMSNorm with $\varepsilon > 0$, without biases, the residual network satisfies it too, and the skip adds at least $d_{\text{model}}(d_{\text{model}}-1)/2$. A learned skip, full or triangular, adds exactly $d_{\text{model}}(d_{\text{model}}+1)/2$ more, the dimension of the triangular $d_{\text{model}} \times d_{\text{model}}$ matrices. With $\varepsilon = 0$ and ReLU, ReLU^2 or ReGLU blocks, both networks satisfy it, and the skip adds exactly d_{model}^2; with the other blocks the skip adds at least $d_{\text{model}}(d_{\text{model}}-1)/2 + 1$, the dimension of $\mathrm{CO}(d_{\text{model}})$.

The proof uses only linear algebra and limits in one variable. The parameters enter through $M = ED$, $A = W_{\text{write}}D$ and $C = EW_{\text{read}}$, as $x \mapsto f(xC)A$ without the skip and $x \mapsto xM + f(xC)A$ with it. A factorization of a rank-d_{model} matrix into two factors of rank d_{model} is unique up to an invertible $d_{\text{model}} \times d_{\text{model}}$ matrix between them, so the skipless network carries two changes of basis, and $M = ED$ ties them into one. It remains to show that a direction that leaves the function unchanged to first order moves M, A and C only along the symmetries of the hidden units, which are isolated by their kinks for the ReLU-type activations, and along rays by how fast they approach their limits for $\tanh$, GELU, SwiGLU and GeGLU. On inputs with two distinct tokens, the output of

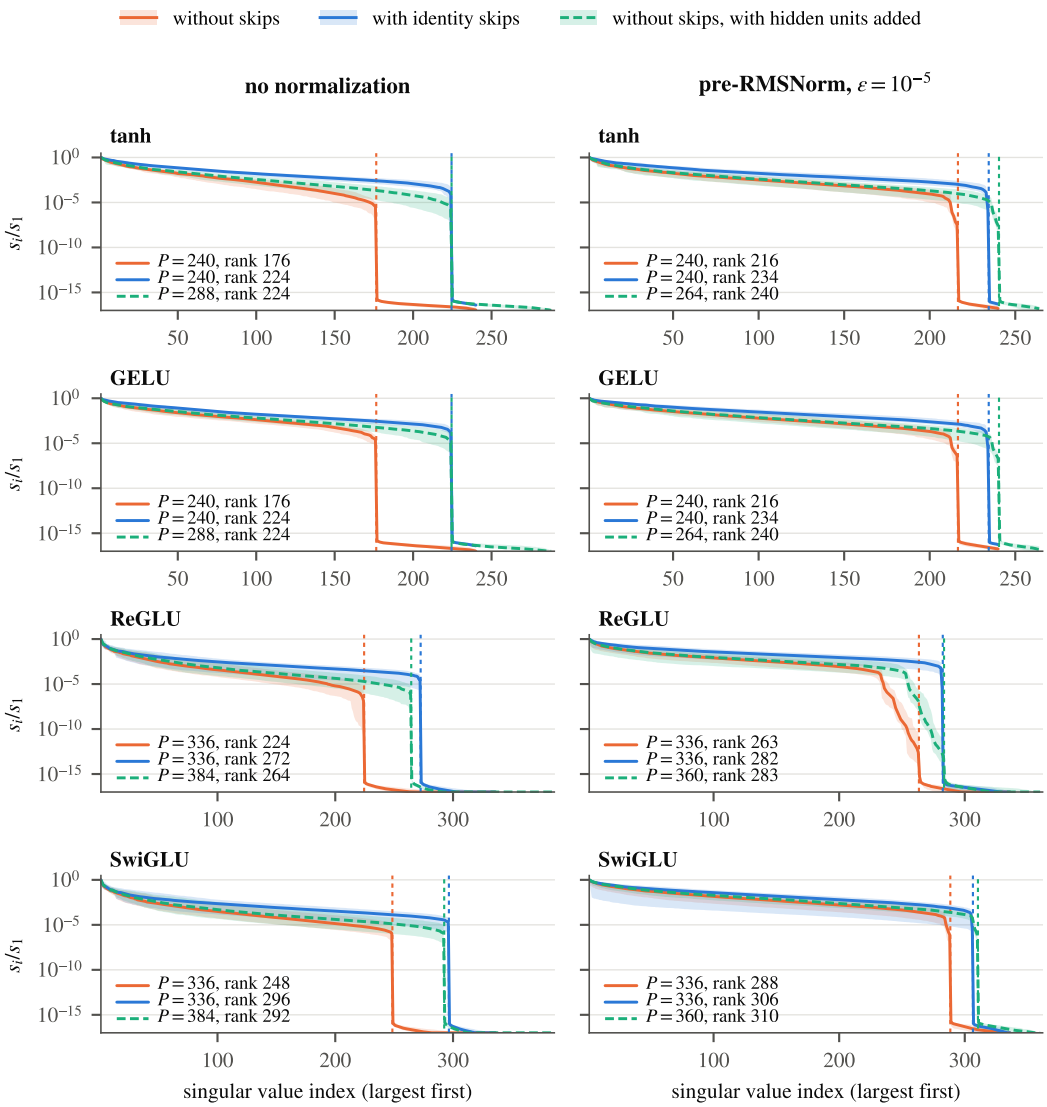


Figure 4: Singular values of the exact Jacobian of the parameter-to-output map on 4000 inputs, for an embedding $\mathbb{R}^6 \to \mathbb{R}^4$, $L = 3$ MLP blocks of width 8 and a readout $\mathbb{R}^4 \to \mathbb{R}^6$, without biases: with and without identity skips, for the same weights, and without skips with hidden units added that bring at least $L \dim \mathcal{G}$ more parameters. Left column: without normalization. Right column: with pre-RMSNorm, $\varepsilon = 10^{-5}$, before every block and the readout. Lines show the median over 20 draws of $\mathcal{N}(0, 1/\text{fan-in})$ weights and $\mathcal{N}(0, 1)$ inputs, and bands the range. Each panel lists the number P of parameters and the rank of each network in the first draw, read at the largest gap, the largest ratio of consecutive singular values, which the dashed lines mark. Except for the skipless ReGLU networks under RMSNorm (Section 6.2), the rank equals the symmetry count in every draw, at a gap of at least 10^4: $P - (L+1)\dim \mathcal{G} - s$ without skips and $P - \dim \mathcal{G} - s$ with them, with $\dim \mathcal{G} = d_{\text{model}}^2 = 16$ without normalization and $d_{\text{model}}(d_{\text{model}} - 1)/2 = 6$ under RMSNorm, and s the rescalings of hidden units.

an attention block at the second token is a $\tanh$ network with one unit per head, since a softmax over two distinct scores is a logistic function of their difference.

6.2 Deeper networks

For deeper networks we compute the rank of the Jacobian of the parameter-to-output map on a large batch of inputs (Appendix F). This rank is a lower bound on the functional dimension, and the

Measured	Depth L	Section	Normalization	$\operatorname{fdim}\mathcal{R} - \operatorname{fdim}\mathcal{S}$
$\tanh$, GELU, GLU variants	3	6.2	none, RMSNorm	$L \dim \mathcal{G}$
$\tanh$, $d_{\text{model}} = 16$	6	6.2	none, RMSNorm	$L \dim \mathcal{G}$
nanoGPT: attention, GELU	2–8	6.2	none, LayerNorm	$L \dim \mathcal{G}$
ReLU^2	3	F	none	Ld_{model}^2
ReLU	3	F	none	$\geq Ld_{\text{model}}^2$

Table 3: Measured at random parameters (Appendix F), without biases; RMSNorm with $\varepsilon = 10^{-2}$. $\dim \mathcal{G}$ is d_{model}^2 without normalization, $d_{\text{model}}(d_{\text{model}} - 1)/2$ with RMSNorm and $(d_{\text{model}} - 1)(d_{\text{model}} - 2)/2$ with LayerNorm.

symmetry count is an upper bound (Theorem 3.3(ii)). Where the two meet, the functional dimension is exactly the count, which is the equality condition.

For residual networks of smooth blocks the equality condition holds at every depth.

Theorem 6.1 (Residual networks at any depth)**.** *Let the blocks be smooth, in any order and of any width.*
(i) *At generic parameters the residual network of L blocks satisfies the equality condition: its functional dimension is $P - \dim \mathcal{G} - s$. Without normalization or attention blocks, it does so with biases too, with a functional dimension $(L + 1)d_{\text{model}}$ lower.*
(ii) *If every block has width at least d_{model}, the skips raise the functional dimension by at least $L \dim \mathcal{G}$, and by exactly $L \dim \mathcal{G}$ when the skipless network satisfies the equality condition too.*

Proof in Appendix A.4.

With small writes, a residual network behaves, to first order, like one with a single wide block that holds all its heads and units, and there its rank is the largest that its symmetries allow (Lemma A.19). An analytic network keeps its largest rank at generic parameters. The argument needs the skip: without it, small writes leave no linear path from the embedding to the readout. For ReLU-type blocks it holds only near small writes (Remark A.20). For the skipless network the equality condition remains measured, with two exceptions. For $\tanh$ blocks, the function of a generic network determines its weights up to changes of basis at its hidden states and permutations and sign flips of its units (Fefferman, 1994). For GELU blocks of width at least $2d_{\text{model}}$, a skipless network can equal a residual network with d_{model} fewer units per block (Theorem 3.6(iv)), and around it the reads of the added units supply the Ld_{model}^2 directions that the residual network lacks.

Theorem 6.2 (Exact gain for $\tanh$ and GELU)**.** *Without normalization, in the two cases below the skipless network satisfies the equality condition too, and the skips raise the functional dimension by exactly Ld_{model}^2.*
(i) $\tanh$ blocks. *At generic parameters the skipless network has functional dimension $P - (L + 1)d_{\text{model}}^2$, and $(L + 1)d_{\text{model}}$ lower with biases.*
(ii) GELU blocks. *If every block has width at least $2d_{\text{model}}$ and there are no biases, then at generic parameters the skipless network has functional dimension $P - (L + 1)d_{\text{model}}^2$.*

Proof in Appendix A.4.

MLPs. Under RMSNorm the networks of Figure 4 use $\varepsilon = 10^{-5}$, the value that nanoGPT uses in its LayerNorm (Karpathy, 2023). In every draw, except in the skipless ReGLU networks under RMSNorm, to which we return below, the two bounds meet: the singular values drop to the floor of double precision exactly at the count, so the skips add $Ld_{\text{model}}^2 = 48$ without normalization and $Ld_{\text{model}}(d_{\text{model}} - 1)/2 = 18$ under RMSNorm. Hidden units added to the skipless network instead raise its rank by one per parameter in $\tanh$ and GELU blocks: 6 units, 48 parameters, reach the rank of the residual network without normalization, and under RMSNorm 3 units, 24 parameters, the fewest whole units that bring at least 18, exceed it by 6. A gated unit brings $3d_{\text{model}} = 12$ parameters and its own rescalings, two for ReGLU and one for SwiGLU, GeGLU and GLU, so 4 gated units, 48 parameters, fall short of the residual network by 8 and 4. The measured redundancy also equals the symmetry count for $\tanh$ networks up to $d_{\text{model}} = 16$ and $L = 6$, with and without RMSNorm, and with biases at $d_{\text{model}} = 4$ (Table 7). After training on four small classification datasets, where

the networks beat a linear classifier, the rank still equals the count, with the same exception and two others, and the skips still add 48 and 18 (Figure 5).

Homogeneous blocks under RMSNorm. Without biases, ReLU, ReLU^2 and ReGLU blocks are positively homogeneous: $B(\lambda h) = \lambda^k B(h)$ for $\lambda > 0$, with $k = 1$ for ReLU and $k = 2$ for ReLU^2 and ReGLU. With $\varepsilon = 0$, where $N(\lambda h) = N(h)$ for $\lambda > 0$, the normalization after such a block would therefore cancel the scaling by the one before it, $N(B(N(h))) = N(B(h))$, and every normalization but the last would drop out of a skipless network. The changes of basis of this network at $h_0, \dots, h_{L-1}$ could then be any invertible matrices, and the one at h_L any matrix in $\mathrm{CO}(d_{\mathrm{model}})$ (Lemmas A.1 and A.2): $Ld_{\mathrm{model}}(d_{\mathrm{model}}+1)/2+1$ more dimensions of symmetry than with $\varepsilon > 0$, 31 for the networks of Figure 4. The residual network keeps one change of basis in $\mathrm{CO}(d_{\mathrm{model}})$ for all its hidden states (Theorem A.3(iii)), so the skips would remove Ld_{model}^2 dimensions of symmetry, as without normalization. With $\varepsilon > 0$ these directions change the function, but they only rescale the output of each input, by a factor that every later normalization brings closer to 1: a normalization passes on a fraction $\varepsilon/(r^2+\varepsilon)$ of a change in the scale of a row with mean square r^2, so along the directions at h_ℓ with $\ell < L$ the output changes by about $(k\varepsilon/r^2)^{L-\ell}$ of its size when $\varepsilon \ll r^2$, and along the scaling at h_L by about ε/r^2 (Lemma A.2). In the skipless ReGLU networks of Figure 4 the singular values of these directions fall in steps from about 10^{-4} to the floor of double precision, which the smallest of them reaches in most draws: the rank read at the largest gap is one below the count in 14 of 20 draws, and in 19 of 20 with hidden units added. With $\varepsilon = 10^{-12}$ the network without added units has rank 31 below the count in every draw and the residual network has rank $P - \dim \mathrm{CO}(d_{\mathrm{model}}) - s$, so the skips add $Ld_{\mathrm{model}}^2 = 48$; with $\varepsilon = 10^{-2}$ both skipless networks have the count in every draw (Appendix F).

Transformers. We take the model of nanoGPT (Karpathy, 2023) with $d_{\mathrm{model}} = 8$, four heads of size two, a vocabulary and a context of 8, and no biases, as its training script sets them, with a readout not tied to the embedding. We keep its LayerNorm or replace it by the identity, and removing both skips of each layer gives the skipless network with the same parameters. With 1 to 4 layers and 4 random draws each, all 64 networks have the predicted redundancy, and the skips add $L(d_{\mathrm{model}}-1)(d_{\mathrm{model}}-2)/2 = 21L$ with LayerNorm and $Ld_{\mathrm{model}}^2 = 64L$ without (Table 8). The rest of the redundancy, the same with and without skips, comes from symmetries inside the heads and from LayerNorm; for one and two layers we construct all of them and find that they span the null space of the Jacobian.

6.3 Skipless networks as limits of residual networks

Every skipless network without biases or normalization is a limit of residual networks with the same widths, so the gain in functional dimension does not come at the price of functions.

Proposition 6.3 (Skipless networks as limits of residual networks)**.** *Let S be a skipless network whose maps f_ℓ are locally Lipschitz.*
(i) Without normalization. *The network S is a limit of residual networks with the same widths and the same maps f_ℓ, uniformly on bounded sets of inputs.*
(ii) Under pre-RMSNorm with $\varepsilon = 0$. *If no hidden state of S has a zero row at any input x whose rows of xE are nonzero, the same holds on bounded sets of such inputs.*

Proof in Appendix A.10.

Blocks that outweigh their skips. The proof takes the residual network F_t, for $t \geq 1$, that keeps the embedding of S and scales the reads of block ℓ by $t^{1-\ell}$, its writes by t^ℓ and the readout by t^{-L}. Its block ℓ computes $h \mapsto t^\ell B_\ell(t^{1-\ell}h)$. Let $s_0 = xE$ and $s_\ell = B_\ell(s_{\ell-1})$ be the hidden states of S. If the hidden state of F_t before block ℓ is $t^{\ell-1}(s_{\ell-1} + r_{\ell-1})$, the block sees $s_{\ell-1} + r_{\ell-1}$ and returns $t^\ell B_\ell(s_{\ell-1} + r_{\ell-1})$, which outweighs the skip by a factor t. The hidden states of F_t are thus $t^\ell(s_\ell + r_\ell)$, with $r_0 = 0$ and

$$r_\ell \;=\; B_\ell(s_{\ell-1} + r_{\ell-1}) - B_\ell(s_{\ell-1}) \;+\; t^{-1}\,(s_{\ell-1} + r_{\ell-1}). \tag{6}$$

The difference is at most a local Lipschitz constant of B_ℓ times $\|r_{\ell-1}\|$, and the last term is $O(1/t)$, so $r_L = O(1/t)$ and $F_t(x) = t^{-L}\,t^L(s_L + r_L)D = S(x) + O(1/t)$ (Remark A.25).

No activation sees a large argument. The large constants sit in the writes, and each read divides by the size $t^{\ell-1}$ that the stream has reached, so block ℓ of F_t evaluates f_ℓ at $(s_{\ell-1} + r_{\ell-1})W^\ell_{\text{read}}$, the pre-activations of S up to $O(1/t)$. A GELU, for instance, is evaluated where the GELU of S is, away from the large arguments at which it would act like a ReLU. Only the stream grows, to order t^L before the readout, which divides by t^L. The construction keeps the widths and uses nothing about the blocks but local Lipschitz continuity, so it applies alike to $\tanh$, GELU, ReLU, ReLU^2, gated and attention blocks. Its weights diverge as $t \to \infty$: the proposition says which functions residual networks reach in the limit, and leaves open how to parametrize them for training. In our checks, at $d_{\text{model}} = 4$ and $L = 3$, the error of F_t relative to the largest output of S is below 10^{-2} at $t = 1000$ for $\tanh$, GELU, ReLU, SiLU, ReLU^2, GLU, ReGLU, SwiGLU, GeGLU and attention blocks, with t times the error nearly constant (Appendix F).

Under RMSNorm. RMSNorm with $\varepsilon = 0$ ignores the size of the stream, so for Proposition 6.3(ii) the reads and the readout of F_t keep their scale and only the writes grow. With $\varepsilon > 0$ the same construction fails: on a stream of size t^ℓ, RMSNorm acts as RMSNorm with $\varepsilon/t^{2\ell}$, so F_t tends to the skipless network that has $\varepsilon = 0$ in every normalization after the first. Whether skipless networks with $\varepsilon > 0$ are limits of residual ones is open.

Weight decay. In training, the gains of RMSNorm are usually exempt from weight decay; nanoGPT, for instance, decays only its matrices (Karpathy, 2023). It penalizes the diverging writes of F_t, and in a pre-RMSNorm network without biases the gains are then the only parameters that can grow without penalty. A gain c multiplies the input of a read, so it puts a large constant before the activation. For positively homogeneous blocks, such as ReLU, ReLU^2 and ReGLU blocks, this only scales the output of the block, which can then outweigh its skip as in F_t. So can gated blocks with any gate, SwiGLU and GeGLU included: if the read splits into gates and values, $W_{\text{read}} = [\, W_{\text{g}} \mid W_{\text{v}} \,]$, the read $[\, W_{\text{g}}/c \mid W_{\text{v}} \,]$ gives the gate its old argument on the input cx and passes c to the values, so the block returns $c\,B(x)$ with a smaller gate read. A vanilla GELU block has no such linear path. Its activation sees large arguments, where $\text{GELU}(cz)/c \to \max(z, 0)$ as $c \to \infty$, so it can outweigh its skip through its gains only by acting as a ReLU block, or with a large write, which weight decay penalizes. This is where GELU breaks, and so does SiLU.

7 Related work

What skips change in training. Residual networks behave like ensembles of relatively shallow networks: deleting a single block from a trained residual network, unless it downsamples, barely changes its accuracy (Veit et al., 2016). By passing shallow features on to deeper layers, skips may also make the learned features less abstract, as Zhang et al. (2026) hypothesize. In a ViT-B masked autoencoder, they scale each skip of the encoder by a factor that decreases with depth and add skips from the encoder to the decoder, which raises the ImageNet-1k accuracy of a linear classifier on the learned features from 67.8 to 72.7 percent. Ji et al. (2026) remove the skips for the same reason.

Changes of basis at hidden layers. Zhao et al. (2023) define a change of basis at every hidden layer of an MLP, which preserves the function where the activation there is linear and, for orthogonal changes, where it is a radial rescaling (their Proposition C.1 with Examples C.3 and C.6). For one hidden layer they compute the dimension of the set of parameters that it reaches (their Theorem 3.3) and the conserved quantities of gradient flow, and use both to study extended flat minima. They remark that Proposition C.1 adapts to networks with skip connections (their Appendix C.5); what the skips do to the symmetry is not worked out there. Written as d_{model} trainable units with the identity as activation, next to the hidden units of a vanilla MLP block without normalization, a skip fits their proposition as it stands, and propagating a change of basis through the skips, the tool of our gauge fix, is their change of basis at the identity units combined with those at the two hidden states that each skip joins (Appendix A.2). We use the same tool for a different result: the tie of Theorem 3.2(iii), the count of Theorem 3.3, and the reductions of Section 5. Godfrey et al. (2022) characterize the linear maps that pass through an elementwise activation, and show that in residual ReLU networks permutations and positive rescalings of hidden units preserve the function when they are equal at all hidden states that the residual connections join (their Remark 3.6 and Appendix E.9.2).

The basis of the residual stream. Elhage et al. (2021) observe that the residual stream has no privileged basis. Ashkboos et al. (2024a) prove that an RMSNorm transformer computes the same function after one orthogonal transformation of every weight matrix touching the stream, a rotation that the skips propagate from the embedding to the readout (their Section 3.1 and Theorem 1). QuaRot and SpinQuant use the same invariance for quantization (Ashkboos et al., 2024b; Liu et al., 2025), model merging aligns residual streams with one global orthogonal matrix (Theus et al., 2025), and Sweeney (2026) studies the residual coordinates when normalization gains are kept as parameters; all of these use the one global change of basis that a residual network keeps. To sparsify trained models, SliceGPT rotates each block by the principal components of the signals between blocks on calibration data, inserts $Q_{\ell-1}^{\top} Q_{\ell}$ into each skip so that neighboring blocks can carry different rotations, and deletes the rows and columns that correspond to the minor principal components (their Sections 3.3 and 3.4). The sparsity is structured: every weight matrix becomes a smaller dense matrix, and the sliced model approximates the original one. We propagate changes of basis through the skips in the opposite direction: SliceGPT inserts matrices into identity skips, while we remove the matrices of learned skips without changing the function. van Nierop (2024, Sections 4.4 and 4.5) counts the symmetries of LayerNorm transformers, with one rotation of the stream for the whole model; what the skips remove is of order Ld_{model}^2 against his $Ld_{\text{model}}d_h + d_{\text{model}}^2$, 7.1M against 1.5M in ViT-B/16 (Appendix E).

Removing weights without skips. The merges of Graef (2024) are the first of our three patterns (Section 5.1). In a skipless network, the change of basis at each hidden state reaches the identity values and projections of He and Hofmann (2024) without normalization, and triangular ones under it (Proposition 5.1(i)); each skip ties the changes of basis of two hidden states, which is consistent with the direction of the effect in their Figure 23. RMNet removes the skips of trained convolutional networks by emulating the identity with ReLU or PReLU units (Meng et al., 2021), at the cost of more units, and LAuReL trains learned skips that differ from the identity by a matrix of low rank (Menghani et al., 2025). The skips of Zhang et al. (2026), scaled by factors that decrease with depth, are scalar multiples of the identity: without normalization, or with normalization and $\varepsilon = 0$, changes of basis turn them into identity skips with the same function (Theorem 3.2(i), Theorem A.24(ii)), so the decay acts through training.

Symmetries and residual networks in theory. Wang and Wang (2025) characterize the symmetries of attention and MLP blocks; their statement for the full model has the counterexamples of Section 5.2. Concurrently with and independently of our first version (7 October 2026, `https://github.com/MarkoKarbevski/skips_fix_gauges/commit/8128617`), Dahlgren et al. (2026) prove that residual networks with layers $f_{\ell} = f_{\ell-1} + \sigma(W_{\ell} f_{\ell-1} + b_{\ell})$ and polynomial activations are generically identifiable: the function determines the weights and biases up to finitely many sign flips (their Theorems 3.1–3.3). Their layers have no write matrix, so the changes of basis that our skips remove do not arise; the skips there break the rescalings and permutations of neurons. Marcotte et al. (2025) show that a block has the same conservation laws of gradient flow with and without its skip (their Proposition 3.2). Likewise, without embedding or readout the skip does not change the derivative of a block with respect to its parameters, so the gain needs junctions outside the block, such as an embedding and a readout. Lin and Jegelka (2018) count the identity map as d_{model} hidden units, the view we take throughout, and MacDonald et al. (2023) show that identity skips around contractive branches keep the Jacobian of the parameter-to-output map of full row rank on linearly independent inputs, which gives global convergence of gradient descent for residual networks with normalized weights; we study the kernel of the same Jacobian.

8 Scope and future work

What is proved, measured and conjectured. The gauge fixes of Theorems 3.2 and 4.1 and Proposition 5.1, the count of Theorem 3.3, the reductions of Section 5 and the limits of Proposition 6.3 hold for every network of the stated kind. That the count is a gain in functional dimension is proved for networks with one block, attention included, and measured in deeper MLPs and small transformers. For networks of smooth blocks, the residual half of the equality condition holds at any depth, so the

gain is at least $L \dim \mathcal{G}$ (Theorem 6.1). Beyond networks of $\tanh$ blocks or of GELU blocks of width at least $2d_{\text{model}}$ without normalization (Theorem 6.2), the exact gain in deeper networks, and the gain for ReLU-type blocks away from small writes (Remark A.20), rest on the equality condition, which can fail: for ReLU networks the functional dimension varies on sets of positive measure (Grigsby et al., 2023), and some skipless networks are more redundant than their symmetries (Appendix F). Under RMSNorm with $\varepsilon = 0$, every normalization but the last drops out of a skipless network of positively homogeneous blocks, which then has more symmetries than the count (Section 6.2). We expect the equality condition for the common blocks:

Conjecture 1 (Equality condition)**.** *Within the scope of Section 2, consider networks whose embedding is linear in continuous inputs and whose blocks are all* $\tanh$*, all GELU, all SwiGLU, all GeGLU or all ReGLU blocks. At generic parameters the skipless and the residual network both satisfy the equality condition, so the skips raise the functional dimension by exactly* $L \dim \mathcal{G}$*.*

For $\tanh$, GELU, SwiGLU and GeGLU blocks, the residual half of the conjecture is Theorem 6.1; without normalization, both halves hold for $\tanh$ blocks and for GELU blocks of width at least $2d_{\text{model}}$ (Theorem 6.2). A ReLU unit that never switches on leaves $2d_{\text{model}} - 1$ directions flat, while a GELU unit is constant on no open set of inputs. Whether trained ReLU networks lose functional dimension this way, and whether this affects how they compare with GELU networks, is left for future work.

Beyond the scope of this paper. Input-dependent skips, such as the hyper-connections of mHC (Xie et al., 2025) and DeepSeek-V4 (DeepSeek-AI, 2026), which mix several streams with weights computed in part from the stream itself, make the junction between blocks nonlinear and are beyond the scope of this paper. So are quantized weights, whose grid a change of basis does not preserve, and streams that pass through an elementwise nonlinearity, as in the original ResNet (He et al., 2016), which keep only permutations and positive rescalings at each hidden state.

Training in the reduced forms. Our claim is about trained networks: in the worst case, the reductions of Section 5 are applied after training, where they lower the memory of the model exactly. Whether they also help during training is left for future work. Two direct tests are to train a skipless network in the reduced parametrization of Proposition 5.1(i) against the full one, and to train learned skips under RMSNorm in their triangular or symmetric form (Theorem 4.1) against full and low-rank ones. Optimizers already use the rotations of the stream: ARO can share one rotation among all matrices that read or write the stream, as the global symmetry allows, or one per transformer layer, a relaxation that it motivates where the output of a layer outweighs its skip (Gong et al., 2026, Sections 6.4 and 6.5).

Orthogonal or symmetric skips, at depth and at width. Under RMSNorm the two parts of a learned skip do different things (Section 4). Orthogonal skips are gauge: a network with orthogonal skips computes the function of a residual network whose hidden states have the same norms (Theorem 4.1(ii)), so they can change only how the network trains. With them, a separate rotation at every hidden state is an exact symmetry; with identity skips it holds only approximately, where the branch outweighs the skip (Section 3.2), the regime in which ARO uses one rotation per layer (Gong et al., 2026, Section 6.4). They also let each block see the stream in a basis of its own, which matters to optimizers that act on each coordinate separately. Symmetric or triangular skips carry what learned skips add to the functional dimension, $d_{\text{model}}(d_{\text{model}} + 1)/2$ per skip under the equality condition (Theorem 4.1(iii)), and they change norms: the output of block ℓ reaches h_L multiplied by $A_{\ell+1} \cdots A_L$, which keeps its norm when the skips are orthogonal and can otherwise damp or amplify it geometrically with depth. At width, both kinds cost about $d_{\text{model}}^2/2$ parameters per skip, and a full skip spends $d_{\text{model}}(d_{\text{model}} - 1)/2$ of its d_{model}^2 parameters, a share that approaches one half, on its orthogonal part. Whether orthogonal or symmetric and triangular skips bring more value, per parameter and per unit of compute, as depth and width grow, is left for future work.

Continuous depth. Geshkovski et al. (2025) model a transformer as a flow in continuous time: the layers become time, the tokens interacting particles, and layer normalization, with its gain set to the identity, keeps them on the sphere. Their view follows the neural ODEs of residual networks, in which the skip is what makes a layer an Euler step, and what the gauge fix becomes in this limit is an

open question. In a flow, a change of basis $h \mapsto hG(t)$ at every time adds the linear term $h\,G^{-1}G'$ to the velocity field, where $G' = \mathrm{d}G/\mathrm{d}t$, and a linear term $hA(t)$ transforms like a non-Abelian gauge field, $A \mapsto G^{-1}AG + G^{-1}G'$, as Hashimoto et al. (2024) show for linear neural ODEs. A field without a linear term thus keeps its form only for constant G, the analogue of the one change of basis that the skips leave, and with an embedding and a readout to absorb $G(0)$ and $G(T)$ every linear term can be removed, the analogue of Theorem 3.2. Geshkovski et al. (2023) remove one in this way: their rescaled tokens $e^{-tV}x_i$, with tokens as columns, take the term Vx_i out of the self-attention field $\sum_j P_{ij}Vx_j = Vx_i + \sum_j P_{ij}V(x_j - x_i)$, at the price of query and key matrices Qe^{tV} and Ke^{tV} that depend on time. In discrete time each layer is $x_i \mapsto Rx_i + \Delta t \sum_j P_{ij}V(x_j - x_i)$ with $R = I + V\Delta t$, and their rescaling R^{-k} at the k-th hidden state (Geshkovski et al., 2023, Remark 3.4) is the change of basis with which the forward version of the gauge fix (Appendix A.2) removes the learned skips R. How the counts, the gain in functional dimension and the reductions carry over to such flows is left for future work, in particular to flows whose parameters change with time, with which Geshkovski et al. (2026) match N input measures to N target measures, and to flows on the sphere, which only orthogonal changes of basis preserve, such as the self-attention model, whose tokens stay near several clusters for an exponentially long time (Geshkovski et al., 2024).

Use of AI

The conception of the study is the authors' own.[1] One author obtained their results, proofs and experiments working with AI assistants, through continuous conversation; the assistants were used to find, verify and simplify proofs and to write the code of the experiments, which the authors validated.

Code availability

The notebook that reproduces the numbers, tables and figures of this note (Appendix F), with its pinned environment and the data from which it draws Figure 4 and Table 8, will be released in a public Git repository, coming soon.

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[1] This note builds on our preprint *Can an MLP Absorb Its Own Skip Connection Exactly?* (https://arxiv.org/abs/2604.23705); Section 3.4 restates its results on absorption and extends some of them.

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A Proofs

Throughout, the tokens of a hidden state are its rows; where every block acts on each token separately, as MLP blocks do, it suffices to take one token, $T = 1$, so that $h \in \mathbb{R}^{1\times d_{\text{model}}}$ is a row vector. Biases, when present, enter as $B_\ell(h) = f_\ell(hW^\ell_{\text{read}} + b^\ell_{\text{read}})W^\ell_{\text{write}} + b^\ell_{\text{write}}$, with row vectors b added to every token. Each subsection recalls the results of the main text named in its title, explains what the proof does, and then proves them; auxiliary results are stated where they are first used. The one-block theorem, Theorem C.1, is proved in Appendix C.

A.1 Junctions (Proposition 3.1)

We recall Proposition 3.1 from Section 3.1.

Proposition 3.1 (Junctions)**. (i) Only UV matters.** *In a mixed block, $(U, V) \mapsto (UK, K^{-1}V)$ preserves the function for every $K \in \mathrm{GL}(d_{\text{model}})$.*
(ii) Hidden states are junctions. *In a skipless network, and in a network of mixed blocks, the map $W \mapsto WG$ on the matrices that write a hidden state and $W \mapsto G^{-1}W$ on those that read it preserves the function for every $G \in \mathcal{G}$. For $\varepsilon = 0$ and $G = cQ$, a read through RMSNorm maps to $Q^\top W$ instead.*
(iii) One group per junction. *The maps of (i) and (ii) commute when they act at different junctions. A skipless network therefore carries one copy of $\mathcal{G}$ per hidden state, and a network of mixed blocks one more copy of $\mathrm{GL}(d_{\text{model}})$ per block.*

Overview. A change of basis at a junction is undone by the reads of that junction, and changes at different junctions multiply the matrices they share from different sides, so they commute; biases add a translation. The only real work is under normalization, where a read sees $N(hG)$ rather than hG: Lemma A.1 determines which changes of basis can be moved through N.

Lemma A.1 (RMSNorm equivariance)**.** *Let $G \in \mathrm{GL}(d_{\text{model}})$.*
(i) *There is a matrix M with $N(hG) = N(h)M$ for all row vectors $h \neq 0$ if and only if $G \in \mathrm{CO}(d_{\text{model}})$ when $\varepsilon = 0$, and $G \in \mathrm{O}(d_{\text{model}})$ when $\varepsilon > 0$. Then $M = Q$, where $G = cQ$ with Q orthogonal.*
(ii) *If $\varepsilon > 0$ and $G = cQ$, then $\|N(hG) - N(h)Q\|/\|N(h)\| = \frac{\varepsilon}{2r^2}|1 - c^{-2}| + O(\varepsilon^2)$, where $r^2 = \|h\|^2/d_{\text{model}}$.*

Proof. Outline. We first take $\varepsilon = 0$, where $N(h)$ depends only on the direction of h, and show that a matrix that can be moved through N is orthogonal up to a positive scale. We then show that $\varepsilon > 0$ removes the scale, and expand the resulting error in ε.

Let $\varepsilon = 0$, so $N(h) = \sqrt{d}_{\text{model}}\, h/\|h\|$. If $G = cQ$ then $N(h\,cQ) = \sqrt{d}_{\text{model}}\, c\, hQ/(c\|h\|) = N(h)Q$. Conversely, if $hG/\|hG\| = hM/\|h\|$ for all $h \neq 0$, taking norms gives $\|hM\| = \|h\|$, so M is orthogonal, and $hGM^\top = (\|hG\|/\|h\|)\, h$: every nonzero row vector is a left eigenvector of $GM^\top$, so $GM^\top = cI$ with $c = \|hG\|/\|h\| > 0$, and $G = cM$. Let $\varepsilon > 0$. For $G = cQ$, $N(h\,cQ) = c\,hQ/\sqrt{c^2\|h\|^2/d_{\text{model}} + \varepsilon}$ and $N(h)Q = hQ/\sqrt{\|h\|^2/d_{\text{model}} + \varepsilon}$; these agree for all h if and only if $c = 1$. If $N(hG) = N(h)M$ for all h with $\varepsilon > 0$, letting $\|h\| \to \infty$ along rays reduces to the case $\varepsilon = 0$, so $G = cM$ with M orthogonal, and the computation just made forces $c = 1$. The expansion of the deviation is the Taylor expansion of $c\sqrt{r^2 + \varepsilon}/\sqrt{c^2r^2 + \varepsilon} - 1$ in ε. □

Lemma A.2 (Homogeneous blocks under RMSNorm)**.** *Let a skipless network with pre-RMSNorm, $h_\ell = B_\ell(N(h_{\ell-1}))$ for $\ell = 1, \ldots, L$ and $y = N(h_L)D$, have blocks that are positively homogeneous of degree k, and let $r_i^2 = \|h_i\|^2/d_{\text{model}}$. Apply the change of basis $G \in \mathrm{GL}(d_{\text{model}})$ at h_ℓ with $\ell < L$, or the scaling $G = cI$ at h_L with its read D unchanged. At every input whose hidden states are nonzero, the output becomes Λy with a scalar $\Lambda > 0$, and*
(i) *$\Lambda = 1$ if $\varepsilon = 0$ or G is orthogonal;*
(ii) *for $G = e^{tS}$ with S symmetric and $\ell < L$,*

$$\frac{d}{dt}\log\Lambda\Big|_{t=0} = -\,k^{L-\ell}\,\frac{h_\ell S h_\ell^\top}{d_{\text{model}}\,(r_\ell^2 + \varepsilon)}\prod_{i=\ell+1}^{L}\frac{\varepsilon}{r_i^2 + \varepsilon},$$

and for the scaling $c = e^t$ at h_L it is $\varepsilon/(r_L^2 + \varepsilon)$.

Proof. Since $N(h)$ is a positive multiple of h,

$$N(h_\ell G)\, G^{-1} = \gamma N(h_\ell), \qquad \gamma = \sqrt{\frac{r_\ell^2 + \varepsilon}{\|h_\ell G\|^2/d_{\text{model}} + \varepsilon}}.$$

Block $\ell + 1$ thus reads $\gamma N(h_\ell)$ and, being homogeneous, writes $\mu_{\ell+1} h_{\ell+1}$ with $\mu_{\ell+1} = \gamma^k$; for the scaling at h_L, set $\mu_L = c$. If a hidden state becomes $\mu_i h_i$ with $\mu_i > 0$, then $N(\mu_i h_i) = \beta_i N(h_i)$ with

$$\beta_i = \mu_i \sqrt{\frac{r_i^2 + \varepsilon}{\mu_i^2 r_i^2 + \varepsilon}},$$

so the next block writes $\mu_{i+1} h_{i+1}$ with $\mu_{i+1} = \beta_i^k$, and the readout gives $\Lambda = \beta_L$.

(i) If $\varepsilon = 0$, every β_i is 1. If G is orthogonal, $\gamma = 1$, so every μ_i and β_i is 1.

(ii) At $\mu_i = 1$, $\partial \log \beta_i / \partial \log \mu_i = \varepsilon/(r_i^2 + \varepsilon)$, and $\frac{d}{dt} \log \gamma\big|_{t=0} = -h_\ell S h_\ell^\top / (d_{\text{model}}(r_\ell^2 + \varepsilon))$, since $\|h_\ell e^{tS}\|^2 = \|h_\ell\|^2 + 2t\, h_\ell S h_\ell^\top + O(t^2)$. The chain rule along $\log \mu_{\ell+1} = k \log \gamma$, $\log \mu_{i+1} = k \log \beta_i$ and $\log \Lambda = \log \beta_L$ gives the product. □

Proof of Proposition 3.1. (i) $(hUK)K^{-1}V = hUV$, and no other part of the network involves U or V. (ii) Fix a hidden state h_ℓ and apply the change of basis G there (Section 2). The writes into h_ℓ produce $h_\ell G$ in place of h_ℓ. A read that sees h_ℓ itself, such as the read of the identity units, computes $(h_\ell G)G^{-1}W = h_\ell W$. With pre-RMSNorm reads, a read sees $N(h_\ell G)$, which by Lemma A.1 equals $N(h_\ell)Q$ exactly when $G = cQ$ with Q orthogonal and $c = 1$ if $\varepsilon > 0$, and is then absorbed by $W \mapsto Q^\top W$. So nothing downstream changes, for every G in $\mathrm{GL}(d_{\text{model}})$ without normalization, in $\mathrm{CO}(d_{\text{model}})$ for $\varepsilon = 0$ and in $\mathrm{O}(d_{\text{model}})$ for $\varepsilon > 0$. With biases and without normalization, the affine change $h_\ell \mapsto h_\ell G + g$ is absorbed by $b_{\text{write}} \mapsto b_{\text{write}} G + g$ on the writes and by $(W, b) \mapsto (G^{-1}W,\ b - gG^{-1}W)$ on the reads, since $h_\ell W + b = (h_\ell G + g)G^{-1}W + b - gG^{-1}W$; this gives $\mathrm{Aff}(d_{\text{model}})$. (iii) The change of basis at the identity units of block ℓ multiplies U_ℓ on the right and V_ℓ on the left. The change at $h_{\ell-1}$ multiplies the reads of block ℓ, U_ℓ included, on the left, and the change at h_ℓ multiplies the writes of block ℓ, V_ℓ included, on the right. Multiplications on the left and on the right commute, and every matrix is multiplied on each side by the change at one junction only, so all these changes commute, and a matrix of the skipless network is touched by one hidden state only. The symmetries inside blocks (Section 2) multiply W_{read} on the right and W_{write} on the left, and commute with all of the above. □

A.2 The gauge fix (Theorems 3.2 and A.3)

Theorem 3.2 collects what the main text uses of the following theorem. We write the parameters of a network of mixed blocks as $\theta = \big(E,\ (U_\ell, V_\ell, W^\ell_{\text{read}}, W^\ell_{\text{write}})_{\ell=1}^L,\ D\big)$, and those of a network with learned skips as $\big(E,\ (A_\ell, W^\ell_{\text{read}}, W^\ell_{\text{write}})_{\ell=1}^L,\ D\big)$. For invertible $G_0, \ldots, G_L$ and $K_1, \ldots, K_L$, let $T_{G,K}$ be the map

$$\begin{aligned} E \mapsto EG_0, \qquad W^\ell_{\text{read}} \mapsto G_{\ell-1}^{-1} W^\ell_{\text{read}}, \qquad W^\ell_{\text{write}} \mapsto W^\ell_{\text{write}} G_\ell, \qquad D \mapsto G_L^{-1} D, \\ U_\ell \mapsto G_{\ell-1}^{-1} U_\ell K_\ell, \qquad V_\ell \mapsto K_\ell^{-1} V_\ell G_\ell, \end{aligned} \tag{7}$$

which applies the change of basis G_ℓ at h_ℓ and K_ℓ at the identity units of block ℓ. On a network with learned skips, T_G acts by $A_\ell \mapsto G_{\ell-1}^{-1} A_\ell G_\ell$ in place of the second line. When every skip $A_\ell = U_\ell V_\ell$ is invertible, let $\Lambda_\ell = A_{\ell+1} \cdots A_L$, with $\Lambda_L = I$, and let

$$\Phi(\theta) = \big(E\Lambda_0,\ (\Lambda_{\ell-1}^{-1} W^\ell_{\text{read}},\ W^\ell_{\text{write}} \Lambda_\ell)_{\ell=1}^L,\ D\big), \tag{8}$$

a parameter of the residual network, with the same formula for learned skips. With biases, Φ also sends the embedding bias b_E to $b_E \Lambda_0$ and each write bias b^ℓ_{write} to $b^\ell_{\text{write}} \Lambda_\ell$, and keeps each read bias. We write F_θ for the function of the network of mixed blocks at θ, and $F^{\text{res}}_{\Phi(\theta)}$ for that of the residual network at $\Phi(\theta)$.

Theorem A.3 (The gauge fix)**.** *Consider networks without normalization in (i) and (ii), and parameters θ whose skips A_ℓ are invertible.*
(i) Same function. *The maps $T_{G,K}$ preserve the function. With $G_\ell = \Lambda_\ell$ and $K_\ell = U_\ell^{-1}\Lambda_{\ell-1}$, $T_{G,K}$ sets every U_ℓ and V_ℓ to I and the other parameters to $\Phi(\theta)$, so $F_\theta = F^{\mathrm{res}}_{\Phi(\theta)}$. For learned skips, T_G with $G_\ell = \Lambda_\ell$ sets every A_ℓ to I in the same way.*
(ii) What the gauge fix identifies. *Every parameter of the residual network is $\Phi(\theta)$ for some θ, and $\Phi(\theta) = \Phi(\theta')$ exactly when $\theta' = T_{G,K}(\theta)$ for some G and K with $G_L = I$. The derivative of Φ is onto at every θ. So the networks of mixed blocks with invertible skips and the residual networks compute the same set of functions, with the same functional dimension at θ and at $\Phi(\theta)$. The same holds for learned skips, without the K_ℓ, and for networks with biases.*
(iii) One change of basis is left. *With or without RMSNorm, let every block be nonzero for some value of its parameters, and let $G_0, \dots, G_L \in \mathcal{G}$. The map T_G of the embedding, the reads, the writes and the readout preserves the function of every residual network if and only if $G_0 = G_1 = \cdots = G_L$. Under RMSNorm with $\varepsilon = 0$, where $G_\ell = c_\ell Q_\ell$ with $c_\ell > 0$ and Q_ℓ orthogonal, T_G multiplies the reads and the readout by $Q_{\ell-1}^\top$ and $Q_L^\top$ in place of $G_{\ell-1}^{-1}$ and G_L^{-1}.*

Overview. Part (i) is an induction along the network: in the coordinates $h'_\ell = h_\ell\Lambda_\ell$ the identity units become identity skips, and we check that the residual recursion with transformed weights produces exactly these coordinates. Part (ii) lifts every direction on the residual side to one with the identity units held fixed, which makes the derivative of Φ onto, and solves for the maps $T_{G,K}$ that relate two parameters with the same image. Part (iii) switches off all branches but one, so that the output compares the bases on the two sides of that block, and a branch that is not identically zero forces them to agree. With biases every hidden state also carries a translation, which the skips do not tie (Lemma A.4).

Proof of Theorem A.3. Part (i), with biases and a skip bias. Merge the identity units, $A_\ell = U_\ell V_\ell$, and write the network as $h_\ell = h_{\ell-1}A_\ell + k_\ell + f_\ell(h_{\ell-1}W^\ell_{\mathrm{read}} + b^\ell_{\mathrm{read}})W^\ell_{\mathrm{write}} + b^\ell_{\mathrm{write}}$ with $h_0 = xE + b_E$, where $k_\ell \in \mathbb{R}^{1\times d_{\mathrm{model}}}$ is a skip bias ($k_\ell = 0$ is allowed). Every A_ℓ is invertible, hence every Λ_ℓ, and $\Lambda_{\ell-1} = A_\ell\Lambda_\ell$. Define new variables and new weights

$$h'_\ell = h_\ell\Lambda_\ell, \qquad E' = E\Lambda_0, \quad b'_E = b_E\Lambda_0, \qquad W'^\ell_{\mathrm{read}} = \Lambda_{\ell-1}^{-1}W^\ell_{\mathrm{read}}, \quad b'^\ell_{\mathrm{read}} = b^\ell_{\mathrm{read}},$$
$$W'^\ell_{\mathrm{write}} = W^\ell_{\mathrm{write}}\Lambda_\ell, \quad b'^\ell_{\mathrm{write}} = (b^\ell_{\mathrm{write}} + k_\ell)\Lambda_\ell, \qquad D' = D.$$

We show by induction that the residual network with primed weights computes h'_ℓ. For $\ell = 0$, $h'_0 = (xE + b_E)\Lambda_0 = xE' + b'_E$. Assume $h'_{\ell-1} = h_{\ell-1}\Lambda_{\ell-1}$. The residual recursion with primed weights gives

$$\begin{aligned} &h'_{\ell-1} + f_\ell(h'_{\ell-1}W'^\ell_{\mathrm{read}} + b'^\ell_{\mathrm{read}})W'^\ell_{\mathrm{write}} + b'^\ell_{\mathrm{write}} \\ &\quad = h_{\ell-1}\Lambda_{\ell-1} + f_\ell(h_{\ell-1}W^\ell_{\mathrm{read}} + b^\ell_{\mathrm{read}})W^\ell_{\mathrm{write}}\Lambda_\ell + (b^\ell_{\mathrm{write}} + k_\ell)\Lambda_\ell, \end{aligned}$$

and since $\Lambda_{\ell-1} = A_\ell\Lambda_\ell$ the right-hand side equals $h_\ell\Lambda_\ell$. Finally $h'_L D' = h_L D$ because $\Lambda_L = I$. This is the map $T_{G,K}$ of (7) with $G_\ell = \Lambda_\ell$ and $K_\ell = U_\ell^{-1}\Lambda_{\ell-1}$: it sends U_ℓ to $G_{\ell-1}^{-1}U_\ell K_\ell = I$ and V_ℓ to $K_\ell^{-1}V_\ell G_\ell = \Lambda_{\ell-1}^{-1}A_\ell\Lambda_\ell = I$, and the other matrices as above. For vanilla MLP blocks, these changes of basis are those of Proposition C.1 of Zhao et al. (2023), transposed to row vectors. Read the network of mixed blocks as a feedforward network whose hidden layers alternate between the hidden states h_ℓ, with the identity as activation, and the hidden layers of the blocks, with the mixed activation $\tilde f\big(\big[\, u \,\big|\, z \,\big]\big) = \big[\, u \,\big|\, f(z) \,\big]$ of Section 2. The identity passes every matrix in $\mathrm{GL}(d_{\mathrm{model}})$ (their Example C.3), and $\tilde f$ passes $\mathrm{diag}(K, I)$ for every $K \in \mathrm{GL}(d_{\mathrm{model}})$, so their proposition gives these changes of basis G_ℓ and K_ℓ; for other blocks the same computation does. Propagating a change of basis through a skip is thus their change of basis at the identity units, chosen together with the changes of basis at the two hidden states that the skip joins. The forward version takes $G_\ell = \Pi_\ell^{-1}$ at h_ℓ, with $\Pi_\ell = A_1 \cdots A_\ell$ and $\Pi_0 = I$. By the same induction it gives the residual network with embedding E, reads $\Pi_{\ell-1}W^\ell_{\mathrm{read}}$, writes $W^\ell_{\mathrm{write}}\Pi_\ell^{-1}$, write biases $(b^\ell_{\mathrm{write}} + k_\ell)\Pi_\ell^{-1}$ and readout $\Pi_L D$. Any linear change of variables $h'_\ell = h_\ell K'_\ell$ that yields identity skips must satisfy

$K'^{-1}_{\ell-1} A_\ell K'_\ell = I$, so $K'_\ell = \Lambda_\ell K'_L$: two such changes, the forward and the backward version among them, differ by a global change of basis and, when biases are present, by a translation at each hidden state (Lemma A.4).

Part (ii). The identity units carry no biases of their own, since these would only add to the write bias. For parameters θ of the mixed network with invertible A_ℓ, the map Φ of (8) is the change of variables of part (i) with $k_\ell = 0$, so $F_\theta = F^{\mathrm{res}}_{\Phi(\theta)}$. Every parameter of the residual network is reached: the θ with the same entries and $U_\ell = V_\ell = I$ has every $\Lambda_\ell = I$, so Φ returns these entries. The Fréchet derivative $\mathrm{D}_\theta \Phi(\theta)$ is onto at every θ. A direction η' of the residual parameters, with the entries η'_E, $\eta'^{\ell}_{\mathrm{read}}$, $\eta'^{\ell}_{\mathrm{write}}$ and η'_D at E', $W'^{\ell}_{\mathrm{read}}$, $W'^{\ell}_{\mathrm{write}}$ and D', is $\mathrm{D}_\theta \Phi(\theta)\, \eta$ for the direction η of θ that is zero at every U_ℓ and V_ℓ and has the entries $\eta'_E \Lambda_0^{-1}$ at E, $\Lambda_{\ell-1} \eta'^{\ell}_{\mathrm{read}}$ at W^{ℓ}_{read}, $\eta'^{\ell}_{\mathrm{write}} \Lambda_\ell^{-1}$ at $W^{\ell}_{\mathrm{write}}$ and η'_D at D, since along it every Λ_ℓ stays fixed; the biases are treated in the same way. The maps $T_{G,K}$ with $G_L = I$ send A_ℓ to $G^{-1}_{\ell-1} A_\ell G_\ell$ and hence Λ_ℓ to $G_\ell^{-1} \Lambda_\ell$, so they leave Φ unchanged. Conversely, let $\Phi(\theta') = \Phi(\theta)$, and set $G_\ell = \Lambda_\ell \Lambda'^{-1}_\ell$, so that $G_L = I$, and $K_\ell = U_\ell^{-1} G_{\ell-1} U'_\ell$. Then $E' = E G_0$, $W'^{\ell}_{\mathrm{read}} = G^{-1}_{\ell-1} W^{\ell}_{\mathrm{read}}$, $W'^{\ell}_{\mathrm{write}} = W^{\ell}_{\mathrm{write}} G_\ell$ and $D' = D$; $U'_\ell = G^{-1}_{\ell-1} U_\ell K_\ell$ by the choice of K_ℓ; and $A'_\ell = \Lambda'_{\ell-1} \Lambda'^{-1}_\ell = G^{-1}_{\ell-1} A_\ell G_\ell$ gives $V'_\ell = U'^{-1}_\ell A'_\ell = K_\ell^{-1} V_\ell G_\ell$. So $\theta' = T_{G,K}(\theta)$. Since $F_\theta = F^{\mathrm{res}}_{\Phi(\theta)}$ and every residual parameter is reached, the two networks compute the same functions. For every batch of inputs, the Jacobian of the mixed network at θ is the Jacobian of the residual network at $\Phi(\theta)$ times the derivative of Φ, by the chain rule; a matrix times a matrix that is onto has the rank of the first, so the two Jacobians have the same rank, and the two networks the same functional dimension. For learned skips the same argument applies without the K_ℓ.

Part (iii). A single global G is a symmetry: the skip carries $h_{\ell-1} G$ to $h_{\ell-1} G$, and the branch becomes $B_\ell(\,\cdot\, G^{-1})\, G$. Conversely, let $G_0, \ldots, G_L$ preserve the function for all parameter values; the change of basis conjugates each branch, $B_\ell \mapsto B_\ell(\,\cdot\, G^{-1}_{\ell-1})\, G_\ell$. First set every write W^k_{write}, and its bias if there is one, to zero. The network computes $x \mapsto xED$ and the transformed network $x \mapsto xEG_0 G_L^{-1} D$, and these agree for all D and E only if $G_L = G_0$. Now suppose $G_0 = \cdots = G_{\ell-1} =: G$ and set every write except that of block ℓ to zero. Then $h_L = h_{\ell-1} + B_\ell(h_{\ell-1})$ with $h_{\ell-1} = xE$, block ℓ reads $h_{\ell-1} G G^{-1}_{\ell-1} = h_{\ell-1}$, and the transformed network outputs $\big(h_{\ell-1} G + B_\ell(h_{\ell-1})\, G\big) G^{-1} D$. Equality with $\big(h_{\ell-1} + B_\ell(h_{\ell-1})\big) D$ for all D gives $B_\ell(h)\,(G_\ell G^{-1} - I) = 0$ for all h and all parameters of block ℓ. As the write of block ℓ ranges over all matrices, the outputs of a block whose computation does not vanish identically span $\mathbb{R}^{d_{\mathrm{model}}}$, so $G_\ell = G$, and by induction all G_ℓ are equal. With pre-RMSNorm reads the argument is the same, with each read transformed by the orthogonal factor of the change of basis (Section 2); when $\varepsilon = 0$ the first step compares only the orthogonal factors, since the readout sees $N(h_L)$, and the second step, in which the write of block ℓ can be scaled freely, equates the scales as well. □

We recall Theorem 3.2 from Section 3.2, where $A_\ell = U_\ell V_\ell$ and $\Lambda_\ell = A_{\ell+1} \cdots A_L$.

Theorem 3.2 (Skip connections are gauge fixes)**. (i) Same function.** *Without normalization, if every A_ℓ is invertible, the network of mixed blocks and the network with learned skips compute the same function as the residual network with the embedding $E\Lambda_0$, the reads $\Lambda^{-1}_{\ell-1} W^{\ell}_{\mathrm{read}}$, the writes $W^{\ell}_{\mathrm{write}} \Lambda_\ell$ and the readout D.*
(ii) *At these parameters the residual network has the same functional dimension as the network of mixed blocks and as the network with learned skips.*
(iii) One change of basis is left. *Without normalization, or under pre-RMSNorm with $\varepsilon > 0$, let every block be nonzero for some value of its parameters, and let $G_0, \ldots, G_L \in \mathcal{G}$. The map $E \mapsto EG_0$, $W^{\ell}_{\mathrm{read}} \mapsto G^{-1}_{\ell-1} W^{\ell}_{\mathrm{read}}$, $W^{\ell}_{\mathrm{write}} \mapsto W^{\ell}_{\mathrm{write}} G_\ell$, $D \mapsto G_L^{-1} D$ preserves the function of every residual network if and only if $G_0 = G_1 = \cdots = G_L$.*

Proof of Theorem 3.2. Parts (i) and (ii) are Theorem A.3(i) and (ii), with $\Phi(\theta)$ written out by (8). Part (iii) is Theorem A.3(iii): without normalization $\mathcal{G} = \mathrm{GL}(d_{\mathrm{model}})$, and under RMSNorm with $\varepsilon > 0$ every G_ℓ is orthogonal, so $Q_\ell = G_\ell$. □

Lemma A.4 (Translations are not tied)**.** *In a residual network with biases and without normalization, a translation $h_\ell \mapsto h_\ell + g$ of any single hidden state preserves the function once the biases are adjusted. The symmetry group generated by changes of basis, translations and the symmetries inside*

blocks therefore has dimension $d_{\text{model}}^2 + (L+1)d_{\text{model}} + s$*, against* $(L+1)(d_{\text{model}}^2 + d_{\text{model}}) + s$ *without skips, and the skips remove* Ld_{model}^2 *dimensions, as without biases. Under RMSNorm no translation is a symmetry.*

Proof. Add g to the write bias of block ℓ, or to the embedding bias when $\ell = 0$; subtract gW_{read}^k from the read bias of every block $k > \ell$; subtract gD from the readout bias. The skips carry g forward unchanged, and every downstream read removes it, so the function is unchanged. Each hidden state therefore keeps its own translation with or without skips, and a skip ties only the linear part of $\mathrm{Aff}(d_{\text{model}})$, which gives the counts. Under RMSNorm a translation changes $N(h)$ by an amount that depends nonlinearly on h, which no read can undo. □

The checks give redundancies of 80 without skips and 32 with them ($\tanh$ networks with biases, $d_{\text{model}} = 4$, $L = 3$; Appendix F), and a translation at a single interior hidden state changes the function by a relative 2×10^{-16}.

A.3 What the skips remove (Theorem 3.3)

We recall Theorem 3.3 from Section 3.3.

Theorem 3.3 (What the skips remove)**.** **(i)** *The changes of basis that preserve the function at all parameters form a group of dimension* $(L+1)\dim \mathcal{G}$ *in a skipless network and* $\dim \mathcal{G}$ *in a residual one, so the* L *skips remove* $L \dim \mathcal{G}$ *dimensions of symmetry.*
(ii) *If in both networks the symmetries act freely at generic parameters, the functional dimension at these parameters is at most* $P - (L+1)\dim \mathcal{G} - s$ *without skips and* $P - \dim \mathcal{G} - s$ *with them.*
(iii) *If in addition both networks satisfy the equality condition, the skips raise the functional dimension by exactly* $L \dim \mathcal{G}$*, which is* Ld_{model}^2 *without normalization.*

Overview. Part (i) counts copies of $\mathcal{G}$: one per hidden state without skips, by Proposition 3.1, and one with them, by Theorem A.3(iii). Part (ii) bounds the rank of every batch Jacobian by the number of parameters minus the number of independent directions in which the symmetries move the parameters, which is the dimension of the symmetries when they act freely; Lemma A.5 proves that they do when the embedding has rank d_{model} and every block has width at least d_{model}. Part (iii) turns both bounds into equalities under the equality condition and subtracts.

Table 4: Parameters, dimension of the symmetries and, under the equality condition, functional dimension of the networks of Section 2, without normalization or biases; P counts the parameters of the skipless network. Each step of the gauge fix removes as many parameters as dimensions of symmetry, while deleting the identity units removes $2Ld_{\text{model}}^2$ parameters but only Ld_{model}^2 dimensions of symmetry. Given the value for the residual network, the others below it hold at corresponding parameters without further assumption (Theorem A.3(ii), Proposition 5.1(ii)).

Network	Parameters	Symmetry	Functional dimension
skipless	P	$(L+1)d_{\text{model}}^2 + s$	$P - (L+1)d_{\text{model}}^2 - s$
mixed, with trainable identity units	$P + 2Ld_{\text{model}}^2$	$(2L+1)d_{\text{model}}^2 + s$	$P - d_{\text{model}}^2 - s$
learned skips	$P + Ld_{\text{model}}^2$	$(L+1)d_{\text{model}}^2 + s$	$P - d_{\text{model}}^2 - s$
residual	P	$d_{\text{model}}^2 + s$	$P - d_{\text{model}}^2 - s$
residual, one block fixed	$P - d_{\text{model}}^2$	s	$P - d_{\text{model}}^2 - s$

Lemma A.5 (Free action)**.** *Consider an MLP network, skipless or residual, without normalization or with pre-RMSNorm, with vanilla MLP blocks with* $\tanh$*, GELU, ReLU or ReLU*^{2} *activations, or with GLU blocks, and the symmetries inside blocks of Section 2. If the embedding has rank* d_{model} *and every block has width at least* d_{model}*, then at generic parameters the changes of basis and the symmetries inside blocks act freely. The same holds if the readout has rank* d_{model} *and every block has width at least* d_{model}*.*

Proof. Permutations and sign flips are isolated, so it suffices to consider changes of basis and rescalings close to the identity. Let changes of basis $G_0, \dots, G_L$ and positive rescalings c_ℓ of hidden units (for ReLU and ReLU^2), of gates and values (for ReGLU) or of values (for the other GLU blocks) fix a generic θ; for $\tanh$ and GELU there are no c_ℓ. The embedding transforms as $E \mapsto EG_0$, and E has rank d_{model}, so $G_0 = I$. Suppose $G_{\ell-1} = I$. The read of block ℓ then transforms as $W^\ell_{\text{read}} \mapsto W^\ell_{\text{read}} \operatorname{diag}(c_\ell)$, with entries 1 on the columns that are not rescaled, and no column of a generic W^ℓ_{read} vanishes, so $c_\ell = \mathbf{1}$. The write then transforms as $W^\ell_{\text{write}} \mapsto W^\ell_{\text{write}} G_\ell$, and W^ℓ_{write} has rank d_{model} because the block has width at least d_{model}, so $G_\ell = I$. By induction only the identity fixes θ. In a residual network the G_ℓ are equal (Theorem A.3(iii)) and the same argument applies. The second statement follows by the mirror induction from the readout, $D \mapsto G_L^{-1} D$: rescalings act freely on the rows of a generic write, except that for ReGLU a row fixes only the product cc' of the gate and value scales of its unit; each read, of rank d_{model} and with generic columns, then forces $G_{\ell-1} = I$ and the remaining scales to be 1. With RMSNorm the changes of basis lie in $\mathrm{CO}(d_{\text{model}})$ or $\mathrm{O}(d_{\text{model}})$, and the argument is unchanged. The width condition is needed: at a hidden state of a skipless network that is written and read at rank $m < d_{\text{model}}$, the changes of basis that fix the parameters form a family of dimension $(d_{\text{model}} - m)^2$, so they move the parameters in only $2d_{\text{model}}m - m^2$ independent directions, as Zhao et al. (2023, Proposition C.3) compute for one hidden layer. □

Remark A.6 (Attention blocks). The lemma extends to transformers. Attention blocks also carry the per-head symmetries of Section 2, whose full group is characterized by Wang and Wang (2025) and which are not our subject. Once the input basis is fixed they act freely on $W_{Q,i}$ and $W_{V,i}$, which have full column rank, and the write W_O has rank d_{model}, so the forward induction goes through; in a mixture of experts the writes of all experts together have rank d_{model}. A numerical check on small skipless transformers confirms free action, including cases beyond the reach of the induction (Section 6 of the notebook, Appendix F).

Proof of Theorem 3.3. Parts (i) and (ii). If a family of transformations of the parameters leaves the function unchanged, then the function does not change to first order along any direction in which the family moves θ, so these directions lie in the kernel of every batch Jacobian at θ, and the functional dimension is at most P minus their number. For a skipless network, Proposition 3.1 gives $L + 1$ independent copies of $\mathcal{G}$, one per hidden state, and the directions in which they move θ are linearly independent of one another and of those of the symmetries inside blocks at generic parameters: the change of basis at h_ℓ multiplies W^ℓ_{write} on the right and $W^{\ell+1}_{\text{read}}$ on the left, the symmetries inside block ℓ multiply W^ℓ_{read} on the right and W^ℓ_{write} on the left, and by hypothesis they act freely at generic parameters (Lemma A.5). The number of independent directions is therefore $(L + 1) \dim \mathcal{G} + s$. For a residual network, Theorem A.3(iii) leaves one copy, and the same argument gives $\dim \mathcal{G} + s$. *Part (iii).* Under the hypothesis, the inequalities of part (ii) are equalities, so the generic functional dimensions are $P - (L + 1) \dim \mathcal{G} - s$ and $P - \dim \mathcal{G} - s$, and the difference is $L \dim \mathcal{G}$, which is d^2_{model} per skip without normalization, and with RMSNorm $d_{\text{model}}(d_{\text{model}} - 1)/2$ when $\varepsilon > 0$ and $d_{\text{model}}(d_{\text{model}} - 1)/2 + 1$ when $\varepsilon = 0$ (Lemma A.1); with biases the translations are redundant in both networks and the difference is again Ld^2_{model} (Lemma A.4). A network with learned skips without normalization has $P + Ld^2_{\text{model}}$ parameters and $L + 1$ independent copies of $\mathrm{GL}(d_{\text{model}})$, since each A_ℓ absorbs $A_\ell \mapsto G^{-1}_{\ell-1} A_\ell G_\ell$; under the same hypothesis its functional dimension is $P + Ld^2_{\text{model}} - (L+1)d^2_{\text{model}} - s = P - d^2_{\text{model}} - s$, the same as the residual network, in agreement with Theorem A.3(ii). □

A.4 No absorption inside a block (Theorems 3.4 to 3.7)

We recall Theorems 3.4 to 3.7 from Section 3.4, for the question (3). Below we write B for B_θ, and B' for $B_{\theta'}$ with $\theta' = (W'_{\text{read}}, W'_{\text{write}})$.

Theorem 3.4 (Homogeneous, gated and bounded blocks)**.** *Let the blocks be positively homogeneous of degree $k \neq 1$, as ReLU^2 and ReGLU blocks are, or gated with a gate that is differentiable at 0 and vanishes there, as SwiGLU and GeGLU blocks are.*

(i) *No block $B_{\theta'}$ of any width satisfies (3).*
(ii) *For gated blocks, and for homogeneous blocks with $k > 1$, no composition of one or more blocks without skips equals a composition of residual blocks, whatever their numbers and widths.*
(iii) *For the blocks of (ii), with an embedding E and a readout D, no skipless network equals a residual network with $ED \neq 0$, also under pre-RMSNorm with $\varepsilon > 0$.*
(iv) *Without normalization, (i), (ii) and (iii) also hold for* $\tanh$ *blocks.*

Theorem 3.5 (ReLU and GELU blocks)**.** *Let σ be ReLU or GELU.*
(i) *If B_θ is nondegenerate, a block $B_{\theta'}$ of the same width satisfies (3) if and only if some set S of hidden units has $W_{\text{read}}[:, S]\, W_{\text{write}}[S, :] = -M$; for generic θ there is none.*
(ii) *If M is invertible and W_{read} has rank d_{model}, a block $B_{\theta'}$ with d_{model} more hidden units satisfies (3), and for generic θ none with fewer does.*

Theorem 3.6 (GELU and smooth blocks at depth)**. (i)** *For GELU blocks and almost every choice of weights, a composition of $L \geq 2$ residual blocks equals no composition of L blocks without skips of the same widths.*
(ii) *With an embedding and a readout, the same holds for networks of smooth blocks of any depth $L \geq 1$, also under pre-RMSNorm with $\varepsilon > 0$.*
(iii) Too few extra units. *For GELU blocks with an embedding and a readout, and almost every choice of weights, a residual network of L blocks equals no skipless network of L blocks with fewer than $Ld_{\text{model}}/2$ more hidden units in all.*
(iv) Enough extra units. *For GELU blocks with an embedding and a readout, a residual network whose reads all have rank d_{model} equals a skipless network with d_{model} more hidden units in every block.*

Theorem 3.7 (ReLU blocks at depth)**.** *Consider networks of ReLU blocks with an embedding and a readout. For almost every choice of weights, a residual network equals no skipless network whose first block has fewer than $2d_{\text{model}}$ hidden units, whatever its depth and its other widths.*

Overview. Theorem 3.4 compares orders. Along rays from the origin a linear skip scales like λ and a block that is homogeneous of degree $k \neq 1$ like λ^k, and near the origin a gated block is of second order, so the two sides of the absorption equation cannot agree (Propositions A.8 and A.9); comparing Jacobians at the origin extends this to any depth (Corollary A.10). Theorem 3.5(i) is the substantial case. The absorption equation splits into its odd and even parts in x. The even part determines the hidden units of the absorbing block up to sign, and for ReLU up to a positive rescaling, through the kinks for ReLU (Theorem A.11) and through linear independence of ridge functions for GELU (Lemma A.12 and Theorem A.13). The odd part of ReLU and GELU is exactly $z/2$, so the odd part of the equation is linear and reads off the set S of units whose signs flip; conversely, flipping the signs of the units in S absorbs the skip whenever the condition holds. The generic clause holds because the condition is a finite union of algebraic sets of positive codimension (Corollary A.16), and Remark A.15 covers singular M. Theorem 3.5(ii) counts the extra units that the matching requires (Proposition A.17). Theorem 3.6(i) and (ii) compare dimensions: a skipless network has a change of basis at every hidden state as a symmetry, and with small writes a residual network is, to first order, one wide block plus a linear map, whose flat directions Lemma C.3 determines (Lemmas A.18 and A.19). Theorem 3.6(iii) adds the parameters of extra units to the same comparison. Theorem 3.7 compares the ranks of the linear pieces at an input and at its negative.

The arguments show that the function of a block determines its parameters, in the manner used for $\tanh$ networks by Sussmann (1992) and Fefferman (1994) and for ReLU networks by Phuong and Lampert (2020), Petzka et al. (2020) and Rolnick and Kording (2020): kink alignment for ReLU, ridge-function independence for GELU. For two-layer ReLU networks with a single output, Petzka et al. (2020, Section 3) already give the corresponding sign flips, show that irreducible networks computing the same function differ only by these flips, permutations and positive rescalings (their Theorem 1), and state the matching criterion for removing a linear term. Lemma A.7 reduces any invertible skip to the identity skip.

Setting. Throughout, $W_{\text{read}} \in \mathbb{R}^{d_{\text{model}} \times m}$ and $W_{\text{write}} \in \mathbb{R}^{m \times d_{\text{model}}}$, σ acts coordinatewise, w_i is column i of W_{read} and $\hat{w}_i$ is row i of W_{write}, and inputs x are row vectors. For $S \subseteq$

$\{1, \dots, m\}$, $W_{\text{read}}[:, S]$ is the submatrix of columns in S, $W_{\text{write}}[S, :]$ the submatrix of rows in S, and Π_S the diagonal projector with $(\Pi_S)_{ii} = 1$ exactly when $i \in S$, so that $W_{\text{read}}\Pi_S W_{\text{write}} = W_{\text{read}}[:, S]W_{\text{write}}[S, :]$. A family of vectors is *pairwise non-collinear* if no one of them is a scalar multiple of another, and a block is *nondegenerate* if the columns of W_{read} are nonzero and pairwise non-collinear and the rows of W_{write} are nonzero. The question is whether there is a block $B'(x) = \sigma(xW'_{\text{read}})W'_{\text{write}}$ of the same width, with $W'_{\text{read}} \in \mathbb{R}^{d_{\text{model}} \times m}$ and $W'_{\text{write}} \in \mathbb{R}^{m \times d_{\text{model}}}$, such that

$$x + \sigma(xW_{\text{read}})W_{\text{write}} = \sigma(xW'_{\text{read}})W'_{\text{write}} \qquad \text{for all } x \in \mathbb{R}^{1 \times d_{\text{model}}}. \tag{9}$$

Lemma A.7 (Invertible reduction)**.** *Let $M \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ be invertible. Then*

$$xM + \sigma(xW_{\text{read}})W_{\text{write}} = \sigma(xW'_{\text{read}})W'_{\text{write}} \quad \textit{for all } x$$

if and only if

$$x + \sigma(xW_{\text{read}})W_{\text{write}}M^{-1} = \sigma(xW'_{\text{read}})W'_{\text{write}}M^{-1} \quad \textit{for all } x.$$

Proof. Right-multiply by M^{-1}, which is invertible. □

Any invertible linear skip therefore reduces to the identity skip, which is the single-block case of Theorem A.3(i).

Proposition A.8 (Homogeneous blocks)**.** *Let B and B' be blocks that are positively homogeneous of degree $k \neq 1$, of any widths, and let $M \neq 0$. Then $xM + B(x) = B'(x)$ fails for some x.*

Examples are ReLU^2 and ReGLU blocks without biases, both of degree 2.

Proof. Outline. We compare the two sides along rays through the origin: a term of degree one and a term of degree $k \neq 1$ cannot agree at every scale.

Suppose it holds for all x and evaluate at λx with $\lambda > 0$: $\lambda xM + \lambda^k B(x) = \lambda^k B'(x)$. Dividing by λ gives $xM = \lambda^{k-1}\big(B'(x) - B(x)\big)$, and $\lambda^{k-1} \to 0$ as $\lambda \to 0$ if $k > 1$ and as $\lambda \to \infty$ if $k < 1$. So $xM = 0$ for all x, which contradicts $M \neq 0$. □

Proposition A.9 (Gated blocks)**.** *Let B and B' be gated blocks without biases, of any widths, whose gates are differentiable at 0 and vanish there, and let $M \neq 0$. Then $xM + B(x) = B'(x)$ fails for some x.*

Examples are SwiGLU and GeGLU blocks, which Shazeer (2020) defines without biases.

Proof. Outline. Near the origin a gated block is of second order in x, while xM is of first order.

A unit of a gated block computes $g(u \cdot x)\,(v \cdot x)$. Since $g(0) = 0$ and g is differentiable at 0, $|g(u \cdot x)| \leq c\,\|x\|$ for small $\|x\|$, so the unit is $O(\|x\|^2)$, and so are B and B'. If $xM + B(x) = B'(x)$ for all x, then $xM = O(\|x\|^2)$ as $x \to 0$. Along a ray $x = tz$ this reads $t\,zM = O(t^2)$, so $zM = 0$ for every z, which contradicts $M \neq 0$. □

Corollary A.10 (Any depth)**.** *Let every block be as in Proposition A.8 with $k > 1$ or as in Proposition A.9, of any widths. Let F be a composition of residual blocks $h \mapsto h + B_\ell(h)$ and H a composition of one or more blocks without skips. Then $F \neq H$. With embeddings and readouts, the residual network $x \mapsto F(xE)D$ differs from the skipless network $x \mapsto H(xE')D'$ whenever $ED \neq 0$, also with pre-RMSNorm and $\varepsilon > 0$. Without normalization the same holds for $\tanh$ blocks.*

Proof. Each such block maps 0 to 0 and has Jacobian 0 there, because it is $O(\|x\|^k)$ with $k > 1$, respectively $O(\|x\|^2)$. By the chain rule F has Jacobian I at the origin and H has Jacobian 0; with embeddings and readouts these become ED and 0. Under pre-RMSNorm with $\varepsilon > 0$, $N(0) = 0$ and N has Jacobian $I/\sqrt{\varepsilon}$ at 0, so every block, read through N, still has Jacobian 0 there, and the residual and the skipless network have Jacobians $ED/\sqrt{\varepsilon}$ and 0 at the origin. For $\tanh$ blocks without normalization, H and $F(h) - h$ are bounded, so $F(xE)D - H(xE')D'$ differs from xED by a bounded function and does not vanish when $ED \neq 0$. □

Parity. The remaining activations have an odd part that is exactly linear. For ReLU, $\mathrm{ReLU}(z) = |z|/2 + z/2$. For GELU, $\mathrm{GELU}(z) = z\Phi(z)$ with Φ the standard normal distribution function, and since $\Phi(-z) = 1 - \Phi(z)$,

$$\mathrm{GELU}(z) = \Psi(z) + z/2, \qquad \Psi(z) := z\big(\Phi(z) - \tfrac{1}{2}\big), \qquad \Psi(-z) = \Psi(z).$$

The even parts, $|z|/2$ and Ψ, match the hidden units of the two blocks; the odd part, $z/2$, turns a sign flip into a linear term and gives the absorption condition.

Theorem A.11 (Identity absorption for ReLU)**.** *Let $\sigma = \mathrm{ReLU}$, $d_{\mathrm{model}} \geq 2$, and let the block be nondegenerate. Then (9) has a solution if and only if*

$$W_{\mathrm{read}}[:, S]\, W_{\mathrm{write}}[S, :] = -I_{d_{\mathrm{model}}} \quad \textit{for some } S. \tag{10}$$

Proof. Outline. Sufficiency is a construction: flipping the signs of the hidden units indexed by S produces the missing identity term. For necessity we split (9) into its odd and even parts in x, use the kinks of the even part to match the hidden units of the two blocks one to one, and read the index set S off the signs of the matching.

Sufficiency. Let $F = I_m - 2\Pi_S$, a diagonal sign matrix, and set $W'_{\mathrm{read}} = W_{\mathrm{read}}F$, $W'_{\mathrm{write}} = W_{\mathrm{write}}$. Since $|xW_{\mathrm{read}}F| = |xW_{\mathrm{read}}|$ coordinatewise,

$$\mathrm{ReLU}(xW'_{\mathrm{read}})W'_{\mathrm{write}} = \tfrac{1}{2}\big(xW_{\mathrm{read}}F + |xW_{\mathrm{read}}|\big)W_{\mathrm{write}},$$
$$x + \mathrm{ReLU}(xW_{\mathrm{read}})W_{\mathrm{write}} = \tfrac{1}{2}\big(xW_{\mathrm{read}} + |xW_{\mathrm{read}}|\big)W_{\mathrm{write}} + x,$$

and the two agree if and only if $\frac{1}{2}xW_{\mathrm{read}}(F - I_m)W_{\mathrm{write}} = x$ for all x, that is $-W_{\mathrm{read}}\Pi_S W_{\mathrm{write}} = I_{d_{\mathrm{model}}}$, which is (10).

Necessity. Suppose (9) holds and split both sides into odd and even parts in x:

$$W'_{\mathrm{read}}W'_{\mathrm{write}} = W_{\mathrm{read}}W_{\mathrm{write}} + 2I_{d_{\mathrm{model}}}, \qquad |xW'_{\mathrm{read}}|\, W'_{\mathrm{write}} = |xW_{\mathrm{read}}|\, W_{\mathrm{write}} \ \text{ for all } x. \tag{11}$$

Step 1, alignment of kinks. The map $x \mapsto |xW_{\mathrm{read}}|\, W_{\mathrm{write}}$ fails to be differentiable exactly on the union of the hyperplanes $\{xw_i = 0\}$; these are m distinct hyperplanes, each carrying a genuine kink since $\hat{w}_i \neq 0$. By the second identity in (11) the map $x \mapsto |xW'_{\mathrm{read}}|\, W'_{\mathrm{write}}$ has the same non-differentiability locus. Each of the m kink hyperplanes $\{x : xw_i = 0\}$ of the target needs at least one unit of B' whose kink lies on it, and B' has only m units; so each hyperplane gets exactly one unit, no two units share a hyperplane, and no kinks cancel. After relabeling, $W'_{\mathrm{read}} = W_{\mathrm{read}}C$ with $C = \mathrm{diag}(c_1, \ldots, c_m)$, all $c_i \neq 0$. *Step 2, coefficients.* Substituting, $|xW_{\mathrm{read}}C| = |xW_{\mathrm{read}}|\,|C|$, so $|xW_{\mathrm{read}}|\,|C|\,W'_{\mathrm{write}} = |xW_{\mathrm{read}}|\,W_{\mathrm{write}}$ for all x. For $d_{\mathrm{model}} \geq 2$ and pairwise non-collinear columns, the functions $|xw_i|$ are linearly independent modulo linear functions, because their kink hyperplanes are distinct; hence $|C|\,W'_{\mathrm{write}} = W_{\mathrm{write}}$ and $W'_{\mathrm{write}} = |C|^{-1}W_{\mathrm{write}}$. *Step 3, signs.* The first identity in (11) becomes $W_{\mathrm{read}}C|C|^{-1}W_{\mathrm{write}} = W_{\mathrm{read}}W_{\mathrm{write}} + 2I_{d_{\mathrm{model}}}$. With $\Sigma = C|C|^{-1} = \mathrm{diag}(\operatorname{sign} c_i)$ this reads $W_{\mathrm{read}}(\Sigma - I_m)W_{\mathrm{write}} = 2I_{d_{\mathrm{model}}}$; the diagonal of $\Sigma - I_m$ has entries 0 or -2, so with $S = \{i : c_i < 0\}$ we get $W_{\mathrm{read}}\Pi_S W_{\mathrm{write}} = -I_{d_{\mathrm{model}}}$, and $|S| \geq d_{\mathrm{model}}$ by the rank of $I_{d_{\mathrm{model}}}$. □

The GELU case needs one classical fact about ridge functions, whose short proof we include.

Lemma A.12 (Vanishing sums of ridge functions)**.** *Let $a_1, \ldots, a_r \in \mathbb{R}^{d_{\mathrm{model}}}$ be nonzero and pairwise non-collinear, and let $g_1, \ldots, g_r : \mathbb{R} \to \mathbb{R}^k$ be $r-1$ times continuously differentiable with $\sum_i g_i(a_i \cdot x) = 0$ for all $x \in \mathbb{R}^{d_{\mathrm{model}}}$. Then every g_i is a polynomial of degree at most $r-2$.*

Proof. Fix i. For each $j \neq i$ choose $u_j \in \mathbb{R}^{d_{\mathrm{model}}}$ with $u_j \cdot a_j = 0$ and $u_j \cdot a_i \neq 0$, which exists because a_i is not a multiple of a_j. The directional derivative ∂_{u_j} annihilates every function of $a_j \cdot x$ and maps a function of $a_k \cdot x$ to a function of $a_k \cdot x$. Applying the product of the commuting operators ∂_{u_j}, $j \neq i$, to the sum therefore leaves $\prod_{j \neq i}(u_j \cdot a_i)\, g_i^{(r-1)}(a_i \cdot x) = 0$. Since $a_i \neq 0$, $a_i \cdot x$ takes every real value, so $g_i^{(r-1)} = 0$. □

Theorem A.13 (Identity absorption for GELU)**.** *Let $\sigma = \mathrm{GELU}$ and let the block be nondegenerate. Then (9) has a solution if and only if (10) holds.*

Proof. Outline. The proof follows that of Theorem A.11. Linear independence of ridge functions replaces the kink argument, and one extra step shows that matched hidden units have the same scale.

Sufficiency. With $F = I_m - 2\Pi_S$, $W'_{\text{read}} = W_{\text{read}}F$ and $W'_{\text{write}} = W_{\text{write}}$, evenness of Ψ gives $\Psi(xW_{\text{read}}F) = \Psi(xW_{\text{read}})$, so

$$\begin{aligned}\mathrm{GELU}(xW'_{\text{read}})W'_{\text{write}} &= \Psi(xW_{\text{read}})W_{\text{write}} + \tfrac{1}{2}xW_{\text{read}}FW_{\text{write}}\\ &= \mathrm{GELU}(xW_{\text{read}})W_{\text{write}} - xW_{\text{read}}\Pi_S W_{\text{write}}\\ &= \mathrm{GELU}(xW_{\text{read}})W_{\text{write}} + x.\end{aligned}$$

Necessity. Splitting (9) into even and odd parts gives

$$\Psi(xW'_{\text{read}})W'_{\text{write}} = \Psi(xW_{\text{read}})W_{\text{write}} \ \text{ for all } x, \qquad W'_{\text{read}}W'_{\text{write}} = W_{\text{read}}W_{\text{write}} + 2I_{d_{\text{model}}}. \tag{12}$$

Step 1, alignment of directions. Ψ is analytic and not a polynomial. Zero columns of W'_{read} contribute $\Psi(0) = 0$; group the other columns of W'_{read} and the columns of W_{read} into classes of collinear vectors. In the first identity of (12) the terms of each class form one ridge function in the direction of the class, and these ridge functions sum to zero, so each is a polynomial by Lemma A.12. The class of a column w_i of W_{read} contains no other column of W_{read}, so if it contained no column of W'_{read} its ridge function would be $-\Psi(xw_i)\,\hat{w}_i$, which is not a polynomial because $\hat{w}_i \neq 0$. Hence each of the m classes of the columns of W_{read} contains a nonzero column of W'_{read}, and since W'_{read} has only m columns, after relabeling $W'_{\text{read}} = W_{\text{read}}C$ with $C = \mathrm{diag}(c_i)$, $c_i \neq 0$. *Step 2, unit scale.* Collecting the terms of direction w_i on both sides gives, for each i, a ridge function $g_i(xw_i) := \Psi(|c_i|\,xw_i)\,(W'_{\text{write}})_{i,:} - \Psi(xw_i)\,(W_{\text{write}})_{i,:}$ with $\sum_i g_i(xw_i) = 0$. By Lemma A.12 each g_i is a polynomial. For $t \to +\infty$, $\Psi(t) = t/2 - t\,\bar{\Phi}(t)$ with $\bar{\Phi}(t) = 1 - \Phi(t)$ exponentially small, so $g_i(t) = \frac{t}{2}\big(|c_i|(W'_{\text{write}})_{i,:} - (W_{\text{write}})_{i,:}\big) + o(1)$; a polynomial with this behavior is linear, and it is even because Ψ is even, so it is constant, and letting $t \to 0$ shows the constant is zero. Hence $|c_i|(W'_{\text{write}})_{i,:} = (W_{\text{write}})_{i,:}$ and the exponentially small remainder $|c_i|\,t\,\bar{\Phi}(|c_i|t)\,(W'_{\text{write}})_{i,:} - t\,\bar{\Phi}(t)\,(W_{\text{write}})_{i,:}$ must vanish, which with $(W_{\text{write}})_{i,:} \neq 0$ forces $\bar{\Phi}(|c_i|t) = \bar{\Phi}(t)$ for all $t > 0$, so $|c_i| = 1$. Therefore $|C| = I_m$ and $W'_{\text{write}} = W_{\text{write}}$. *Step 3, signs.* Identical to the ReLU case: the second identity in (12) becomes $W_{\text{read}}(C - I_m)W_{\text{write}} = 2I_{d_{\text{model}}}$ with $C = \mathrm{diag}(\pm 1)$, so $S = \{i : c_i = -1\}$ satisfies (10) and $|S| \geq d_{\text{model}}$. □

Remark A.14. Any activation whose even part ψ is analytic and not a polynomial matches the hidden units in the same way: Steps 1 and 2 of the proof of Theorem A.13 apply with ψ in place of Ψ (zero columns of W'_{read} now contribute a constant, which is a polynomial), except that the unit scale in Step 2 follows from a general fact: if $\sum_k a_k\psi(\alpha_k t)$ is a polynomial for distinct $\alpha_k > 0$, then every $a_k = 0$. Indeed, the Taylor coefficients $\psi_n \sum_k a_k\alpha_k^n$ vanish for all large n, infinitely many ψ_n are nonzero, and a generalized Vandermonde matrix with distinct positive nodes is invertible. Linear independence of the ridge functions alone does not suffice: for $\sigma(z) = z/2 + |z|^3$ the even part is homogeneous, so a unit can be rescaled together with its output row, and the unit scale is not forced. The odd part o of σ enters only through the flips: with the units matched, both directions reduce to $x = -2\sum_{i\in S} o(xw_i)\,\hat{w}_i$, with S the flipped units. For $o(z) = z/2$, that is $\sigma(z) - \sigma(-z) = z$, as for SiLU, whose even part is $\frac{z}{2}\tanh\frac{z}{2}$, this is (10), and $o(z) = \alpha z$ with $\alpha \neq 0$ only rescales it; a polynomial o of higher degree adds the condition that the units in S cancel its terms of degree three and higher, and a smooth o that is not a polynomial admits no S (Lemma A.12), so the skip is never absorbed at the same width.

Remark A.15 (Singular linear skips). Nothing in the proofs of Theorems A.11 and A.13 uses that the linear term is the identity. With xM in place of x on the left of (9), the odd part of the equation reads $W'_{\text{read}}W'_{\text{write}} = W_{\text{read}}W_{\text{write}} + 2M$, the even part is unchanged, and the sign step yields $W_{\text{read}}\Pi_S W_{\text{write}} = -M$. So for any $M \neq 0$, invertible or not, absorption holds if and only if $W_{\text{read}}[:, S]W_{\text{write}}[S, :] = -M$ for some S, with $|S| \geq \mathrm{rank}\,M$, which is a measure-zero condition on the weights for each fixed M. This gives Theorem 3.5(i) for singular M.

Corollary A.16 (Generic disjointness for one block)**.** *Let σ be ReLU or GELU and $d_{\text{model}} \geq 2$. For generic weights,* (9) *has no solution.*

Proof. Outline. The condition of Theorems A.11 and A.13 is a finite union of algebraic sets of positive codimension, and such a union has measure zero.

For each of the finitely many S, (10) is d_{model}^2 polynomial equations in the weights and defines a set of measure zero; the hypotheses of Theorems A.11 and A.13 hold outside another set of measure zero. So a generic residual block is not a block of the same width without a skip, and, by the same argument with the two sides of (9) exchanged, a generic block without a skip is not a residual block of the same width: neither class contains the other. □

Proposition A.17 (d_{model} more units)**.** *Let σ be ReLU or GELU, $d_{\text{model}} \geq 2$ and $M \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ invertible.*
(i) *If W_{read} has rank d_{model}, a block B' of width $m + d_{\text{model}}$ satisfies $xM + \sigma(xW_{\text{read}})W_{\text{write}} = B'(x)$ for all x.*
(ii) *For generic weights no block of width less than $m + d_{\text{model}}$ does.*

Proof. Outline. Two units with opposite reads produce a linear term of rank one, so d_{model} extra units produce xM. Conversely, the matching of the units in Theorems A.11 and A.13 shows that every extra unit adds at most one such term.

(i) As $\sigma(z) - \sigma(-z) = z$, a column w and rows a, b give $\sigma(xw)(a + b) + \sigma(-xw)(-b) = \sigma(xw)\, a + (xw)\, b$. Choose d_{model} units whose columns $w_{i_1}, \dots, w_{i_{d_{\text{model}}}}$ form an invertible matrix W_K, and let $b_1, \dots, b_{d_{\text{model}}}$ be the rows of $W_K^{-1}M$. Adding b_k to the row $\hat{w}_{i_k}$, and a unit with column $-w_{i_k}$ and row $-b_k$, for each k, gives a block of width $m + d_{\text{model}}$ that computes $\sigma(xW_{\text{read}})W_{\text{write}} + xW_K W_K^{-1}M$.

(ii) Outside a set of measure zero the block is nondegenerate, and for each of the finitely many sets S of units the determinant of $M + W_{\text{read}}[:, S]W_{\text{write}}[S, :]$ is nonzero, since it is a polynomial in the weights equal to $\det M \neq 0$ at $W_{\text{write}} = 0$. Let a block B' of width m', with columns w'_j and rows $\hat{w}'_j$, satisfy $xM + B(x) = B'(x)$. Its odd part gives $W'_{\text{read}}W'_{\text{write}} = W_{\text{read}}W_{\text{write}} + 2M$, and its even part $|xW'_{\text{read}}|\, W'_{\text{write}} = |xW_{\text{read}}|\, W_{\text{write}}$ for ReLU and $\Psi(xW'_{\text{read}})W'_{\text{write}} = \Psi(xW_{\text{read}})W_{\text{write}}$ for GELU. Group the nonzero columns w'_j and the columns w_i into classes of collinear vectors, write $w'_j = s_j u$ with a unit vector u of the class, $u = w_i/\|w_i\|$ in the class of w_i, and let $P_u = \|w_i\|\hat{w}_i$ there and $P_u = 0$ in a class without a column of W_{read}. The even part then forces, class by class, $\sum_j |s_j|\, \hat{w}'_j = P_u$ for ReLU, since the functions $|xu|$ of distinct classes are linearly independent (Step 2 of Theorem A.11), and for GELU, by Lemma A.12 and the independence of the functions $\Psi(\alpha t)$ for distinct $\alpha > 0$ modulo polynomials (Remark A.14), $\sum_{|s_j|=\alpha} \hat{w}'_j = \hat{w}_i$ if w_i lies in the class and $\alpha = \|w_i\|$, and 0 otherwise. With $Q_u = \sum_j s_j \hat{w}'_j$ the odd part reads $\sum_u u\,(Q_u - P_u) = 2M$. Call a class mixed if it contains w'_j of both signs, for GELU of both signs and equal $|s_j|$. A class that is not mixed has $Q_u = P_u$ or $Q_u = -P_u$; let S collect the units i whose class is not mixed and has $Q_u = -P_u \neq 0$. Then

$$2\big(M + W_{\text{read}}[:, S]W_{\text{write}}[S, :]\big) = \sum_{u \text{ mixed}} u\,(Q_u - P_u).$$

Every class of a column w_i contains some w'_j, since $P_u \neq 0$, and every mixed class contains at least two, so $m' \geq m$ plus the number of mixed classes. If $m' < m + d_{\text{model}}$, the right side has rank less than d_{model}, so the left side is singular, which the weights exclude. □

Proof of Theorem 3.4. Part (i) is Proposition A.8 for homogeneous blocks and Proposition A.9 for gated ones, and parts (ii) and (iii) are Corollary A.10. For part (iv), B and B' are bounded for $\tanh$ blocks and xM is not, which gives (i), and (ii) and (iii) are again Corollary A.10. □

Proof of Theorem 3.5. Part (i) is Theorems A.11 and A.13, with Lemma A.7 for invertible M and Remark A.15 for singular M; for each M the condition holds only on a set of weights of measure zero (Corollary A.16 and Remark A.15). Part (ii) is Proposition A.17. □

Depth. Theorem 3.6 and Theorem 6.1 rest on two lemmas. The first turns a gap in functional dimension into disjointness. The second shows that with small writes the Jacobian of a residual network reaches the largest rank that its symmetries allow, by the argument of Theorem C.1 for one block.

Lemma A.18 (A rank criterion)**.** *Let $\theta \mapsto F_\theta$, $\theta \in \mathbb{R}^P$, and $\eta \mapsto G_\eta$, $\eta \in \mathbb{R}^{P'}$, be two families of networks with the same inputs and outputs, whose outputs are real-analytic in the parameters. Let a Lie group act smoothly on $\mathbb{R}^{P'}$ without changing G_η, with orbits of dimension at least k at generic η. If at some θ_0 the Jacobian of $\theta \mapsto (F_\theta(x))_{x\in X}$ has rank greater than $P' - k$ on some finite batch X, then for almost every θ no η satisfies $F_\theta = G_\eta$.*

Proof. Write $F_X(\theta), G_X(\eta) \in \mathbb{R}^n$ for the outputs on X. At generic η the map G_X is constant along an orbit of dimension at least k, so its Jacobian has rank at most $r_0 = P' - k$; the rank is lower semicontinuous, so the bound holds at every η. The Jacobian of F_X reaches its largest rank r^*, which exceeds r_0 by hypothesis, outside the zero set of the sum of the squares of its minors of size r^*. This function is analytic and not identically zero, so its zero set has measure zero. Take θ outside it. By the constant rank theorem, F_X maps a neighborhood U of θ onto an embedded manifold Y of dimension r^*, and after shrinking U a linear projection $\pi : \mathbb{R}^n \to \mathbb{R}^{r^*}$ maps Y diffeomorphically onto an open set. The map $\pi \circ G_X$ has rank at most $r_0 < r^*$, so all its values are critical, and by Sard's theorem its image has measure zero. So $G_X(\mathbb{R}^{P'}) \cap Y$ has measure zero in Y, and so does its preimage under F_X in U, which contains every θ' in U with $F_{\theta'} = G_\eta$ for some η. Countably many such U cover the complement of the zero set. □

Lemma A.19 (Small writes)**.** *Let $d_{\text{model}} \geq 2$, let E and D have rank d_{model}, and let each block, in any order and of any width, be an attention block with heads of width at most d_{model}, causal or not, on sequences of at least two tokens, or an MLP block, with all MLP blocks* tanh*, all GELU, all SwiGLU or all GeGLU blocks. At generic parameters, multiply every write, and every write bias, by τ. For all small $\tau \neq 0$ the residual network without biases has functional dimension $P - d_{\text{model}}^2 - s$ without normalization and $P - d_{\text{model}}(d_{\text{model}} - 1)/2 - s$ under pre-RMSNorm with $\varepsilon > 0$. Without normalization and attention blocks, biases lower the first value by $(L + 1)d_{\text{model}}$. Without an embedding, a readout, biases and normalization it has functional dimension $P - s$.*

Proof. Write the write of block ℓ, for an attention block its output projection, as τV_ℓ. Let $W = [\, W_{\text{read}}^1 \mid \cdots \mid W_{\text{read}}^L \,]$ place the reads of the MLP blocks side by side and V stack their V_ℓ, so that $f(nW)V$ is one wide block that holds all units of the network, and let the heads of all attention blocks form one attention block. Divide the columns of the Jacobian that belong to the reads, for attention blocks the queries, keys and values, and to the read biases by τ, which keeps its rank, and let $\tau \to 0$. On a batch of inputs the limit is a linear map J_0 on $\mathbb{R}^P$; as in Appendix C, a direction η is flat if $J_0\,\eta = 0$ on every input.

MLP blocks without normalization. The stream is $h_\ell = xE + b_E + O(\tau)$, so the output is

$$y = xED + b_E D + b_D + \tau \sum_{\ell=1}^{L} \Big(f\big((xE + b_E)W_{\text{read}}^\ell + b_{\text{read}}^\ell\big)V_\ell + b_{\text{write}}^\ell \Big) D + O(\tau^2),$$

with the write biases written as $\tau b_{\text{write}}^\ell$. Hence $J_0\,\eta = \mathrm{D}_\phi G(\phi, x)\,\psi$ for the one-block function G of Lemma C.3 at $M = ED$, $\gamma = b_E D + b_D$, the columns c_i of EW, the entries β_i of $b_E W + b_{\text{read}}$ and the rows a_i of VD, in the direction ψ with

$$\psi_M = \eta_E D + E\eta_D, \qquad \psi_\gamma = \eta_{b_E} D + b_E \eta_D + \eta_{b_D} + \textstyle\sum_\ell \eta_{b_{\text{write}}^\ell} D,$$
$$\psi_C = E\eta_W, \qquad \psi_\beta = b_E \eta_W + \eta_{b_{\text{read}}}, \qquad \psi_A = \eta_V D,$$

where ψ_C, ψ_β and ψ_A collect the ψ_{c_i}, ψ_{β_i} and ψ_{a_i}, and the values of the gated units enter as the gates do. At generic parameters the hypotheses of Lemma C.3 hold, as for one block (Appendix C). For a flat η the lemma gives $\psi_M = 0$, $\psi_\gamma = 0$ and a rescaling at every unit. As E and D have rank d_{model}, $\psi_M = 0$ gives $\eta_E = EX$ and $\eta_D = -XD$ for some $X \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$; ψ_C and ψ_A make

η_W and η_V a rescaling of the units, which fixes $\eta_{b_{\mathrm{read}}}$ through ψ_β; and $\psi_\gamma = 0$ fixes η_{b_D} once η_{b_E} and the L write biases are chosen. So the flat directions form a space of dimension $d_{\mathrm{model}}^2 + s$, and $d_{\mathrm{model}}^2 + (L+1)d_{\mathrm{model}} + s$ with biases. Without an embedding and a readout, $M = I$ is fixed, and only the rescalings are left.

Attention blocks without normalization. Without biases, $y = xED + \tau \sum_\ell g_\ell(xE)D + O(\tau^2)$, where g_ℓ is block ℓ with the write V_ℓ. So $J_0\,\eta$ is the derivative of the residual network with one block that holds all heads and units, in the direction with the entries ψ_M, ψ_C and ψ_A above and, for each head i,

$$\psi_{B_i} = E\big(\eta_{Q,i} W_{K,i}^\top + W_{Q,i}\eta_{K,i}^\top\big)E^\top/\sqrt{d_h}, \qquad \psi_{C_i} = E\big(\eta_{V,i}W_{O,i} + W_{V,i}\eta_{O,i}\big)D,$$

at $B_i = EW_{Q,i}W_{K,i}^\top E^\top/\sqrt{d_h}$ and $C_i = EW_{V,i}W_{O,i}D$ as in Appendix C, where $W_{O,i}$ holds the rows of V_ℓ that belong to head i of block ℓ. On inputs whose tokens all equal x, head i gives every token xC_i, and its derivative is $x\psi_{C_i}$, since all its scores move alike. There the equation is that of the units with the linear part $M + \sum_i C_i$, and Lemma C.3 gives a rescaling at every unit. The units then drop out on every input. On inputs whose tokens other than the second equal the first, head i gives the second token the output of (14) with $k = 1$ if its block is causal and $k = T - 1$ otherwise; the constants $-\frac{1}{2}\log k$ act as biases of the $\tanh$ units, which Lemma C.3 allows, so Step 2 for attention blocks in Appendix C gives $\psi_M = 0$, $\psi_{B_i} = 0$ and $\psi_{C_i} = 0$. Cancelling E on the left and $E^\top$ or D on the right, the last two leave, as in Step 1 there, the directions of the changes of basis between the queries and keys and between the values and outputs of each head. So the flat directions form a space of dimension $d_{\mathrm{model}}^2 + s$, and without an embedding and a readout of dimension s.

Under pre-RMSNorm. The stream is $h_\ell = xE + O(\tau)$, every block reads $N(xE) + O(\tau)$, and the output is

$$y = N(xE)D + \tau\,\mathrm{D}N(xE)\Big(\textstyle\sum_\ell g_\ell\big(N(xE)\big)\Big)D + O(\tau^2),$$

with N applied to every token. So $J_0\,\eta = \mathrm{D}N(h)\big(x\eta_E + k(n)\big)D + N(h)\eta_D$, with $h = xE$, $n = N(h)$ and $k(n)$ the derivative of the wide block along η at the input n. This is the equation of the proof of Theorem C.1(ii) with $h_1 = xE$ and $\mathrm{D}_\theta h_1\,\eta = x\eta_E + k(n)$, and its Steps 1 to 5 go through, with $k(n)$ bounded in Step 2 since n stays in the ball, and with the units handled first, through equal tokens, in Step 4. In Step 3 the direction η^Y now has the entries EY and $-YD$ at E and D and no others, as $\mathrm{D}N(h)\,(hY) = N(h)Y$ for antisymmetric Y. So the flat directions are these, the rescalings of the units and the changes of basis inside the heads, a space of dimension $d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 + s$.

In each case the kernel of J_0 shrinks as the batch grows and becomes stationary at the flat directions, so on some finite batch J_0 has rank P minus this dimension. By lower semicontinuity the Jacobian of the residual network on this batch has at least this rank for all small $\tau \neq 0$. Its symmetries bound the rank by the same number, since they move the parameters in independent directions when E has rank d_{model} and the queries and values of every head have full column rank: the global change of basis (Theorem A.3(iii)), the translations (Lemma A.4), the rescalings of the units and the changes of basis inside the heads. □

Proof of Theorem 3.6(i) and (ii). By Lemma A.19, a residual network without biases has functional dimension $P - d_{\mathrm{model}}^2 - s$ at some parameters, where s counts the changes of basis inside the heads and the rescalings of the values of gated units, and a network of GELU blocks without an embedding and a readout has functional dimension P. In a skipless network of the same widths, with embedding E' and readout D', the changes of basis at h_0 and h_L, those inside the heads and the rescalings have orbits of dimension $2d_{\mathrm{model}}^2 + s$ at generic parameters, where E' and D' have rank d_{model} and the queries and values of every head have full column rank. As $P - d_{\mathrm{model}}^2 - s > P - 2d_{\mathrm{model}}^2 - s$, Lemma A.18 gives the claim. Without an embedding, a readout and attention blocks, for $L \geq 2$ the changes of basis at h_1 have orbits of positive dimension, since every block has at least one unit, so Lemma A.18 applies again. Under pre-RMSNorm with $\varepsilon > 0$ the networks stay analytic, Lemma A.19 gives the residual network functional dimension $P - d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 - s$, and the orthogonal changes of basis at h_0 and h_L, which commute with the normalization, have orbits

of dimension $d_{\text{model}}(d_{\text{model}}-1)+s$ in the skipless network; as $d_{\text{model}} \geq 2$, Lemma A.18 applies once more. □

Proof of Theorem 3.6(iii) and (iv). Let the residual network have blocks with $m_1, \ldots, m_L$ hidden units and P parameters. A skipless network whose blocks have $m'_\ell \geq d_{\text{model}}$ hidden units has $P' = P + 2d_{\text{model}} \sum_\ell (m'_\ell - m_\ell)$ parameters, and its changes of basis at $h_0, \ldots, h_L$ have orbits of dimension $(L+1)d_{\text{model}}^2$ at generic parameters (Lemma A.5). By Lemma A.19 the residual network has functional dimension $P - d_{\text{model}}^2$ at some parameters, which exceeds $P' - (L+1)d_{\text{model}}^2$ exactly when $\sum_\ell (m'_\ell - m_\ell) < L d_{\text{model}}/2$. For each of the finitely many such choices of widths, Lemma A.18 gives a set of weights of measure zero, and so does their union. For (iv), Theorem 3.5(ii) with $M = I$ turns each residual block, whose read has rank d_{model}, into a block without a skip with d_{model} more units, and composing these gives a skipless network that equals the residual one. □

Proof of Theorem 3.7. MLP blocks act on each token separately, so we compare the networks on rows $x \in \mathbb{R}^{1\times d_{\text{in}}}$. Outside a set of weights of measure zero, E and D have rank d_{model}, no read of a unit vanishes, and $I + W^\ell_{\text{read}}[:, S]\, W^\ell_{\text{write}}[S, :]$ is invertible for every block ℓ and every set S of its units, since each determinant is a polynomial in the weights that equals 1 when the writes vanish. On an open set of inputs on which no unit of the residual network switches, with active sets S_ℓ, the network is the linear map $x \mapsto xEMD$ with $M = \prod_\ell \left(I + W^\ell_{\text{read}}[:, S_\ell]\, W^\ell_{\text{write}}[S_\ell, :]\right)$ invertible, of rank d_{model}. By induction over the blocks, on such a set the pre-activation of each unit of the next block is x times a nonzero vector, so the inputs at which some unit switches have measure zero, and the inputs near which the network is linear of rank d_{model} form an open dense set Ω.

Let a skipless network F' have embedding E' and a first block with reads $w'_1, \ldots, w'_{m'}$, $m' < 2d_{\text{model}}$, and set $a_i = E' w'_i$. Take x_0 with $x_0, -x_0 \in \Omega$ and $x_0 a_i \neq 0$ whenever $a_i \neq 0$. The units active near x_0, those with $x_0 a_i > 0$, and those active near $-x_0$ are disjoint, so one of the two sets, say the first after replacing x_0 by $-x_0$ if needed, has fewer than d_{model} units. Near x_0 the other units of the first block output zero, so F' depends on x only through the numbers xa_i of the active units, and $F'(x_0 + u) = F'(x_0)$ for every small u in a subspace of dimension greater than $d_{\text{in}} - d_{\text{model}}$. The residual network F changes by $uEMD$, which vanishes only on a subspace of dimension $d_{\text{in}} - d_{\text{model}}$. So some small u changes F and leaves F' unchanged, and $F' \neq F$. □

Proof of Theorem 6.1. Part (i). Let r be P minus the dimension of the symmetries of the residual network, the value in Lemma A.19. At generic parameters the symmetries move the parameters in independent directions, so the rank of the Jacobian is at most r on every batch. Lemma A.19 gives parameters and a finite batch on which it equals r. On this batch the rank is therefore at least r outside the zero set of the sum of the squares of the minors of size r, an analytic function that is not identically zero, so this zero set has measure zero. At generic parameters the functional dimension is thus r, which is the equality condition.

Part (ii). When every block has width at least d_{model}, the functional dimension of the skipless network is at most $P - (L+1)\dim \mathcal{G} - s$, and $(L+1)d_{\text{model}}$ lower with biases (Theorem 3.3(ii), Remark A.6, Lemma A.4); with biases, the induction in the proof of Lemma A.5 also makes the translations act freely, through the entries at b_E and at the write biases. This gives the gain of at least $L \dim \mathcal{G}$, with equality when the skipless network satisfies the equality condition. □

Proof of Theorem 6.2. By Theorem 6.1(i) the residual network satisfies the equality condition, so it remains to show it for the skipless network; the gain is then $L d_{\text{model}}^2$, with or without biases.

Part (i), $\tanh$. With read biases b_ℓ and write biases c_ℓ, which may vanish, the skipless network computes

$$x^0 = x, \qquad x^\ell = \tanh\left(x^{\ell-1} A_\ell + \beta_\ell\right) \ (1 \leq \ell \leq L), \qquad y = x^L A_{L+1} + \beta_{L+1},$$

with $A_1 = EW_1$, $A_\ell = V_{\ell-1} W_\ell$ and $A_{L+1} = V_L D$, where W_ℓ and V_ℓ are the read and the write of block ℓ, and $\beta_1 = b_E W_1 + b_1$, $\beta_\ell = c_{\ell-1} W_\ell + b_\ell$ and $\beta_{L+1} = c_L D + b_D$. As $\sigma(z) = \tanh(z/2)$ is injective, two such networks have the same output exactly when the nets of Fefferman (1994) with the hidden layers $x^\ell = \sigma(2x^{\ell-1} A_\ell + 2\beta_\ell)$ and the output layer $\sigma(x^L A_{L+1} + \beta_{L+1})$ do, and then so do, for every input coordinate and every output, the nets with one input and one output that keep

only these. At generic parameters every entry of every A_ℓ is nonzero, and no ratio of the weights that one unit sends to two units of a layer of width m is a fraction p/q with $1 \le q \le 100m^2$. These conditions are open, and they are those of his Theorem I.B.9, on nets with one input and one output, which puts no condition on the thresholds. So near generic parameters, two networks with the same output give, for every input coordinate and every output, nets with one input and one output that agree up to permutations of the units of each hidden layer and sign flips. These are the identity: at the first layer where a unit is moved or flipped, the weights it receives would move by a fixed amount, since none vanishes and the weights that one unit sends to two units differ in absolute value. So the two networks have the same A_ℓ and β_ℓ. When E, D and every read and write have rank d_{model}, the parameters with the same A_ℓ and β_ℓ are those reached by the changes of basis at the $L+1$ hidden states, and with biases by the translations there.

Take generic parameters and a batch on which the rank r of the Jacobian equals the functional dimension. Near these parameters the rank on this batch stays r, so its kernel is the space of directions that are flat for every input, and the parameters with the same outputs on the batch form a manifold of dimension $P - r$ along which the function does not change. So $P - r$ is at most the dimension of the symmetries, and the symmetries give the reverse inequality (Theorem 3.3(ii), Lemma A.4).

Part (ii), GELU. Let block ℓ have $m_\ell + d_{\text{model}}$ units, with $m_\ell \ge d_{\text{model}}$, and take a residual network with blocks of m_ℓ units at generic parameters, with every write multiplied by τ as in Lemma A.19. By the proof of Proposition A.17(i), each of its blocks equals a block without skip with d_{model} more units: for d_{model} units i whose reads w_i form an invertible matrix, add the row γ_i of its inverse to the write of unit i, and add a unit with the read $-w_i$ and the write $-\gamma_i$. This construction is smooth in the residual parameters near these, where the chosen reads stay invertible, and gives a skipless network that computes the residual one. By the chain rule, the columns of the skipless Jacobian therefore span those of the residual Jacobian, which, with the read columns divided by τ, tend to the limit J_0 of Lemma A.19, of rank $P_R - d_{\text{model}}^2$ on some finite batch, where $P_R = P - 2Ld_{\text{model}}^2$ counts the parameters of the residual network. As $\sigma(z) - \sigma(-z) = z$ gives $\sigma'(-z) = 1 - \sigma'(z)$, and every block is the identity at $\tau = 0$, the reads of the added units give the further columns

$$\big(\sigma'(xEw_i) - 1\big)(xE\delta_i)\,\gamma_i D, \qquad \delta_i \in \mathbb{R}^{d_{\text{model}}},$$

in the limit. Suppose that $J_0\,\eta$ plus a sum of these vanishes for all x. Each added term has the Gaussian part of the term of ψ_{c_i} in Lemma C.3, with $E\delta_i$ in place of ψ_{c_i} and the row $\gamma_i D$ in place of a_i. In the proof of that lemma for GELU, the coefficient of t^2 for the slowest unit j becomes $u_j\big((n\psi_{c_j})\,a_j + (nE\delta_j)\,\gamma_j D\big)$. At generic parameters $\gamma_j D$ is not parallel to a_j, so $nE\delta_j = 0$ for all n in an open set, and $\delta_j = 0$ as E has rank d_{model}; the proof then goes on as before. So, on some finite batch, the added reads raise the rank in the limit by Ld_{model}^2, to $P_R - d_{\text{model}}^2 + Ld_{\text{model}}^2 = P - (L+1)d_{\text{model}}^2$. By lower semicontinuity the skipless network has this rank for small τ, which is the most its symmetries allow, and an analytic network keeps its largest rank at generic parameters. □

Remark A.20 (ReLU-type blocks). For ReLU, ReLU^2 and ReGLU blocks without normalization, the proof of Lemma A.19 goes through on a finite batch on which no pre-activation of the limit vanishes: Lemma C.3 covers these units, and for small τ every pre-activation keeps its sign on the batch, so the Jacobian is polynomial in the weights near these parameters. The residual network therefore has functional dimension $P - d_{\text{model}}^2 - s$ on an open set of weights that contains all small enough multiples of generic writes, and at almost every point of this set the skips add at least Ld_{model}^2 when every block has width at least d_{model}. The count fails on another open set. For $d_{\text{model}} = 2$, the block $h \mapsto h + 2\big(\text{ReLU}(-h_1), \text{ReLU}(-h_2)\big) = \big(|h_1|, |h_2|\big)$ maps the plane onto a quadrant, and a unit of the next block with the read $-(1,1)^\top$ never switches on. Its read and its write then leave $2d_{\text{model}} - 1$ more directions flat, and small changes of the weights keep this. Which weights keep the count is left for future work.

A.5 No absorption in attention (Theorem 3.8, Proposition 3.9)

Theorem 3.8 collects what the main text uses of the following theorem, for the attention blocks B_θ of (4).

Theorem A.21 (Attention and linear terms)**.** *Let θ and θ' be parameters of attention blocks with any numbers and widths of heads, and let $M \neq 0$ be a $d_{\text{model}} \times d_{\text{model}}$ matrix.*
(i) *For almost every $h \in \mathbb{R}^{T \times d_{\text{model}}}$, $hM + B_\theta(h) \neq B_{\theta'}(h)$.*
(ii) *Some h whose rows have norm at most 1 has a row in which $hM + B_\theta(h)$ and $B_{\theta'}(h)$ differ by at least $\|M\|/3$, where $\|M\|$ is the spectral norm.*
(iii) *With an embedding E and a readout D with $ED \neq 0$, the residual network $x \mapsto \big(xE + B_\theta(xE)\big)D$ differs from every skipless network $x \mapsto B_{\theta'}(xE')D'$.*

Overview. Two kinds of input suffice. If all tokens are equal, attention returns them unchanged, so the value–output products of all heads must add up to M. If one token is zero, all its scores are zero, so every head gives it the average of the tokens, while the skip gives it zero; this asks the same sum to vanish. Part (ii) makes the conflict quantitative, and analyticity turns it into part (i). With an embedding and a readout, the same two inputs give part (iii).

Proof of Theorem A.21. Let $F(h) = hM + B_\theta(h) - B_{\theta'}(h)$, and let θ' have H' heads. The difference $B_{\theta'} - B_\theta$ is one attention block with $H + H'$ heads,

$$B_{\theta'}(h) - B_\theta(h) = \sum_{k=1}^{H+H'} R_k(h)\, h\, C_k,$$

where the $R_k(h)$ are the attention maps of the heads of θ' and of θ, and the C_k are their value–output products, with a minus sign for the heads of θ. Let $C = \sum_k C_k$.

Step 1: equal tokens. Let every row of h be x. Each row of $R_k(h)$ sums to 1, so every row of $R_k(h)h$ is x. Hence every row of $F(h)$ is $x(M - C)$.

Step 2: one zero token. Let the last row of h be 0 and the other rows y. Every score in the last row of $R_k(h)$ is a product with the last row of h, which is 0. So that row of $R_k(h)$ is $(1/T, \dots, 1/T)$, also with a causal mask, since the last token sees all T tokens. Hence the last row of $F(h)$ is $-\frac{T-1}{T}\, yC$.

Proof of (ii). Over $\|x\|, \|y\| \leq 1$, Step 1 gives rows of norm up to $\|M - C\|$, and Step 2 up to $\frac{T-1}{T}\|C\| \geq \frac{1}{2}\|C\|$. If $\|M - C\| < \|M\|/3$, then $\|C\| > \frac{2}{3}\|M\|$, so Step 2 reaches more than $\|M\|/3$.

Proof of (i). F is real-analytic, because the softmax is, and by (ii) some entry of F is not identically zero. The zeros of a real-analytic function that is not identically zero form a set of measure zero: in one variable they are isolated, and the general case follows by induction on the dimension with Fubini's theorem.

Proof of (iii). Let C_θ and $C_{\theta'}$ be the sums of the value–output products of the heads of θ and of θ'. On inputs whose rows all equal a row vector u, Step 1 gives the rows $uE(I + C_\theta)D$ in the residual network and $uE'C_{\theta'}D'$ in the network without a skip. On inputs whose last row is 0 and whose other rows equal v, Step 2 gives the last rows $\frac{T-1}{T}\, vEC_\theta D$ and $\frac{T-1}{T}\, vE'C_{\theta'}D'$. If the two networks are equal, then $E(I + C_\theta)D = E'C_{\theta'}D' = EC_\theta D$, so $ED = 0$. □

We recall Theorem 3.8 from Section 3.4.

Theorem 3.8 (No absorption in attention)**.** *Let θ and θ' be parameters of attention blocks with any numbers and widths of heads, and let $M \neq 0$ be a $d_{\text{model}} \times d_{\text{model}}$ matrix.*
(i) *For almost every $h \in \mathbb{R}^{T \times d_{\text{model}}}$, $hM + B_\theta(h) \neq B_{\theta'}(h)$.*
(ii) *With an embedding E and a readout D with $ED \neq 0$, the residual network $x \mapsto \big(xE + B_\theta(xE)\big)D$ differs from every skipless network $x \mapsto B_{\theta'}(xE')D'$.*

Proof of Theorem 3.8. Parts (i) and (ii) are Theorem A.21(i) and (iii). □

Proposition 3.9 follows from a lemma on attention heads.

Lemma A.22 (Attention heads are identifiable)**.** *Let $T \geq 3$. Let $\Gamma_1, \ldots, \Gamma_m \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ be distinct, let $C_1, \ldots, C_m, M \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$, and write $P_\Gamma(h) = \operatorname{softmax}(h\Gamma h^\top)$, with the softmax taken row by row. If*

$$hM + \sum_{k=1}^{m} P_{\Gamma_k}(h)\, h\, C_k = 0$$

for all h in a nonempty open subset of $\mathbb{R}^{T \times d_{\text{model}}}$, then $M = 0$ and $C_1 = \cdots = C_m = 0$.

Overview. Move one token along a complex line. Each head becomes a meromorphic function of one complex variable, whose poles lie on a vertical line that its score matrix determines, so heads with distinct score matrices have disjoint poles. At a pole of one head the other terms are regular, so its residue must vanish, and the residues force its value–output product to vanish. What is left is the skip and a uniform head, which two simple inputs separate.

Proof of Lemma A.22. One moving token. Fix h in the open set, with rows $x_1, \ldots, x_T$, and fix $v \in \mathbb{R}^{1 \times d_{\text{model}}}$. Replace the second row by $x_2 + tv$ with $t \in \mathbb{C}$, and look at the first row. For a head with score matrix Γ, the first row has scores $z_r = x_1 \Gamma x_r^\top$ for $r \neq 2$, and $a + bt$ for $r = 2$, where $a = x_1 \Gamma x_2^\top$ and $b = x_1 \Gamma v^\top$. Its output is

$$g_\Gamma(t) = \frac{e^{a+bt}\,(x_2 + tv) + Z\bar{x}}{e^{a+bt} + Z}, \qquad Z = \sum_{r \neq 2} e^{z_r}, \qquad \bar{x} = \frac{1}{Z} \sum_{r \neq 2} e^{z_r} x_r,$$

where $\bar{x}$ is the average of the other tokens under the weights of the head. The first row of hM is $x_1 M$, which does not depend on t.

Poles. If $b \neq 0$, then g_Γ is meromorphic on $\mathbb{C}$, with simple poles at

$$t_n = c + \frac{(2n+1)\pi i}{b}, \quad n \in \mathbb{Z}, \qquad \text{where } c = \frac{\log Z - a}{b},$$

so all its poles lie on the vertical line $\operatorname{Re} t = c$. The residue at t_n is $(x_2 + t_n v - \bar{x})/b$. If $\Gamma = 0$, then g_Γ is affine in t and has no poles.

The poles tell the heads apart. For h and v outside a closed set with empty interior, $b \neq 0$ for every $\Gamma_k \neq 0$, and distinct $\Gamma_k \neq 0$ give distinct lines $\operatorname{Re} t = c$. The first condition fails only on the zero sets of nonzero polynomials. For the second, suppose that two heads $\Gamma \neq \Gamma'$ have $c = c'$ for all h and v near some point. Clearing denominators and comparing the terms bilinear in (x_2, v) gives $x_1 \Gamma' = \lambda\, x_1 \Gamma$ for some number $\lambda \neq 0$. Comparing the terms linear in v then gives $\log Z' = \lambda \log Z$, that is,

$$\log \sum_{r \neq 2} e^{\lambda z_r} \;=\; \lambda \log \sum_{r \neq 2} e^{z_r}.$$

Because $T \geq 3$, the sum has at least two terms, $r = 1$ and $r = 3$, and z_3 varies with x_3. Differentiating in z_3 gives $\operatorname{softmax}(\lambda z)_3 = \operatorname{softmax}(z)_3$ on an interval of z_3, which forces $\lambda = 1$. So $x_1 \Gamma' = x_1 \Gamma$ for all x_1 near a point, and $\Gamma' = \Gamma$, a contradiction. With two tokens the sum has the single term $r = 1$, and this step fails (Section 3.4).

Residues. The identity holds for real t near 0, so by analytic continuation it holds on all of $\mathbb{C}$ away from the poles. At a pole t_n of a head $\Gamma_k \neq 0$, no other term is singular, so the residue times C_k vanishes: $(x_2 + t_n v - \bar{x})\, C_k = 0$ for every $n \in \mathbb{Z}$. Subtracting two consecutive ones gives $vC_k = 0$. This holds for all v in an open set, so $C_k = 0$.

The skip. What remains is $hM + P_0(h)\, h\, C_0 = 0$, where C_0 is the coefficient of $\Gamma = 0$ (or 0 if no Γ_k vanishes). This is linear in h, so it holds for all h. Take the first row of h equal to x and the other rows 0: the first two rows give $x(M + C_0/T) = 0$ and $xC_0/T = 0$. So $C_0 = 0$ and $M = 0$. □

We recall Proposition 3.9 from Section 3.4.

Proposition 3.9 (The softmaxes must match)**.** *Let $T \geq 3$, let θ and θ' be parameters of attention blocks, and suppose that $hM + B_\theta(h) = B_{\theta'}(h)$ for all h in a nonempty open set. Then $M = 0$, and for every matrix Γ, the heads with score matrix Γ have the same sum of value–output products in B_θ as in $B_{\theta'}$.*

Proof of Proposition 3.9. Group the heads of $B_{\theta'}$ and B_θ by score matrix, add their value–output products, with a minus sign for the heads of θ, and apply Lemma A.22. □

A.6 One free block per hidden state (Proposition 5.1)

We recall Proposition 5.1 from Section 5.1, where W_1 is an invertible $d_{\text{model}} \times d_{\text{model}}$ block formed by d_{model} columns of a matrix that reads h_ℓ, or by d_{model} rows of a matrix that writes it.

Proposition 5.1 (One free block per hidden state)**.** **(i)** *In a skipless network or a network of mixed blocks, the map $W \mapsto WG$ on the matrices that write h_ℓ and $W \mapsto G^{-1}W$ on those that read it turns W_1 into the identity for $G = W_1$ if W_1 is part of a read, and for $G = W_1^{-1}$ if it is part of a write. Under RMSNorm with $\varepsilon > 0$, $G = Q$ from a QR factorization $W_1 = QT$, or $W_1^\top = QT$ for a write, turns W_1 into a triangular matrix. Done at every hidden state, this removes d_{model}^2, respectively $d_{\text{model}}(d_{\text{model}} - 1)/2$, parameters per hidden state without changing the function.*
(ii) *In a residual network, the same map with one G at all hidden states preserves the function and turns one such block, in any read or write of the network, into the identity or a triangular matrix.*

Overview. The proposition spends the change of basis of Proposition 3.1 at each hidden state on one square block of a read or of a write: the change of basis given by that block, or by its orthogonal factor under normalization, turns it into the identity or into a triangular matrix.

Proof of Proposition 5.1. By Proposition 3.1, a change of basis $G \in \mathcal{G}$ at h_ℓ preserves the function and replaces every read W of h_ℓ by $G^{-1}W$ and every write W into h_ℓ by WG. So the columns of a read that form W_1 become $G^{-1}W_1$, and the rows of a write that form W_1 become W_1G. Without normalization $\mathcal{G} = \mathrm{GL}(d_{\text{model}})$, and $G = W_1$ for a read, or $G = W_1^{-1}$ for a write, turns W_1 into I. Under RMSNorm with $\varepsilon > 0$, $\mathcal{G} = \mathrm{O}(d_{\text{model}})$ (Lemma A.1). For a read, let $W_1 = QT$ be a QR factorization, with T upper triangular; $G = Q$ gives $Q^\top W_1 = T$. For a write, let $W_1^\top = Q_0 R$ be a QR factorization; then $Q = Q_0^\top$ is orthogonal, $T = R^\top$ is lower triangular, $W_1 = TQ$, and $G = Q^\top$ gives $W_1 Q^\top = T$. The identity fixes all d_{model}^2 entries of W_1, and the triangular form fixes the $d_{\text{model}}(d_{\text{model}} - 1)/2$ entries on one side of its diagonal. The change of basis at h_ℓ touches only the writes into h_ℓ and the reads of h_ℓ, and a matrix is touched on each side by one hidden state only (Proposition 3.1(iii)), so the reductions at all hidden states can be made together, one block per hidden state. In a residual network, one change of basis $G \in \mathcal{G}$ applied at every hidden state preserves the function (Theorem A.3(iii)); it replaces every read W by $G^{-1}W$ and every write W by WG, and leaves the identity skips unchanged, since $G^{-1}IG = I$. The same choice of G then turns one block W_1, of any one layer, into the identity or into a triangular matrix. □

A.7 LayerNorm reduces to RMSNorm (Proposition A.23)

The reduction of LayerNorm to RMSNorm is standard: the centering folds into the adjacent weights (Elhage et al., 2021), and centering every write gives an RMSNorm network whose hidden states are mean-free (Ashkboos et al., 2024a; Theus et al., 2025). We state it with the constants that the counts need. It is used in Sections 2 and 4 and for ViT-B in Section 5.1 and Appendix E.

Proposition A.23 (LayerNorm reduces to RMSNorm)**.** *Let $Z = I - \frac{1}{d_{\text{model}}}\mathbf{1}\mathbf{1}^\top$ and $\mathcal{V} = \mathbf{1}^\perp$, and consider a pre-LayerNorm network, skipless or residual, whose reads and readout see $\mathrm{LN}(h) = \gamma \odot hZ/\sqrt{\|hZ\|^2/d_{\text{model}} + \varepsilon} + \beta$, with h, γ and β row vectors.*
(i) *Replacing each read W by $\mathrm{diag}(\gamma)W$ with the added bias βW, and each write W, write bias b and embedding E by WZ, bZ and EZ, gives a network that computes the same function, whose hidden states lie in $\mathcal{V}$, and whose reads see $N(h) = h/\sqrt{\|h\|^2/d_{\text{model}} + \varepsilon}$.*
(ii) *In an orthonormal basis of $\mathcal{V}$, this network is a pre-RMSNorm network of width $d_{\text{model}} - 1$, with every read multiplied by $\sqrt{d_{\text{model}}/(d_{\text{model}} - 1)}$ and ε replaced by $\varepsilon d_{\text{model}}/(d_{\text{model}} - 1)$.*
(iii) *If the gains γ have nonzero entries, the difference in redundancy between the skipless and the residual version of the LayerNorm network equals the difference for the reduced networks of (ii).*

Overview. LayerNorm sees only the mean-free part of the stream, so the mean can be removed from every write without changing the function. On the mean-free subspace LayerNorm is RMSNorm up to constants. The map from the parameters of the LayerNorm network to those of the reduced network has a derivative that is onto, and it is the same for the skipless and the residual version, so it preserves the difference in redundancy.

Proof. (i) Since $\mathrm{LN}(h) = \gamma \odot N(hZ) + \beta$, a read satisfies $\mathrm{LN}(h)\, W = N(hZ)\,\mathrm{diag}(\gamma)W + \beta W$. Every hidden state is a sum of writes, directly in a skipless network and through the skips in a residual one, so replacing every write, write bias and the embedding by its image under Z replaces every hidden state h by hZ. Since $Z^2 = Z$, every read then sees the same $N(hZ)$ as before, and the function is unchanged; the hidden states lie in $\mathcal{V}$. (ii) Let the columns of $\bar{U} \in \mathbb{R}^{d_{\mathrm{model}} \times (d_{\mathrm{model}}-1)}$ be an orthonormal basis of $\mathcal{V}$ and $h = y\bar{U}^\top$ with a row vector y. Then $\|h\|^2/d_{\mathrm{model}} + \varepsilon = \frac{d_{\mathrm{model}}-1}{d_{\mathrm{model}}}\big(\|y\|^2/(d_{\mathrm{model}} - 1) + \varepsilon d_{\mathrm{model}}/(d_{\mathrm{model}} - 1)\big)$, so $N(y\bar{U}^\top) = \sqrt{d_{\mathrm{model}}/(d_{\mathrm{model}} - 1)}\, N'(y)\bar{U}^\top$, where N' is RMSNorm of dimension $d_{\mathrm{model}} - 1$ with constant $\varepsilon d_{\mathrm{model}}/(d_{\mathrm{model}} - 1)$. A read $N(y\bar{U}^\top)W$ is the read $\sqrt{d_{\mathrm{model}}/(d_{\mathrm{model}} - 1)}\,\bar{U}^\top W$ of $N'(y)$, and a write WZ is the write $W\bar{U}$ in the coordinates y. (iii) The reduced parameters are $\sqrt{d_{\mathrm{model}}/(d_{\mathrm{model}} - 1)}\,\bar{U}^\top \mathrm{diag}(\gamma)W$ and $b + \beta W$ for each read, $W\bar{U}$ and $b\bar{U}$ for each write, $E\bar{U}$ for the embedding, and the analogous expressions for the readout. For γ with nonzero entries each is linear and onto in W, b or E, so the derivative of the map from the parameters of the LayerNorm network to those of the reduced network is onto. The LayerNorm network computes the reduced network's function composed with this map, so by the chain rule every batch Jacobian of the LayerNorm network is that of the reduced network times this derivative, and at corresponding parameters both have the same functional dimension, and the redundancy of the LayerNorm network is that of the reduced network plus the difference of the parameter counts. That difference is the same for the skipless and the residual versions, which have the same shapes. □

Section 6 of the notebook (Appendix F) checks each step on a residual pre-LayerNorm network with $d_{\mathrm{model}} = 6$: folding the gains changes the function by a relative 2×10^{-16}, centering the writes by 2×10^{-16}, and an orthogonal change of basis on $\mathcal{V}$ by 8×10^{-16}, while a generic orthogonal change that moves $\mathbf{1}$ changes it by 0.75.

A.8 RMSNorm: a gauge fix and a constraint (Theorems 4.1 and A.24)

Theorem 4.1 is the case $\varepsilon > 0$ of the following theorem, without the converse in (ii). As in Section 4, $\Pi_\ell = A_1 \cdots A_\ell$, with $\Pi_0 = I$.

Theorem A.24 (RMSNorm with learned skips)**.** *Consider pre-RMSNorm networks with $\varepsilon \geq 0$, and the network with learned skips of (5), whose skips A_ℓ are invertible in (i) and (ii).*
(i) Triangular or symmetric skips. *Let $Q_0 = I$, and for $\ell = 1, \ldots, L$ let $Q_{\ell-1}^\top A_\ell = R_\ell Q_\ell^\top$ be an RQ factorization, with R_ℓ upper triangular with positive diagonal and Q_ℓ orthogonal. The network with the skips R_ℓ, the reads $Q_{\ell-1}^\top W_{\mathrm{read}}^\ell$, the writes $W_{\mathrm{write}}^\ell Q_\ell$ and the readout $Q_L^\top D$ computes the same function. So does the network built in the same way from the polar decompositions $Q_{\ell-1}^\top A_\ell = S_\ell Q_\ell^\top$, with S_ℓ symmetric positive definite in place of R_ℓ.*
(ii) Orthogonal skips are gauge. *If every Π_ℓ lies in $\mathcal{G}$, write $\Pi_\ell = c_\ell P_\ell$ with $c_\ell > 0$ and P_ℓ orthogonal, where $c_\ell = 1$ if $\varepsilon > 0$. Then the network computes the same function as the residual network with the embedding E, the reads $P_{\ell-1} W_{\mathrm{read}}^\ell$, the writes $W_{\mathrm{write}}^\ell \Pi_\ell^{-1}$ and the readout $P_L D$. Conversely, let the reads and the readout have rank d_{model}, and let invertible $G_0, \ldots, G_L$ turn every skip into the identity, $G_{\ell-1}^{-1} A_\ell G_\ell = I$. If the network, written in the variables $h_\ell G_\ell$, is a pre-RMSNorm residual network, then every Π_ℓ lies in $\mathcal{G}$.*
(iii) What each skip adds. *Under the equality condition for the networks with learned skips, with identity skips and without skips, the functional dimension of the network with learned skips exceeds that of the residual network by $L\,(d_{\mathrm{model}}^2 - \dim \mathcal{G})$, and that of the residual network exceeds that of the skipless network by $L \dim \mathcal{G}$. These are $L\, d_{\mathrm{model}}(d_{\mathrm{model}} + 1)/2$ and $L\, d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2$ for $\varepsilon > 0$, and $L\,(d_{\mathrm{model}}(d_{\mathrm{model}} + 1)/2 - 1)$ and $L\,(d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 + 1)$ for $\varepsilon = 0$.*

Overview. Part (i) runs the sweep in the proof of Theorem A.3(i) forward, with orthogonal matrices only, which remove the orthogonal factor of each skip and leave its triangular or symmetric factor. Part (ii) reruns the change of variables of that proof and locates the only step that can fail, the read through N; Lemma A.1 says exactly when a matrix can be moved through N. Part (iii) counts the symmetries with Proposition 3.1 and Theorem 3.3.

Proof of Theorem A.24. Part (i). For every $\varepsilon \geq 0$, multiplying the writes into h_ℓ, the skip A_ℓ included, on the right by an orthogonal Q_ℓ, and the reads of h_ℓ on the left by $Q_\ell^\top$, preserves the function (Proposition 3.1(ii)): the raw read $A_{\ell+1}$ directly, and the reads through N because $N(hQ_\ell) = N(h)Q_\ell$ (Lemma A.1). With $Q_0 = I$ and the Q_ℓ of the statement, the skip A_ℓ becomes $Q_{\ell-1}^\top A_\ell Q_\ell = R_\ell$; the RQ factorization of the invertible matrix $Q_{\ell-1}^\top A_\ell$ exists, and is unique. The same computation with the polar decompositions gives $Q_{\ell-1}^\top A_\ell Q_\ell = S_\ell$: every invertible matrix X is $X = SQ^\top$ with $S = (XX^\top)^{1/2}$ symmetric positive definite and $Q^\top = S^{-1}X$ orthogonal, uniquely. The readout absorbs the last factor, $D \mapsto Q_L^\top D$. The recursion can also start at any h_k with any orthogonal Q_k, running forward as above for $\ell > k$ and backward for $\ell \leq k$ with the QR factorization $A_\ell Q_\ell = Q_{\ell-1}R_\ell$ or the polar decomposition $A_\ell Q_\ell = Q_{\ell-1}S_\ell$; the embedding absorbs Q_0. With Q_k the rotation of Proposition 5.1(i) for one block of a read or a write of h_k, that block becomes triangular too, and $(L+1)d_{\text{model}}(d_{\text{model}}-1)/2$ parameters are removed in all. If Q_0 and Q_L are fixed, as with a tied readout, the recursions from h_0 forward and from h_L backward meet at any one skip, which keeps its full matrix; without normalization the same holds in $\mathrm{GL}(d_{\text{model}})$, with identities in place of the R_ℓ.

Part (ii). Let the network be $h_\ell = h_{\ell-1}A_\ell + f_\ell(N(h_{\ell-1})W_{\text{read}}^\ell)W_{\text{write}}^\ell$ and set $h'_\ell = h_\ell \Pi_\ell^{-1}$, the forward version of the change of variables in the proof of Theorem A.3(i), which turns every skip into the identity. The read of block ℓ in the new variables is $N(h'_{\ell-1}\Pi_{\ell-1})W_{\text{read}}^\ell$. If every Π_ℓ, $\ell = 0, \dots, L$, lies in $\mathcal{G}$, Lemma A.1 gives $N(h'\Pi_{\ell-1}) = N(h')P_{\ell-1}$, so the reads $P_{\ell-1}W_{\text{read}}^\ell$, the writes $W_{\text{write}}^\ell \Pi_\ell^{-1}$ and the readout $P_L D$, since the readout reads $N(h_L)$, reproduce the induction of that proof, and the result is a pre-RMSNorm residual network. Conversely, $G_{\ell-1}^{-1}A_\ell G_\ell = I$ for every ℓ gives $G_\ell = \Pi_\ell^{-1}G_0$, so in the variables $h'_\ell = h_\ell G_\ell$ block ℓ reads $N(h'_{\ell-1}G_0^{-1}\Pi_{\ell-1})W_{\text{read}}^\ell$, and the readout reads $N(h'_L G_0^{-1}\Pi_L)$. In a pre-RMSNorm residual network every read has the form $N(h')W$; as the reads and the readout have rank d_{model}, this gives matrices M_ℓ with $N(h'G_0^{-1}\Pi_\ell) = N(h')M_\ell$ for all h', and Lemma A.1 puts every $G_0^{-1}\Pi_\ell$ in $\mathcal{G}$. For $\ell = 0$ this puts G_0 in $\mathcal{G}$, and so every Π_ℓ lies in $\mathcal{G}$.

Part (iii). Apply Proposition 3.1 to the pre-RMSNorm network with learned skips: the skip A_ℓ absorbs any change of basis, and the normalized reads restrict it to $\mathcal{G}$, which is $\mathrm{CO}(d_{\text{model}})$ when $\varepsilon = 0$ and $\mathrm{O}(d_{\text{model}})$ when $\varepsilon > 0$ (Lemma A.1), so each of the $L+1$ hidden states carries its own copy of $\mathcal{G}$. The network with learned skips has Ld_{model}^2 more parameters than the residual network and L more copies of $\mathcal{G}$ in its symmetry group. If both satisfy the equality condition, the difference of their functional dimensions is therefore $L\,(d_{\text{model}}^2 - \dim \mathcal{G})$, which is $L\,d_{\text{model}}(d_{\text{model}}+1)/2$ when $\varepsilon > 0$ and $L\,(d_{\text{model}}(d_{\text{model}}+1)/2 - 1)$ when $\varepsilon = 0$. The difference between the residual and the skipless network is $L \dim \mathcal{G}$ by Theorem 3.3(iii), which is $L\,d_{\text{model}}(d_{\text{model}}-1)/2$ when $\varepsilon > 0$ and $L\,(d_{\text{model}}(d_{\text{model}}-1)/2 + 1)$ when $\varepsilon = 0$. □

We recall Theorem 4.1 from Section 4.

Theorem 4.1 (RMSNorm: a gauge fix and a constraint)**.** *Consider pre-RMSNorm networks with $\varepsilon > 0$, and the network with learned skips of (5).*
(i) Triangular skips. *If every A_ℓ is invertible, there are orthogonal $Q_1, \dots, Q_L$, with $Q_0 = I$, such that every $Q_{\ell-1}^\top A_\ell Q_\ell$ is upper triangular with positive diagonal, and the network with the skips $Q_{\ell-1}^\top A_\ell Q_\ell$, the reads $Q_{\ell-1}^\top W_{\text{read}}^\ell$, the writes $W_{\text{write}}^\ell Q_\ell$ and the readout $Q_L^\top D$ computes the same function. Other orthogonal Q_ℓ make every skip symmetric positive definite instead.*
(ii) Orthogonal skips are gauge. *If every A_ℓ is orthogonal, the network computes the same function as the residual network with the embedding E, the reads $\Pi_{\ell-1}W_{\text{read}}^\ell$, the writes $W_{\text{write}}^\ell \Pi_\ell^\top$ and the readout $\Pi_L D$.*
(iii) What each skip adds. *Under the equality condition for the networks with learned skips, with*

identity skips and without skips, the functional dimension of the network with learned skips exceeds that of the residual network by $L\, d_{\mathrm{model}}(d_{\mathrm{model}}+1)/2$, and that of the residual network exceeds that of the skipless network by $L\, d_{\mathrm{model}}(d_{\mathrm{model}}-1)/2$.

Proof of Theorem 4.1. For $\varepsilon > 0$, $\mathcal{G} = \mathrm{O}(d_{\mathrm{model}})$. Part (i) is Theorem A.24(i). In (ii), every Π_ℓ is orthogonal exactly when every A_ℓ is, and then $P_\ell = \Pi_\ell$ and $\Pi_\ell^{-1} = \Pi_\ell^\top$, so (ii) is the first half of Theorem A.24(ii). Part (iii) is Theorem A.24(iii). □

A.9 Absorption under RMSNorm with $\varepsilon = 0$ (Proposition 4.2)

We recall Proposition 4.2 from Section 4.

Proposition 4.2 (Absorption under RMSNorm with $\varepsilon = 0$)**.** *Consider pre-RMSNorm networks with $\varepsilon = 0$, with blocks of any kind and width.*
(i) *Every skipless network satisfies $F(cx) = F(x)$ for all $c > 0$.*
(ii) *A residual network with this property computes $x \mapsto N(xE)D$.*
(iii) *The residual networks that equal a skipless network have parameters in a set of measure zero.*

Proof. (i) The skipless network reads its input only through $N(xE)$, and $N(cxE) = N(xE)$ for $c > 0$ when $\varepsilon = 0$.

(ii) Fix an input x whose rows of xE are nonzero, and let $c \to \infty$. Every block reads rows of length $\sqrt{d_{\mathrm{model}}}$, so its output stays bounded, and the hidden states of the residual network at cx are $h_\ell = cxE + O(1)$. Hence $N(h_L) \to N(xE)$ row by row, and $F(cx) \to N(xE)D$. If $F(cx) = F(x)$ for all $c > 0$, then $F(x) = N(xE)D$.

(iii) By (i) and (ii), a residual network that equals a skipless network computes $x \mapsto N(xE)D$; we show that the set Z of such parameters has measure zero. Outside a set of measure zero, E and D have rank d_{model}, so $vD = v'D$ only for $v = v'$. Block ℓ adds $a_\ell(x)W^\ell_{\mathrm{write}}$ to the stream, where $a_\ell(x)$ is the input of its write: the activations of its hidden units at $N(h_{\ell-1})$, and for attention the rows $P_{\Gamma_i}(N(h_{\ell-1}))\, N(h_{\ell-1})\, W_{V,i}$ of its heads, with the output projections as the write. If a_ℓ is not identically zero but every a_k with $k > \ell$ is, the blocks after ℓ add nothing, so the parameters lie in the set Y_ℓ where a_ℓ is not identically zero and $N(h_\ell(x)) = N(xE)$ for all x. Fix all parameters but W^ℓ_{write}, which leaves $h_{\ell-1}$ and a_ℓ unchanged, and an input x with a row t in which $a_\ell(x) \neq 0$. In that row the condition asks $a_{\ell,t}(x)W^\ell_{\mathrm{write}}$ to lie on the ray $\{\lambda x_t E - h_{\ell-1,t}(x) : \lambda > 0\}$. As W^ℓ_{write} varies, $a_{\ell,t}(x)W^\ell_{\mathrm{write}}$ takes every value in $\mathbb{R}^{1\times d_{\mathrm{model}}}$, so this happens on a set of measure zero, since $d_{\mathrm{model}} \geq 2$, and by Fubini's theorem Y_ℓ has measure zero. On the rest of Z every a_ℓ vanishes identically, in particular a_1, whose argument $N(xE)$ takes every value on the sphere of radius $\sqrt{d_{\mathrm{model}}}$. For each block of Section 2 this forces a column of the read of block 1, or of one of its value matrices, to vanish, which again happens on a set of measure zero. □

A.10 Skipless networks as limits of residual networks (Proposition 6.3)

Proposition 6.3 (Skipless networks as limits of residual networks)**.** *Let S be a skipless network whose maps f_ℓ are locally Lipschitz.*
(i) Without normalization. *The network S is a limit of residual networks with the same widths and the same maps f_ℓ, uniformly on bounded sets of inputs.*
(ii) Under pre-RMSNorm with $\varepsilon = 0$. *If no hidden state of S has a zero row at any input x whose rows of xE are nonzero, the same holds on bounded sets of such inputs.*

Let S have the embedding E, the blocks $B_\ell(h) = f_\ell(hW^\ell_{\mathrm{read}})W^\ell_{\mathrm{write}}$ and the readout D, and let $s_0 = xE$ and $s_\ell = B_\ell(s_{\ell-1})$ be its hidden states, so that $S(x) = s_L D$. For $t \geq 1$, let F_t be the residual network with the embedding E, the reads $t^{1-\ell}W^\ell_{\mathrm{read}}$, the writes $t^\ell W^\ell_{\mathrm{write}}$ and the readout $t^{-L}D$; under pre-RMSNorm with $\varepsilon = 0$, let F_t keep the embedding, the reads and the readout of S and have the writes $t^\ell W^\ell_{\mathrm{write}}$. The proofs bound the remainder of the recurrence (6), which shows that F_t tends to S. Norms are Frobenius norms for hidden states and operator norms for matrices.

Proof of (i). Block ℓ of F_t maps h to $f_\ell(t^{1-\ell}hW^\ell_{\text{read}})\,t^\ell W^\ell_{\text{write}} = t^\ell B_\ell(t^{1-\ell}h)$. We show by induction that its hidden states are $h_\ell = t^\ell(s_\ell + r_\ell)$, with $r_0 = 0$ and r_ℓ given by (6). For $\ell = 0$, $h_0 = xE = s_0$. If $h_{\ell-1} = t^{\ell-1}(s_{\ell-1} + r_{\ell-1})$, then

$$h_\ell \;=\; h_{\ell-1} + t^\ell B_\ell(t^{1-\ell}h_{\ell-1}) \;=\; t^\ell\big(B_\ell(s_{\ell-1} + r_{\ell-1}) + t^{-1}(s_{\ell-1} + r_{\ell-1})\big) \;=\; t^\ell(s_\ell + r_\ell).$$

Let X be bounded and $\sigma_\ell = \sup_{x\in X}\|s_\ell\|$, which is finite because the blocks are continuous. Suppose that $\|r_{\ell-1}\| \le C_{\ell-1}/t$ for all $x \in X$ and $t \ge 1$, which holds for $\ell = 1$ with $C_0 = 0$. Then $s_{\ell-1}$ and $s_{\ell-1} + r_{\ell-1}$ lie in the ball of radius $\sigma_{\ell-1} + C_{\ell-1}$, on which B_ℓ is Lipschitz with some constant Λ_ℓ, since f_ℓ is locally Lipschitz. Hence

$$\|r_\ell\| \;\le\; \Lambda_\ell\|r_{\ell-1}\| + t^{-1}\big(\sigma_{\ell-1} + \|r_{\ell-1}\|\big) \;\le\; C_\ell/t, \qquad C_\ell = \Lambda_\ell C_{\ell-1} + \sigma_{\ell-1} + C_{\ell-1}.$$

Finally $F_t(x) = t^{-L}h_L D = S(x) + r_L D$, so $\|F_t(x) - S(x)\| \le C_L\|D\|/t$ for all $x \in X$ and $t \ge 1$. □

Proof of (ii). Let X be a bounded set of inputs whose rows of xE are nonzero. Here N divides each row of h by its norm and multiplies it by $\sqrt{d_{\text{model}}}$, so $N(ch) = N(h)$ for $c > 0$, and $\|N(y') - N(y)\| \le 2\sqrt{d_{\text{model}}}\,\|y' - y\|/\|y\|$ for nonzero rows y, y'. The hidden states of S are $s_0 = xE$ and $s_\ell = B_\ell(N(s_{\ell-1}))$, and $S(x) = N(s_L)D$. For $\ell \ge 1$, s_ℓ depends on x only through $N(xE)$, whose rows lie on the sphere of radius $\sqrt{d_{\text{model}}}$; as no row of s_ℓ vanishes, continuity and compactness give $c > 0$ below the norm of every row of every s_ℓ, $\ell \ge 1$. Since N removes the factor $t^{\ell-1}$, the hidden states of F_t are again $h_0 = s_0$ and $h_\ell = t^\ell(s_\ell + r_\ell)$, now with $r_0 = 0$ and

$$r_\ell \;=\; B_\ell\big(N(s_{\ell-1} + r_{\ell-1})\big) - B_\ell\big(N(s_{\ell-1})\big) \;+\; t^{-1}\,(s_{\ell-1} + r_{\ell-1}),$$

as long as no row of $s_{\ell-1} + r_{\ell-1}$ vanishes. The maps B_ℓ are Lipschitz on the bounded image of N, with some constants Λ_ℓ, and $\sigma_\ell = \sup_{x\in X}\|s_\ell\|$ is finite. If every row of $r_{\ell-1}$ is shorter than c, then $\|r_\ell\| \le 2\sqrt{d_{\text{model}}}\,\Lambda_\ell\|r_{\ell-1}\|/c + t^{-1}(\sigma_{\ell-1} + \|r_{\ell-1}\|)$, so by induction $\|r_\ell\| \le C_\ell/t$ on X, and every row of r_ℓ is shorter than c once $t \ge t_0 = \max_\ell 2C_\ell/c$. Finally $\|F_t(x) - S(x)\| = \|(N(s_L + r_L) - N(s_L))D\| \le 2\sqrt{d_{\text{model}}}\,\|D\|\,C_L/(ct)$ for all $x \in X$ and $t \ge t_0$. □

Remark A.25 (The rate). The proofs give the rate $1/t$. Without normalization, for every bounded set X of inputs there is C such that $\|F_t(x) - S(x)\| \le C/t$ for all $x \in X$ and $t \ge 1$. Under pre-RMSNorm with $\varepsilon = 0$, the same holds for all $t \ge t_0$, for every bounded set X of inputs whose rows of xE are nonzero, with t_0 depending on X. The constants depend on X and on S.

B Universal approximation: width and biases

Lin and Jegelka (2018) show that residual ReLU networks whose blocks have a single hidden unit, $h \mapsto h + \mathrm{ReLU}(hw + c)\,v$ with $w \in \mathbb{R}^{d\times 1}$, $v \in \mathbb{R}^{1\times d}$ and a bias $c \in \mathbb{R}$, approximate every integrable function on $\mathbb{R}^d$ in L^1 as the depth grows, while ReLU networks without skips whose hidden layers have width at most the input dimension d cannot (their Proposition 2.1). The separation is confined to this narrow regime: with hidden layers of width $d + 1$, ReLU networks without skips already approximate every continuous function uniformly on compact sets (Hanin and Sellke, 2017).

The construction also needs the biases, for the shifts and the cut-offs at constants of its blocks (their Proposition 3.2). Without biases, a ReLU network, with or without skips, is positively homogeneous of degree one, $F(\lambda x) = \lambda F(x)$ for $\lambda > 0$, since ReLU and the linear maps are. On every ray $\{t\theta : t > 0\}$ such a network is a multiple of t, and so is every limit of such networks, uniformly or in L^1 on a compact set, on almost every ray: the multiples of t form a closed space, and by Fubini L^1 convergence passes to almost every ray. In one dimension, for instance, every such network is $x \mapsto a\,\mathrm{ReLU}(x) - b\,\mathrm{ReLU}(-x)$, which stays at L^1 distance at least $(2 - \sqrt{2})/3$ from x^2 on $[-1, 1]$. On all of $\mathbb{R}^d$, no integrable function other than 0 can be approximated in L^1, since a positively homogeneous function other than 0 is not integrable.

C The one-block theorem

This appendix states and proves the result of Section 6.1. With one block, the skipless and the residual network are

$$x \mapsto f(xEW_{\text{read}})W_{\text{write}}D \qquad \text{and} \qquad x \mapsto \big(xE + f(xEW_{\text{read}})W_{\text{write}}\big)D$$

without normalization, and under pre-RMSNorm

$$x \mapsto N\big(f(N(xE)W_{\text{read}})W_{\text{write}}\big)D \qquad \text{and} \qquad x \mapsto N\big(xEA + f(N(xE)W_{\text{read}})W_{\text{write}}\big)D,$$

with the identity skip $A = I$. A learned skip $A \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ is full or upper triangular.

Theorem C.1 (One block)**.** *Let $d_{\text{model}} \geq 2$, let the input and output dimensions be at least d_{model}, and consider networks with one block: a* $\tanh$*, GELU, ReLU, ReLU^2, ReGLU, SwiGLU or GeGLU block, or an attention block of H heads of width $d_h \leq d_{\text{model}}$, causal or not, on sequences of at least two tokens.*
(i) *Let the network have no normalization, with or without biases, and let the block have width at least d_{model}, where the width of an attention block is Hd_h. For generic parameters the skipless network has functional dimension $P - 2d_{\text{model}}^2 - s$ and the residual network $P - d_{\text{model}}^2 - s$; with biases on the embedding, the block and the readout, counted in P, both values are $2d_{\text{model}}$ lower, and $2d_{\text{model}} + 2Hd_h$ lower for attention. Both networks satisfy the equality condition, and the skip raises the functional dimension by exactly d_{model}^2.*
(ii) *Let the network be pre-RMSNorm with $\varepsilon > 0$, without biases. For generic parameters the residual network has functional dimension $P - d_{\text{model}}(d_{\text{model}} - 1)/2 - s$, which is the equality condition. If the block has width at least d_{model}, the skip raises the functional dimension by at least $d_{\text{model}}(d_{\text{model}} - 1)/2$.*
(iii) *Let the network be pre-RMSNorm without biases, with a learned skip A, full or upper triangular. For generic parameters and $\varepsilon > 0$, its functional dimension exceeds that of the residual network of part (ii) by exactly $d_{\text{model}}(d_{\text{model}} + 1)/2$, and it satisfies the equality condition. For $\varepsilon = 0$ it exceeds that of the residual network with $\varepsilon = 0$ by at most $d_{\text{model}}(d_{\text{model}} + 1)/2 - 1$, with equality for ReLU, ReLU^2 and ReGLU blocks.*
(iv) *Let the network be pre-RMSNorm with $\varepsilon = 0$, without biases, with a ReLU, ReLU^2 or ReGLU block of width at least d_{model}. For generic parameters the skipless network has functional dimension $P - d_{\text{model}}^2 - d_{\text{model}}(d_{\text{model}} - 1)/2 - 1 - s$ and the residual network $P - d_{\text{model}}(d_{\text{model}} - 1)/2 - 1 - s$. Both networks satisfy the equality condition, and the skip raises the functional dimension by exactly d_{model}^2.*
(v) *Let the network be pre-RMSNorm with $\varepsilon = 0$, without biases, with a* $\tanh$*, GELU, SwiGLU, GeGLU or attention block. For generic parameters the skip raises the functional dimension by at least $d_{\text{model}}(d_{\text{model}} - 1)/2 + 1$, the dimension of* $\mathrm{CO}(d_{\text{model}})$*.*

In part (ii), for the skipless network we prove only the bound $P - d_{\text{model}}(d_{\text{model}} - 1) - s$ of Theorem 3.3(ii), which our checks attain (Appendix F). With parts (ii) and (iii), a learned skip under RMSNorm with $\varepsilon > 0$ adds at least d_{model}^2 over no skip if the block has width at least d_{model}, as an identity skip does without normalization. Part (i) extends to a learned absolute position embedding, $h_0 = xE + E_{\text{pos}}$, for an attention block without biases, with P counting E_{pos} (*Position embeddings* below). Rotary encodings act inside the heads, where they reduce the change of basis between queries and keys; we leave them to future work.

What we use. The proof needs linear algebra and limits of functions of one real variable. We write $F(\theta, x)$ for the output of the network at the parameters $\theta \in \mathbb{R}^P$ and the input x, and $\mathrm{D}_\theta F(\theta, x)$ for the Fréchet derivative of $F(\cdot, x)$ at θ, a linear map on $\mathbb{R}^P$. The derivative is written with an upright D so that it is not confused with the readout matrix D of the network. For a direction $\eta \in \mathbb{R}^P$, $\mathrm{D}_\theta F(\theta, x)\,\eta$ is the derivative of $\tau \mapsto F(\theta + \tau\eta, x)$ at $\tau = 0$. We write η_E, η_D, η_{read}, η_{write}, ... for the entries of η at the positions of E, D, W_{read}, W_{write}, ..., and $\mathrm{D}_\theta Q\,\eta$ for the derivative along η of a matrix Q that depends on the parameters; for instance $\mathrm{D}_\theta M\,\eta = \eta_E D + E\eta_D$ for $M = ED$, by the product rule. Call η *flat* if $\mathrm{D}_\theta F(\theta, x)\,\eta = 0$ for every input x, and let $\mathcal{K}(\theta)$ be the space of flat

directions. For the ReLU-type activations, $F(\theta, x)$ is differentiable in θ at the inputs x at which no hidden unit sits at its kink, which form an open set of full measure, and "every input" means every such input. For a batch of inputs $x_1, \dots, x_n$, the Jacobian of $\theta \mapsto (F(\theta, x_1), \dots, F(\theta, x_n))$ is the matrix of the linear map $\eta \mapsto (\mathrm{D}_\theta F(\theta, x_1)\,\eta, \dots, \mathrm{D}_\theta F(\theta, x_n)\,\eta)$; it has P columns, and its kernel consists of the directions with $\mathrm{D}_\theta F(\theta, x_k)\,\eta = 0$ for the inputs of the batch. Adding inputs can only shrink this kernel, and a decreasing chain of subspaces of $\mathbb{R}^P$ stops, so some finite batch has kernel exactly $\mathcal{K}(\theta)$. By rank–nullity, the functional dimension at θ is therefore

$$P - \dim \mathcal{K}(\theta).$$

The theorem thus amounts to finding the flat directions. A statement holds for *generic* parameters if it fails only where some nonzero polynomial in the parameters vanishes, a set of measure zero.

Overview. For an MLP block, the parameters enter the function only through $M = ED$, $A = W_{\mathrm{write}}D$ and $C = EW_{\mathrm{read}}$. Step 1 finds the directions that keep these matrices fixed: they are the directions in which the changes of basis move the parameters, two of them without the skip and one with it. Step 2 shows that a flat direction changes M, A and C only along the symmetries of the hidden units: each unit is isolated, by its kink for the ReLU-type activations, and for $\tanh$, GELU, SwiGLU and GeGLU by how fast it approaches its limit on rays $x = tn$ with $t \to \infty$. Step 3 counts. For an attention block the same steps run with one factorization per head, and inputs of two tokens turn Step 2 into the case of $\tanh$. For part (ii), large inputs first remove the final normalization and then separate the skip from the block. Part (iii) runs the same steps with a learned skip, which leaves a symmetric change of basis at h_0 in the input of the block; homogeneity or rays remove it.

Part (i) for MLP blocks

Reduction. With biases b_E, b_{read}, b_{write} and b_D, the skipless network computes $F(\theta, x) = f(xC + \beta)A + \gamma_0$ and the residual network $F(\theta, x) = xM + f(xC + \beta)A + \gamma$, where

$$M = ED, \qquad C = EW_{\mathrm{read}}, \qquad A = W_{\mathrm{write}}D,$$
$$\beta = b_E W_{\mathrm{read}} + b_{\mathrm{read}}, \qquad \gamma_0 = b_{\mathrm{write}}D + b_D, \qquad \gamma = b_E D + \gamma_0.$$

Without biases $\beta = 0$ and $\gamma = \gamma_0 = 0$. For a vanilla block let c_i be the columns of C, one per hidden unit, a_i the rows of A, β_i the entries of β, and $z_i = xc_i + \beta_i$, so that $F(\theta, x) = xM + \gamma + \sum_i \sigma(z_i)\, a_i$. For generic parameters E, D, W_{read} and W_{write} have rank d_{model}, every a_i is nonzero, and the c_i are nonzero and pairwise non-collinear: they lie in the column space of E, of dimension $d_{\mathrm{model}} \geq 2$, where two generic vectors are not collinear. For the gated blocks ReGLU, SwiGLU and GeGLU, in the same way, the value weights (e_i, β_i') of each unit, introduced below, have $e_i \neq 0$ and are not proportional to its gate weights (c_i, β_i).

Step 1: changes of basis. We need the following lemma.

> **Lemma C.2** (The directions that keep M, A and C)**.** *Let E, D, W_{read} and W_{write} have rank d_{model}. A direction η keeps C and A fixed to first order, $\mathrm{D}_\theta C\,\eta = 0$ and $\mathrm{D}_\theta A\,\eta = 0$, if and only if*
>
> $$\eta_E = EX, \qquad \eta_{\mathrm{read}} = -XW_{\mathrm{read}}, \qquad \eta_{\mathrm{write}} = W_{\mathrm{write}}Y, \qquad \eta_D = -YD$$
>
> *for some $X, Y \in \mathbb{R}^{d_{\mathrm{model}} \times d_{\mathrm{model}}}$, and it keeps M fixed as well, $\mathrm{D}_\theta M\,\eta = 0$, if and only if moreover $X = Y$. With biases, it keeps β and γ_0, respectively β and γ, fixed as well if and only if moreover $\eta_{b_E} = b_E X + g_0$, $\eta_{b_{\mathrm{read}}} = -g_0 W_{\mathrm{read}}$, $\eta_{b_{\mathrm{write}}} = b_{\mathrm{write}}Y + g_1$ and $\eta_{b_D} = -g_1 D$, respectively $\eta_{b_D} = -(g_0 + g_1)D$, for some row vectors $g_0, g_1 \in \mathbb{R}^{1 \times d_{\mathrm{model}}}$. The matrices X, Y and vectors g_0, g_1 are determined by the direction, so these directions form a space of dimension $2d_{\mathrm{model}}^2$ without the skip and d_{model}^2 with it, and $2d_{\mathrm{model}}$ more with biases.*
>
> *Proof.* Since W_{read} has rank d_{model}, it has a right inverse R, with $W_{\mathrm{read}}R = I$, and E, of rank d_{model} with d_{model} columns, has a left inverse. From $\mathrm{D}_\theta C\,\eta = \eta_E W_{\mathrm{read}} + E\eta_{\mathrm{read}} = 0$, multiplying on the right by R gives $\eta_E = EX$ with $X = -\eta_{\mathrm{read}}R$; then $E\eta_{\mathrm{read}} = -EXW_{\mathrm{read}}$,

and cancelling E on the left gives $\eta_{\text{read}} = -XW_{\text{read}}$. In the same way, D has a right inverse and W_{write} a left inverse, and $\mathrm{D}_\theta A\,\eta = \eta_{\text{write}} D + W_{\text{write}}\eta_D = 0$ gives $\eta_{\text{write}} = W_{\text{write}} Y$ and $\eta_D = -YD$. Conversely these directions give $\mathrm{D}_\theta C\,\eta = EXW_{\text{read}} - EXW_{\text{read}} = 0$ and $\mathrm{D}_\theta A\,\eta = 0$. In the residual network $\mathrm{D}_\theta M\,\eta = \eta_E D + E\eta_D = E(X-Y)D$, which vanishes if and only if $X = Y$, by cancelling E on the left and D on the right. With biases, set $g_0 = \eta_{b_E} - b_E X$ and $g_1 = \eta_{b_{\text{write}}} - b_{\text{write}} Y$. Then $\mathrm{D}_\theta \beta\,\eta = \eta_{b_E} W_{\text{read}} + b_E \eta_{\text{read}} + \eta_{b_{\text{read}}} = g_0 W_{\text{read}} + \eta_{b_{\text{read}}}$, which vanishes exactly when $\eta_{b_{\text{read}}} = -g_0 W_{\text{read}}$, and in the same way $\mathrm{D}_\theta \gamma_0\,\eta = g_1 D + \eta_{b_D}$ and, with $Y = X$, $\mathrm{D}_\theta \gamma\,\eta = (g_0 + g_1)D + \eta_{b_D}$. Finally $\eta_E = EX$ determines X, $\eta_D = -YD$ determines Y, and then η_{b_E} and $\eta_{b_{\text{write}}}$ determine g_0 and g_1. □

These are the directions in which the changes of basis $I + \tau X$ at h_0 and $I + \tau Y$ at h_1 move the parameters, and, with biases, the translations at the two hidden states (Lemma A.4).

Step 2: one hidden layer. We need a lemma on the function

$$G(\phi, x) = xM + \gamma + \textstyle\sum_i \sigma(z_i)\, a_i, \qquad z_i = xc_i + \beta_i,$$

of the matrices $\phi = \big(M, \gamma, (c_i, \beta_i, a_i)_i\big)$ themselves. For a direction ψ of ϕ, with entries ψ_M, ψ_γ, ψ_{c_i}, ψ_{β_i} and ψ_{a_i}, the Fréchet derivative of $G(\cdot, x)$ at ϕ gives

$$\mathrm{D}_\phi G(\phi, x)\,\psi = x\psi_M + \psi_\gamma + \sum_i \big[\sigma(z_i)\,\psi_{a_i} + \sigma'(z_i)\,(x\psi_{c_i} + \psi_{\beta_i})\,a_i\big]. \tag{13}$$

A gated unit computes $\sigma(z_i)\, w_i\, a_i$ with a gate $z_i = xc_i + \beta_i$ and a value $w_i = xe_i + \beta_i'$, where $\sigma(z) = \max(z, 0)$ for ReGLU, $z/(1 + e^{-z})$ for SwiGLU and $z\Phi(z)$ for GeGLU, so that ϕ also contains the e_i and β_i', and the terms of the unit in $\mathrm{D}_\phi G(\phi, x)\,\psi$ are $\sigma(z_i)\, w_i\, \psi_{a_i} + \sigma'(z_i)\,(x\psi_{c_i} + \psi_{\beta_i})\, w_i\, a_i + \sigma(z_i)\,(x\psi_{e_i} + \psi_{\beta_i'})\, a_i$.

Lemma C.3 (A flat direction only rescales units)**.** *Let every $a_i \neq 0$ and let the c_i be nonzero and pairwise non-collinear; for the gated blocks let also every $e_i \neq 0$, with (e_i, β_i') not proportional to (c_i, β_i). If $\mathrm{D}_\phi G(\phi, x)\,\psi = 0$ for all x, then $\psi_M = 0$, $\psi_\gamma = 0$, and for every unit i:*
for $\tanh$ *and GELU, $\psi_{a_i} = 0$, $\psi_{c_i} = 0$ and $\psi_{\beta_i} = 0$;*
for ReLU, $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i (c_i, \beta_i)$ and $\psi_{a_i} = -\lambda_i a_i$ for some number λ_i;
for ReLU^2, $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i (c_i, \beta_i)$ and $\psi_{a_i} = -2\lambda_i a_i$;
for ReGLU, $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i (c_i, \beta_i)$, $(\psi_{e_i}, \psi_{\beta_i'}) = -\alpha_i (e_i, \beta_i')$ and $\psi_{a_i} = (\alpha_i - \lambda_i) a_i$ for some numbers λ_i and α_i;
for SwiGLU and GeGLU, $\psi_{c_i} = 0$, $\psi_{\beta_i} = 0$, $(\psi_{e_i}, \psi_{\beta_i'}) = -\alpha_i (e_i, \beta_i')$ and $\psi_{a_i} = \alpha_i a_i$ for some number α_i.

Proof. Let $H_i = \{x : z_i = 0\}$, a hyperplane since $c_i \neq 0$. Since the c_i are pairwise non-collinear, the H_i are pairwise non-parallel, so every H_i contains points that lie on no other H_k. We use one fact from linear algebra: an affine function that vanishes on an open piece of H_i is a multiple of z_i, since it vanishes on all of H_i and H_i has codimension one. We write $J(x) = \mathrm{D}_\phi G(\phi, x)\,\psi$, a function of x alone, which vanishes by assumption.

ReLU. Here $\sigma' = \mathbf{1}[z > 0]$ jumps at 0. Near a point x_0 of H_i that lies on no other H_k, every term of (13) is continuous except $\mathbf{1}[z_i > 0](x\psi_{c_i} + \psi_{\beta_i})\, a_i$, which jumps by $(x_0\psi_{c_i} + \psi_{\beta_i})\, a_i$ as x crosses H_i at x_0. As $J = 0$ is continuous, this jump vanishes, and since $a_i \neq 0$, the affine function $x\psi_{c_i} + \psi_{\beta_i}$ vanishes on an open piece of H_i, so $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i (c_i, \beta_i)$. The terms of unit i then add up to $\max(z_i, 0)\,(\psi_{a_i} + \lambda_i a_i)$, since $z\,\mathbf{1}[z > 0] = \max(z, 0)$. Across H_i, the gradient of J with respect to x jumps by the matrix $c_i(\psi_{a_i} + \lambda_i a_i)$, which must vanish, so $\psi_{a_i} = -\lambda_i a_i$. With every unit removed in this way, $J(x) = x\psi_M + \psi_\gamma = 0$ for all x gives $\psi_M = 0$ and $\psi_\gamma = 0$.

ReLU^2. The same argument one derivative higher: $\sigma'(z) = 2\max(z, 0)$ is continuous, and across H_i the gradient of $2\max(z_i, 0)(x\psi_{c_i} + \psi_{\beta_i})\, a_i$ jumps by $2c_i(x_0\psi_{c_i} + \psi_{\beta_i})\, a_i$, which forces $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i (c_i, \beta_i)$. The terms of unit i then add up to $\max(z_i, 0)^2(\psi_{a_i} + 2\lambda_i a_i)$, whose second derivative jumps across H_i, so $\psi_{a_i} = -2\lambda_i a_i$.

ReGLU. As (e_i, β_i') is nonzero and not proportional to (c_i, β_i), the affine function w_i does not vanish on all of H_i, so it is nonzero on an open piece of H_i that meets no other H_k; we take x_0 there. Across H_i the terms of unit i jump by $(x_0\psi_{c_i} + \psi_{\beta_i})\, w_i(x_0)\, a_i$, so $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i(c_i, \beta_i)$ as before. The terms of unit i then add up to $\max(z_i, 0)\, V_i(x)$ with the affine function $V_i(x) = w_i(x)\,(\psi_{a_i} + \lambda_i a_i) + (x\psi_{e_i} + \psi_{\beta_i'})\, a_i$. The jump of their gradient across H_i is $c_i V_i(x_0)$, so V_i vanishes on an open piece of H_i, and $V_i(x) = z_i\, \nu_i$ for a row vector ν_i. The terms of unit i then add up to $\max(z_i, 0)^2\, \nu_i$, whose second derivative jumps across H_i, so $\nu_i = 0$ and $V_i = 0$ for all x: $w_i(x)\,(\psi_{a_i} + \lambda_i a_i) = -(x\psi_{e_i} + \psi_{\beta_i'})\, a_i$. Both sides are products of an affine function of x and a fixed row vector, and $a_i \neq 0$, so $\psi_{a_i} + \lambda_i a_i = \alpha_i a_i$ and $(\psi_{e_i}, \psi_{\beta_i'}) = -\alpha_i(e_i, \beta_i')$ for some number α_i. Then $\psi_M = 0$ and $\psi_\gamma = 0$ as for ReLU.

$\tanh$. We restrict (13) to a ray $x = tn$, $t > 0$, with a direction n such that the numbers $u_k = nc_k$ are all nonzero and $|u_j| < |u_k|$ for one unit j and all $k \neq j$. For each j and each sign $s = \pm 1$, the directions n with $su_j > 0$ and this property form a nonempty open set: start from a point of the hyperplane $nc_j = 0$ where no other nc_k vanishes, and move a little to the side s. For $z > 0$,

$$\tanh z = 1 - 2e^{-2z} + r(z), \quad 0 \le r(z) \le 2e^{-4z},$$
$$\tanh' z = 4e^{-2z} - q(z), \quad 0 \le q(z) \le 8e^{-4z},$$

from $\tanh z = (1 - e^{-2z})/(1 + e^{-2z})$ and $\tanh' = 1 - \tanh^2$, and $\tanh$ is odd. On the ray $z_k = u_k t + \beta_k$, so for large t the terms of unit k in (13) are

$$s_k\psi_{a_k} + e^{-2|u_k|t}\, e^{-2s_k\beta_k}\big[-2s_k\psi_{a_k} + 4\,(t\, n\psi_{c_k} + \psi_{\beta_k})\, a_k\big] + O\big((1+t)\, e^{-4|u_k|t}\big),$$

with s_k the sign of u_k. All terms of $J(tn)$ are therefore $t\, n\psi_M$, a constant, the bracket of unit j times $e^{-2|u_j|t}$, and terms that are $O\big((1+t)e^{-\rho t}\big)$ with $\rho = \min\big(4|u_j|, \min_{k\neq j} 2|u_k|\big) > 2|u_j|$. As $J(tn) = 0$ for all t, dividing by t and letting $t \to \infty$ gives $n\psi_M = 0$; then the constant vanishes; then multiplying by $e^{2|u_j|t}$ and letting $t \to \infty$ shows that the bracket of unit j, a polynomial of degree one in t, tends to zero, so both of its coefficients vanish: $(n\psi_{c_j})\, a_j = 0$ and $2\psi_{\beta_j} a_j = s\,\psi_{a_j}$. As $a_j \neq 0$, $n\psi_{c_j} = 0$ for all n in an open set, so $\psi_{c_j} = 0$; and the two signs $s = \pm 1$ give $\psi_{a_j} = 2\psi_{\beta_j} a_j = -2\psi_{\beta_j} a_j$, so $\psi_{\beta_j} = 0$ and $\psi_{a_j} = 0$. This holds for every unit j, and then $J(x) = x\psi_M + \psi_\gamma = 0$ gives $\psi_M = 0$ and $\psi_\gamma = 0$.

GELU. Here $\sigma(z) = z\Phi(z)$, with Φ the standard normal distribution function and $\varphi = \Phi'$ its density. The tail $\bar\Phi = 1 - \Phi$ satisfies $\varphi(z)(1/z - 1/z^3) \le \bar\Phi(z) \le \varphi(z)/z$ for $z > 0$, as one checks by differentiating, and $\Phi(-z) = 1 - \Phi(z)$ carries these bounds to $z < 0$. From these, as $|z| \to \infty$ on either side,

$$\sigma(z) = \max(z, 0) - \varphi(z) + O\big(\varphi(z)/z^2\big),$$
$$\sigma'(z) = \mathbf{1}[z > 0] + \varphi(z)\,(z - 1/z) + O\big(\varphi(z)/|z|^3\big).$$

Take a ray $x = tn$ as for $\tanh$. The terms of unit k in (13) split into a polynomial of degree at most one in t, from $\max(z, 0)$ and $\mathbf{1}[z > 0]$, and the Gaussian part

$$\varphi(z_k)\Big[u_k\,(n\psi_{c_k})\, t^2 + \big(u_k\psi_{\beta_k} + \beta_k\, n\psi_{c_k}\big)\, t + \beta_k\psi_{\beta_k} - (n\psi_{c_k})/u_k\Big] a_k \\ - \ \varphi(z_k)\,\psi_{a_k} \ + \ O\big(\varphi(z_k)/t\big).$$

Gaussian parts tend to zero faster than any power of t, so as $J(tn) = 0$ for all t, the polynomial part vanishes, and so does the sum of the Gaussian parts. For $k \neq j$, $\varphi(z_k)/\varphi(z_j) = \exp\big(-\frac{1}{2}(u_k^2 - u_j^2)\, t^2 + O(t)\big)$ tends to zero faster than any power of t, since $u_k^2 > u_j^2$. Dividing the sum of the Gaussian parts by $\varphi(z_j)$ and letting $t \to \infty$ therefore makes the coefficients of t^2, t and 1 in the bracket of unit j vanish, in this order: $u_j(n\psi_{c_j})\, a_j = 0$, so $n\psi_{c_j} = 0$; then $u_j\psi_{\beta_j}\, a_j = 0$, so $\psi_{\beta_j} = 0$; then $\psi_{a_j} = 0$. As for $\tanh$, $n\psi_{c_j} = 0$ on an open set gives $\psi_{c_j} = 0$, and finally $\psi_M = 0$ and $\psi_\gamma = 0$.

SwiGLU and GeGLU. Take a ray $x = tn$ as for $\tanh$, and write $v_k = ne_k$, so that $w_k = v_k t + \beta'_k$; as $e_j \neq 0$, also $v_j \neq 0$ on an open set of such n. For SwiGLU, $\sigma(z) = z - z/(1+e^z)$ gives, as $|z| \to \infty$ with s the sign of z,

$$\sigma(z) = \max(z, 0) - |z|e^{-|z|} + O\big(|z|e^{-2|z|}\big),$$
$$\sigma'(z) = \mathbf{1}[z > 0] + s\,(|z| - 1)\,e^{-|z|} + O\big(|z|e^{-2|z|}\big).$$

The terms of unit k in (13) then split into a polynomial in t, from $\max(z, 0)$ and $\mathbf{1}[z > 0]$, and a rest that is $e^{-|z_k|}$ times a polynomial in t, up to $O(t^3 e^{-2|z_k|})$; for GeGLU, with the expansions of GELU, the rest is $\varphi(z_k)$ times a polynomial in t and $1/z_k$, up to $O(\varphi(z_k)/t)$. As for $\tanh$ and GELU, the polynomial parts vanish, and dividing the sum of the rests by $e^{-|z_j|}$, respectively $\varphi(z_j)$, and letting $t \to \infty$ shows that the rest of the slowest unit j vanishes. For SwiGLU this rest is $e^{-|z_j|}$ times

$$s\,(|z_j| - 1)\,(t\,n\psi_{c_j} + \psi_{\beta_j})\,w_j\,a_j - |z_j|\big((t\,n\psi_{e_j} + \psi_{\beta'_j})\,a_j + w_j\,\psi_{a_j}\big),$$

with s the sign of u_j and $|z_j| = |u_j|t + s\beta_j$, a polynomial of degree three in t, whose coefficients therefore vanish. The coefficient of t^3 is $s|u_j|\,(n\psi_{c_j})\,v_j\,a_j$, so $\psi_{c_j} = 0$ as for $\tanh$. The coefficient of t^2 is then $-|u_j|\big((n\psi_{e_j})\,a_j + v_j(\psi_{a_j} - s\psi_{\beta_j}a_j)\big)$. As it vanishes for all n in an open set, $\psi_{e_j} = -\alpha e_j$ and $\psi_{a_j} - s\psi_{\beta_j}a_j = \alpha a_j$ for a number α, the same for both signs since ψ_{e_j} is; the two signs then give $\psi_{\beta_j} = 0$. The coefficient of t is then $-|u_j|(\psi_{\beta'_j} + \alpha\beta'_j)\,a_j$, so $\psi_{\beta'_j} = -\alpha\beta'_j$. For GeGLU the rest of unit j is $\varphi(z_j)$ times the same bracket with $z_j - 1/z_j$ in place of $s(|z_j| - 1)$ and 1 in place of $|z_j|$, and its coefficients of t^3, t^2, t and 1 give, in this order, $\psi_{c_j} = 0$, $\psi_{\beta_j} = 0$, $(\psi_{e_j}, \psi_{a_j}) = (-\alpha e_j, \alpha a_j)$ and $\psi_{\beta'_j} = -\alpha\beta'_j$. In both cases the terms of unit j then vanish; this holds for every unit, and $\psi_M = 0$ and $\psi_\gamma = 0$ as for $\tanh$. □

The directions left by the lemma are exactly those of the rescalings of the units: for ReLU, scaling the input weights and the bias of unit i by $1 + \tau\lambda_i$ and its output weights by $1 - \tau\lambda_i$ changes the function only at second order in τ.

Step 3: counting. Let η be flat. The network depends on θ only through ϕ, its M, γ and the c_i, β_i and a_i formed from C, β and A, so by the chain rule $\mathrm{D}_\theta F(\theta, x)\,\eta = \mathrm{D}_\phi G(\phi, x)\,(\mathrm{D}_\theta\phi\,\eta)$, and Lemma C.3 applies to $\psi = \mathrm{D}_\theta\phi\,\eta$. The derivatives of M, A, C, β and γ along η are thus those of a rescaling of the hidden units. A rescaling of the units by numbers λ_i is also a direction in parameter space: it scales column i of W_{read} and entry i of b_{read} by λ_i and row i of W_{write} by $-\lambda_i$ for ReLU, by $-2\lambda_i$ for ReLU^2, and similarly for ReGLU and for the values of SwiGLU and GeGLU, and along it C, β and A have the same derivatives. Subtracting it from η leaves a direction along which M, A, C, β and γ have derivative zero, which Lemma C.2 describes. So every flat direction is the sum of a direction of the changes of basis and translations of Lemma C.2 and a rescaling of the units, and conversely all of these are flat, since they come from symmetries. The two kinds are independent: a rescaling that is also a direction of Lemma C.2 has $\eta_E = EX = 0$, so $X = 0$, and then $W_{\mathrm{read}}\,\mathrm{diag}(\lambda) = 0$, so every $\lambda_i = 0$, because no column of W_{read} vanishes. Hence $\dim \mathcal{K}(\theta) = 2d_{\mathrm{model}}^2 + s$ without the skip and $d_{\mathrm{model}}^2 + s$ with it, with $2d_{\mathrm{model}}$ more for each with biases, where s is the number of rescalings: m for ReLU, ReLU^2, SwiGLU and GeGLU, $2m$ for ReGLU, none for $\tanh$ and GELU. The functional dimension $P - \dim \mathcal{K}(\theta)$ has the values of the theorem. The symmetries account for all of $\mathcal{K}(\theta)$, which is the equality condition, and the difference between the two networks is d_{model}^2.

Part (i) for attention blocks

Reduction. Let $W_{Q,i}, W_{K,i}, W_{V,i} \in \mathbb{R}^{d_{\mathrm{model}} \times d_h}$ and $W_{O,i} \in \mathbb{R}^{d_h \times d_{\mathrm{model}}}$ be the projections of head i, and let

$$M = ED, \qquad B_i = EW_{Q,i}W_{K,i}^\top E^\top/\sqrt{d_h}, \qquad C_i = EW_{V,i}W_{O,i}D, \qquad S = \textstyle\sum_i C_i,$$

with M only in the residual network. Head i gives token t the scores $x_t B_i x_s^\top$, and the output at token t is $x_t M + \sum_i \sum_s a_{ts}^{(i)}\, x_s C_i$, so the parameters enter only through these matrices. We use inputs of two kinds: with a causal mask, any sequence, whose second token sees only the first two; without a mask, sequences whose tokens other than the second all equal the first. Let k be the number of tokens other than the second that the second token sees, $k = 1$ with a mask and $k = T - 1$ without. With $u = x_2$ and $v = x_2 - x_1$, head i puts the weight $1/(1 + k\, e^{-uB_i v^\top})$ on the second token and the rest on the copies of the first, and $1/(1 + e^{-z}) = \frac{1}{2} + \frac{1}{2}\tanh(z/2)$ gives the output at the second token

$$y_2 = uK - \tfrac{1}{2}\, vS + \tfrac{1}{2} \sum_i \tanh\left(\tfrac{1}{2}(uB_i v^\top - \log k)\right) vC_i, \qquad K = M + S. \tag{14}$$

With biases b_E on the embedding, $b_{Q,i}$, $b_{K,i}$, $b_{V,i}$ on the queries, keys and values, b_O on the output projection and b_D on the readout, $uB_i v^\top$ becomes $(uB_i + \beta_i)v^\top$ with $\beta_i = (b_E W_{Q,i} + b_{Q,i}) W_{K,i}^\top E^\top / \sqrt{d_h}$, and the constant $\gamma = \sum_i (b_E W_{V,i} + b_{V,i}) W_{O,i} D + b_O D + b_D$ is added, plus $b_E D$ in the residual network. The key biases drop out, since they shift all scores of a token equally, and the value biases enter only through γ, since the weights of a token sum to one.

Step 1: the directions that keep M, B_i and C_i. For generic parameters E and D have rank d_{model}; every $W_{Q,i}$, $W_{K,i}$, $W_{V,i}$ and $W_{O,i}^\top$ has rank d_h, since $d_h \le d_{\text{model}}$; and the queries $[W_{Q,1}, \ldots, W_{Q,H}] \in \mathbb{R}^{d_{\text{model}} \times H d_h}$ and the output projection $W_O \in \mathbb{R}^{H d_h \times d_{\text{model}}}$ of all heads have rank d_{model}, since $H d_h \ge d_{\text{model}}$. Write $\eta_{Q,i}$, $\eta_{K,i}$, $\eta_{V,i}$ and $\eta_{O,i}$ for the entries of a direction η at the positions of $W_{Q,i}$, $W_{K,i}$, $W_{V,i}$ and $W_{O,i}$, and consider a direction with $\mathrm{D}_\theta M\, \eta = 0$ and, for every head, $\mathrm{D}_\theta B_i\, \eta = 0$ and $\mathrm{D}_\theta C_i\, \eta = 0$. Multiplying $\mathrm{D}_\theta B_i\, \eta = 0$ on the left by the projection $I - EE^+$ onto the orthogonal complement of the column space of E, where E^+ is the pseudo-inverse, leaves $(I - EE^+)\, \eta_E\, W_{Q,i} W_{K,i}^\top E^\top = 0$. As $W_{K,i}^\top E^\top$ has full row rank, $(I - EE^+)\, \eta_E\, W_{Q,i} = 0$ for every head, and as the queries of all heads have rank d_{model}, $\eta_E = EX$ with $X = E^+ \eta_E$. In the same way, multiplying $\mathrm{D}_\theta C_i\, \eta = 0$ on the right by the projection $I - D^+ D$ gives $\eta_D = -YD$ with $Y = -\eta_D\, D^+$, because $E W_{V,i}$ has full column rank and the output projection has rank d_{model}. Substituting, and cancelling E on the left and $E^\top$ or D on the right,

$$(\eta_{Q,i} + X W_{Q,i}) W_{K,i}^\top + W_{Q,i} (\eta_{K,i} + X W_{K,i})^\top = 0,$$
$$(\eta_{V,i} + X W_{V,i}) W_{O,i} + W_{V,i} (\eta_{O,i} - W_{O,i} Y) = 0.$$

If P_1 and P_2 have full column rank and $Z_1 P_2^\top + P_1 Z_2^\top = 0$, then multiplying on the right by a right inverse of $P_2^\top$ gives $Z_1 = P_1 N$ for a square matrix N, and then $Z_2 = -P_2 N^\top$. Hence $\eta_{Q,i} = -X W_{Q,i} + W_{Q,i} N_i$, $\eta_{K,i} = -X W_{K,i} - W_{K,i} N_i^\top$, $\eta_{V,i} = -X W_{V,i} + W_{V,i} N_i'$ and $\eta_{O,i} = W_{O,i} Y - N_i' W_{O,i}$: the directions of the changes of basis $I + \tau X$ at h_0 and $I + \tau Y$ at h_1, and of $I + \tau N_i$ between the queries and keys and $I + \tau N_i'$ between the values and outputs of head i. In the residual network $\mathrm{D}_\theta M\, \eta = E(X - Y)D = 0$ gives $X = Y$. These directions are independent: $EX = 0$ forces $X = 0$, $YD = 0$ forces $Y = 0$, and then $W_{Q,i} N_i = 0$ and $W_{V,i} N_i' = 0$ force $N_i = N_i' = 0$. With biases, $\mathrm{D}_\theta \beta_i\, \eta = 0$ gives $\eta_{b_{Q,i}} = b_{Q,i} N_i - g W_{Q,i}$ with $g = \eta_{b_E} - b_E X$, since $W_{K,i}^\top E^\top$ has full row rank, and $\mathrm{D}_\theta \gamma\, \eta = 0$ determines η_{b_D}. This leaves g, η_{b_O} and the entries of η at the key and value biases free, $2 d_{\text{model}} + 2 H d_h$ dimensions, which the translations at h_0 and h_1, the shifts of the key biases and the value biases moved into the output bias span.

Step 2: two tokens. For generic parameters and generic v the vectors $B_i v^\top$ are nonzero and pairwise non-collinear, since $d_{\text{model}} \ge 2$, and every vC_i is nonzero. For such a v, (14) is, as a function of u, a residual network with one hidden layer of $\tanh$ units, one per head, with input weights $\frac{1}{2} B_i v^\top$, biases $\frac{1}{2}(\beta_i v^\top - \log k)$, output weights $\frac{1}{2} vC_i$, linear part K and constant term $\gamma - \frac{1}{2} vS$. Along a flat direction η these have the derivatives $\frac{1}{2}(\mathrm{D}_\theta B_i\, \eta) v^\top$, $\frac{1}{2}(\mathrm{D}_\theta \beta_i\, \eta) v^\top$, $\frac{1}{2} v (\mathrm{D}_\theta C_i\, \eta)$, $\mathrm{D}_\theta K\, \eta$ and $\mathrm{D}_\theta \gamma\, \eta - \frac{1}{2} v (\mathrm{D}_\theta S\, \eta)$, and Lemma C.3 for $\tanh$ shows that all of these vanish. This holds for all v in an open set, so $\mathrm{D}_\theta B_i\, \eta = 0$, $\mathrm{D}_\theta \beta_i\, \eta = 0$ and $\mathrm{D}_\theta C_i\, \eta = 0$; hence $\mathrm{D}_\theta S\, \eta = 0$, $\mathrm{D}_\theta M\, \eta = \mathrm{D}_\theta K\, \eta - \mathrm{D}_\theta S\, \eta = 0$ and $\mathrm{D}_\theta \gamma\, \eta = 0$.

Step 3: counting. Every direction of Step 1 is flat, since it comes from a symmetry, and by Step 2 every flat direction, already on the inputs of (14), is one of them. So $\dim \mathcal{K}(\theta)$ is $2 d_{\text{model}}^2 + 2 H d_h^2$ without the skip and $d_{\text{model}}^2 + 2 H d_h^2$ with it, and $2 d_{\text{model}} + 2 H d_h$ more with biases.

Grouped-query attention. If the H query heads share G heads of keys and values, the same proof applies with B_i and C_i formed from the keys and values of the group of head i. Step 1 uses only the queries and the output projection of all heads, and in its last part a direction that keeps the products $W_{Q,i}W_{K,g}^\top$ over the heads i of group g has one N for the whole group, since $W_{K,g}$ has full column rank; likewise for the values. So $s = 2Gd_h^2$, and the biases add $2d_{\text{model}} + 2Gd_h$.

Position embeddings. Let a learned position embedding add the row e_t of E_{pos} to token t of h_0, in a network with an attention block and no biases, and write η_{e_t} for the entry of a direction η at the position of e_t. As E has rank d_{model}, choose ϵ_t with $\epsilon_t E = e_t$: the network computes the network without position embedding on the tokens $\tilde{x}_t = x_t + \epsilon_t$. Along η, $h_{0,t} = \tilde{x}_t E$ has the derivative $\tilde{x}_t\eta_E + \nu_t$ with $\nu_t = \eta_{e_t} - \epsilon_t\eta_E$. This is the derivative of the network without position embedding along the other entries of η together with a shift κ_t of each token $\tilde{x}_t$, where $\kappa_t E = \nu_t$. Fix a token t that sees at least one other token, and inputs for which the tokens $\tilde{x}_s$, $s \neq t$, that token t sees all equal one vector w. The output at token t is symmetric in these tokens, so its derivative sees their shifts only through their mean $\bar{\kappa}$, and it is (14) with $u = \tilde{x}_t$, $v = u - w$ and k the number of these tokens. The shifts move u by κ_t and v by $\kappa_t - \bar{\kappa}$, which adds $\frac{1}{2}B_i(\kappa_t - \bar{\kappa})^\top$ to the derivatives of the input weights, $\frac{1}{2}\kappa_t B_i v^\top$ to those of the biases and $\frac{1}{2}(\kappa_t - \bar{\kappa})C_i$ to those of the output weights of the $\tanh$ units, and Step 2 again makes each of these derivatives vanish. For all v this gives $\mathrm{D}_\theta B_i\,\eta = 0$, $\mathrm{D}_\theta C_i\,\eta = 0$, $\kappa_t B_i = 0$ and $B_i(\kappa_t - \bar{\kappa})^\top = 0$ for every head, and since the queries and the keys of all heads have rank d_{model}, $\nu_t = \kappa_t E = 0$ and $\bar{\kappa}E = 0$. This holds for every token $t \geq 2$, and for every token without a mask; with a mask, $\bar{\kappa} = \kappa_1$ at $t = 2$, so $\nu_1 = 0$ as well. So $\eta_{e_t} = \epsilon_t\eta_E = e_t X$ by Step 1, the change of basis at h_0 applied to E_{pos}, and the values of part (i) hold with P counting E_{pos}. A fixed encoding, which is not trained, must instead be preserved by the change of basis at h_0; if its rows span a space of dimension r, this removes rd_{model} dimensions of symmetry from both networks, and the skip still adds d_{model}^2.

Part (ii)

Write $h_1 = xE + g(x)$, with $g(x) = f(N(xE)W_{\text{read}})W_{\text{write}}$ the output of the block, and $y = N(h_1)D$, with N applied to every token; for attention, x has T tokens, and for an MLP block one suffices. Let η be flat, and write $\mathrm{D}_\theta h_1\,\eta$ and $\mathrm{D}_\theta y\,\eta$ for the derivatives of h_1 and y along η at a given input, and $\mathrm{D}N(h)$ for the Fréchet derivative of N at h.

Step 1: the final normalization. Split $\eta_D = -YD + Z$, where $Y \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ and the rows of Z are orthogonal to the row space of D, which has dimension d_{model}. At every token, by the chain rule,

$$0 \;=\; \mathrm{D}_\theta y\,\eta \;=\; \big(\mathrm{D}N(h_1)\,(\mathrm{D}_\theta h_1\,\eta) - N(h_1)Y\big)D + N(h_1)Z,$$

and the two terms lie in complementary subspaces, so both vanish. As x varies, the rows $N(h_1)$ span $\mathbb{R}^{d_{\text{model}}}$, since at large inputs they approach $\sqrt{d_{\text{model}}}\,xE/\|xE\|$ and xE takes every direction, so $Z = 0$; and since D has full row rank, $\mathrm{D}N(h_1)\,(\mathrm{D}_\theta h_1\,\eta) = N(h_1)Y$. With $\rho^2 = \|h\|^2/d_{\text{model}} + \varepsilon$, the derivative of N at h is $\mathrm{D}N(h)\,k = k/\rho - h\,\langle h, k\rangle/(d_{\text{model}}\rho^3)$, which vanishes only for $k = 0$ when $\varepsilon > 0$, and the unique $\mathrm{D}_\theta h_1\,\eta$ with $\mathrm{D}N(h_1)\,(\mathrm{D}_\theta h_1\,\eta) = N(h_1)Y$ is

$$\mathrm{D}_\theta h_1\,\eta = h_1 Y + \frac{h_1 Y_s h_1^\top}{d_{\text{model}}\varepsilon}\,h_1, \qquad Y_s = \tfrac{1}{2}(Y + Y^\top),$$

as one checks by substituting $k = h_1 Y + \mu h_1$ and solving for the number μ.

Step 2: large inputs. Fix w with $wE \neq 0$, for attention one such row for every token, and let $x = tw$ with $t \to \infty$. The block reads $N(twE)$, which stays in the ball of radius $\sqrt{d_{\text{model}}}$, so $g(tw)$ stays bounded, and for attention the scores are bounded too. The derivative of $g(tw)$ along η is bounded as well: by the formula of Step 1, $\|\mathrm{D}N(h)\,k\| \leq 2\sqrt{d_{\text{model}}}\,\|k\|/\|h\|$, which is $O(1)$ for $h = twE$ and $k = tw\eta_E$. Hence $\mathrm{D}_\theta h_1\,\eta = t\,w\eta_E + O(1)$ at the input tw, while $h_1(tw) = t\,wE + O(1)$ and the right-hand side of Step 1 is $t^3\,(wEY_sE^\top w^\top)\,wE/(d_{\text{model}}\varepsilon) + O(t^2)$. So $wEY_sE^\top w^\top = 0$ for all w in an open set, hence $EY_sE^\top = 0$, and $Y_s = 0$ because E has rank d_{model}. The matrix Y is antisymmetric, and $\mathrm{D}_\theta h_1\,\eta = h_1 Y$.

Step 3: the skip. An antisymmetric Y is the derivative at $\tau = 0$ of the orthogonal changes of basis $I + \tau Y + O(\tau^2)$ that the residual network shares between h_0 and h_1 (Lemma A.1). They define a direction η^Y with the entries EY, $-YW_{\text{read}}$, $W_{\text{write}}Y$ and $-YD$ at E, W_{read}, W_{write} and D, along which the stream h_1 has the derivative $h_1 Y$. The direction $\eta' = \eta - \eta^Y$ is flat, with $\eta'_D = 0$ and $\mathrm{D}_\theta h_1\, \eta' = 0$ at every input. At the inputs tw, $\mathrm{D}_\theta h_1\, \eta' = t\, w\eta'_E + O(1)$, so $w\eta'_E = 0$ for all w in an open set, and $\eta'_E = 0$.

Step 4: the block. With $\eta'_E = 0$, the condition $\mathrm{D}_\theta h_1\, \eta' = 0$ says that the derivative of $n \mapsto f(nW_{\text{read}})W_{\text{write}}$ along the entries of η' at the weights of the block vanishes for every n in the image of $x \mapsto N(xE)$, which is the open ball of radius $\sqrt{d_{\text{model}}}$, since E has rank d_{model} and N maps $\mathbb{R}^{d_{\text{model}}}$ onto that ball for $\varepsilon > 0$. For $\tanh$, GELU, SwiGLU, GeGLU and attention this derivative is, along every line through the ball, a function of one variable given by a convergent power series around every point; it vanishes on an interval, hence on the whole line, so it vanishes for all inputs. For the ReLU-type activations without biases the kinks lie on hyperplanes through the center of the ball, and the arguments of Lemma C.3 are local. So Lemma C.3, with E and D replaced by the identity and $M = 0$, makes the entries of η' at each hidden unit a rescaling. For attention, each token $N(x_t E)$ ranges over the ball, and Step 2 for attention blocks, with E and D replaced by the identity and $M = 0$, keeps every $W_{Q,i}W_{K,i}^\top$ and $W_{V,i}W_{O,i}$ fixed to first order; by Step 1 there, η' is a direction of the changes of basis inside the heads.

Step 5: counting. The flat directions are thus those of the orthogonal change of basis and of the symmetries inside the block. An antisymmetric Y with $EY = 0$ vanishes, since E has rank d_{model}, and the symmetries inside the block move generic weights in independent directions, so $\dim \mathcal{K}(\theta) = d_{\text{model}}(d_{\text{model}} - 1)/2 + s$. If the block has width at least d_{model}, the skipless network with the same shapes has functional dimension at most $P - d_{\text{model}}(d_{\text{model}} - 1) - s$ by Theorem 3.3(ii), Lemma A.5 and, for attention, Remark A.6, so the skip adds at least $d_{\text{model}}(d_{\text{model}} - 1)/2$.

Part (iii)

Write $B(n) = f(nW_{\text{read}})W_{\text{write}}$ for the block as a function of its input n and of its weights, applied to every token for attention, so that $h_1 = xEA + g(x)$ with $g(x) = B(N(xE))$, and $y = N(h_1)D$. For a direction ψ of the weights of the block, $\mathrm{D}_\psi B(n)$ is the derivative of $B(n)$ along ψ at fixed n, and $\mathrm{D}B(n)\, k$ is the derivative of B at n along k at fixed weights. Let η be flat.

Steps 1 and 2 of part (ii) run with EA in place of E, which has rank d_{model} since A is invertible for generic parameters: $\eta_D = -YD$ with Y antisymmetric, and $\mathrm{D}_\theta h_1\, \eta = h_1 Y$ at every input.

Step 3: the skip. At the inputs tw of Step 2, $\mathrm{D}_\theta h_1\, \eta = t\, w(\eta_E A + E\eta_A) + O(1)$ and $h_1 Y = t\, wEAY + O(1)$, so $\eta_E A + E\eta_A = EAY$. Multiplying on the left by $I - EE^+$ gives $(I - EE^+)\, \eta_E = 0$, as A is invertible, so $\eta_E = EX$ with $X = E^+\eta_E$, and then $\eta_A = AY - XA$. At every input $\mathrm{D}_\theta h_1\, \eta = xEAY + \mathrm{D}_\theta g\, \eta$ and $h_1 Y = xEAY + g(x)Y$, so $\mathrm{D}_\theta g\, \eta = g(x)Y$. By the formula for $\mathrm{D}N$ in Step 1 of part (ii), which gives the same for $\varepsilon = 0$, the input $n = N(xE)$ of the block has the derivative

$$\mathrm{D}N(xE)\,(xEX) = nX - q(n)\, n, \qquad q(n) = nX_s n^\top / d_{\text{model}}, \quad X_s = \tfrac{1}{2}(X + X^\top).$$

Step 4: the symmetric part of X. Write $X = X_a + X_s$ with X_a antisymmetric. The rotations of h_0 and h_1 with the derivatives X_a and Y are symmetries. They define a flat direction η^{rot} with the entries EX_a at E, $AY - X_a A$ at A, $-X_a W_{\text{read}}$ at W_{read}, $W_{\text{write}}Y$ at W_{write} and $-YD$ at D. The flat direction $\eta' = \eta - \eta^{\text{rot}}$ has the entries EX_s, $-X_s A$ and 0 at E, A and D, so Step 3 applies to it with X_s in place of X and 0 in place of Y, and $\mathrm{D}_\theta g\, \eta' = 0$. The change nX_s of the input moves the block as the entries $X_s W_{\text{read}}$ at W_{read} do; for attention, X_s moves the queries, keys and values in the same way. So, with ψ the entries of η' at the weights of the block plus these,

$$\mathrm{D}_\psi B(n) = q(n)\, \mathrm{D}B(n)\, n \tag{15}$$

for every n in the image of N, the open ball of radius $\sqrt{d_{\text{model}}}$. For attention, $q(n)\, n$ stands for the matrix with the rows $q(n_t)\, n_t$, one per token.

We show that $q = 0$, hence $X_s = 0$. For ReLU, ReLU^2 and ReGLU, $B(\lambda n) = \lambda^k B(n)$ for $\lambda > 0$, with $k = 1$ or 2, so the right side of (15) is $k\, q(n) B(n)$. Replacing n by λn with $0 < \lambda \leq 1$

multiplies the left side by λ^k and the right side by λ^{k+2}, so both vanish, and $q = 0$ wherever $B \neq 0$. For the other blocks both sides are analytic in n, so (15) holds for all n, and we take $n = tm$ with $t \to \infty$, as in the proof of Lemma C.3. For GELU the right side is $t^3 q(m) \max(mW_{\text{read}}, 0) W_{\text{write}}$ up to terms that decay faster than any power of t, and the left side is a polynomial of degree at most one in t up to such terms, so $q(m) \max(mW_{\text{read}}, 0) W_{\text{write}} = 0$. SwiGLU and GeGLU units $\sigma(nc_i)\,(ne_i)\,a_i$ give $2t^4 q(m) \sum_{i:\, mc_i > 0} (mc_i)(me_i)\,a_i$ against a polynomial of degree two. For $\tanh$ the right side is $4t^3 q(m) \sum_i (mc_i)\, e^{-2|mc_i|t} a_i$ up to terms $O(t^3 e^{-4|mc_i|t})$; for the unit j with the smallest $|mc_j|$, the left side contributes a polynomial of degree one times $e^{-2|mc_j|t}$ (proof of Lemma C.3), so $q(m)(mc_j)\,a_j = 0$. For attention, on the two-token inputs of Step 2 for attention blocks with $u = tm$ and v fixed, the right side is $t^3 q(m)\, mS + O(t^2)$ and the left side is $O(t)$. In every case q vanishes on an open set for generic weights, so $X_s = 0$.

Step 5: counting. With $X_s = 0$, the direction η' moves only the weights of the block, and by (15) $B(n)$ has derivative zero along it on the whole ball. So η' lies in the space $\mathcal{K}_B$ of such directions, which is the space found in Step 4 of part (ii), and every direction of $\mathcal{K}_B$ is flat. Hence $\mathcal{K}(\theta)$ is spanned by the directions η^{rot} and $\mathcal{K}_B$, as for the residual network with its single rotation, and for generic weights of the block $\mathcal{K}_B$ has the same dimension in both networks. The directions η^{rot} are independent of $\mathcal{K}_B$ and of each other, since $\eta_E = EX_a$ and $\eta_D = -YD$ determine X_a and Y. For a full A they have dimension $d_{\text{model}}(d_{\text{model}} - 1)$. For an upper triangular A, $AY - X_a A$ must be upper triangular, which determines Y from X_a: for generic A the map from antisymmetric Y to the entries of AY below the diagonal is invertible, as it is for $A = I$. They then have dimension $d_{\text{model}}(d_{\text{model}} - 1)/2$. So $\mathcal{K}(\theta)$ is larger than for the residual network by $d_{\text{model}}(d_{\text{model}} - 1)/2$, respectively 0, while the network has d_{model}^2, respectively $d_{\text{model}}(d_{\text{model}} + 1)/2$, more parameters. In both cases the functional dimension is larger by $d_{\text{model}}(d_{\text{model}} + 1)/2$. For the blocks of the theorem, $\mathcal{K}_B$ consists of the symmetries inside the block (Step 4 of part (ii)), so the equality condition holds.

The case $\varepsilon = 0$. Now N maps onto the sphere of radius $\sqrt{d_{\text{model}}}$, and the scalings of h_0 and h_1 are symmetries too, since $N(ch) = N(h)$ for $c > 0$. In Step 1, $\mathrm{D}N(h)$ vanishes along h, so $\mathrm{D}_\theta h_1\, \eta = h_1 Y + \mu\, h_1$ with a number μ at each input. Y is again antisymmetric, since $N(h_1)Y$ must be orthogonal to h_1, which is $t\, wEA + O(1)$ at the inputs tw. At these inputs g and its derivative along η do not depend on t, since $N(twE) = N(wE)$. The terms of order t give $w\Lambda = \mu_w\, wEA$ with $\Lambda = \eta_E A + E\eta_A - EAY$ and a number μ_w, for all w in an open set. Adding to w a vector w_0 with $w_0 E = 0$ leaves wEA unchanged, so $w_0\Lambda$ is a multiple of wEA for every such w, hence $w_0\Lambda = 0$, and $\Lambda = E\Lambda'$ with $\Lambda' = E^+\Lambda$. Every vector $v = wE$ of an open set is then a left eigenvector of $\Lambda' A^{-1}$, which is therefore cI for a number c, as in the proof of Lemma A.1: $\eta_E A + E\eta_A = EA(Y + cI)$. The constant terms then give $\mu = c$ wherever wEA and $g(w)$ are linearly independent. Steps 3 and 4 run with $Y + cI$ in place of Y and with $X_s = c_0 I + X_0$, where X_0 has trace zero; the scalings, with the derivatives c_0 at h_0 and c at h_1, join the rotations in η^{rot}; and (15) holds with X_0 in place of X_s on the sphere. For the residual network the same steps give $\eta_E = E(Y + cI)$, so $X_s = cI$ and $X_0 = 0$. Its flat directions are the rotation and the scaling of all hidden states, and the space $\mathcal{K}_B$ of directions of the block along which B has derivative zero on the sphere. For the learned skip, η^{rot} and $\mathcal{K}_B$ are flat, which gives at least $d_{\text{model}}(d_{\text{model}} - 1) + 2$, respectively $d_{\text{model}}(d_{\text{model}} - 1)/2 + 2$, dimensions besides $\mathcal{K}_B$, against $d_{\text{model}}(d_{\text{model}} - 1)/2 + 1$ for the residual network. So the difference is at most $d_{\text{model}}(d_{\text{model}} + 1)/2 - 1$. For ReLU, ReLU^2 and ReGLU, both sides of (15) are positively homogeneous, of degrees k and $k + 2$, so $\|n\|^2\, \mathrm{D}_\psi B(n) = k d_{\text{model}}\, q(n) B(n)$ for all $n \neq 0$. On each region where no nc_k changes sign, both sides are polynomials in n. Across the hyperplane $nc_i = 0$, at a point of no other such hyperplane, only the terms of unit i change, so they satisfy the identity on their own: $\|n\|^2$ times a polynomial equals $k d_{\text{model}}\, q(n)$ times $(nc_i)\, a_i$, $(nc_i)^2 a_i$ or $(nc_i)(ne_i)\, a_i$. Since $\|n\|^2$ has no real linear factor, it divides $q(n)$, so X_0 is a multiple of I, and $X_0 = 0$ because its trace vanishes. Then Step 5 gives equality.

Part (iv)

With $\varepsilon = 0$, $N(h) = \sqrt{d_{\text{model}}}\, h/\|h\|$ depends only on the direction of h, and $B(\lambda n) = \lambda^k B(n)$ for $\lambda > 0$, so $N(B(N(h))) = N(B(h))$: the skipless network computes $x \mapsto N(B(xE))D$. Its

normalization before the block drops out, which leaves the changes of basis $\mathrm{GL}(d_{\mathrm{model}})$ at h_0 and $\mathrm{CO}(d_{\mathrm{model}})$ at h_1, of dimension $d_{\mathrm{model}}^2 + d_{\mathrm{model}}(d_{\mathrm{model}}-1)/2 + 1$, and the residual network keeps one $\mathrm{CO}(d_{\mathrm{model}})$ for both hidden states (Theorem A.3(iii)).

The residual network. The case $\varepsilon = 0$ of part (iii), with $A = I$, shows that a flat direction is the sum of a direction of the rotation and scaling of all hidden states and a direction ψ of the weights of the block along which B has derivative zero on the sphere of radius $\sqrt{d_{\mathrm{model}}}$. This derivative is positively homogeneous of degree k in n, so it vanishes for all $n \neq 0$, and Lemma C.3, with E and D replaced by the identity and $M = 0$, makes ψ a rescaling of the units. So $\dim \mathcal{K}(\theta) = d_{\mathrm{model}}(d_{\mathrm{model}}-1)/2 + 1 + s$.

The skipless network. Write $h_1 = B(xE) = f(xC)W_{\mathrm{write}}$ with $C = EW_{\mathrm{read}}$, and let η be flat. As in the case $\varepsilon = 0$ of part (iii), $\eta_D = -YD$ with Y antisymmetric, since h_1 takes all values in an open set, and $\mathrm{D}_\theta h_1\, \eta = h_1 Y + \mu\, h_1$ with a number μ at each input. On each open region of inputs where no unit changes sign, h_1 and $J = \mathrm{D}_\theta h_1\, \eta - h_1 Y$ are homogeneous polynomials in x of degree k, and for generic weights two entries of h_1 have no common factor; from $J_a h_b = J_b h_a$ for any two entries, h_a divides J_a, so $J = c\, h_1$ on the region for a number c. Across the hyperplane $xc_i = 0$, at a point x_0 of no other such hyperplane, h_1 is continuous, and J is continuous for ReLU^2 and jumps by a multiple of a_i for ReLU and ReGLU; as $h_1(x_0)$ is not parallel to a_i for generic weights, the numbers c on the two sides agree. So $\mathrm{D}_\theta h_1\, \eta = h_1(Y + cI)$ at every input. Subtracting the change of basis $I + \tau(Y + cI)$ at h_1, with the entries $W_{\mathrm{write}}(Y + cI)$ at W_{write} and $-YD$ at D, leaves a direction along which $f(xC)W_{\mathrm{write}}$ has derivative zero, which moves C and W_{write} only along a rescaling of the units, by Lemma C.3 with $M = 0$. After subtracting that rescaling as well, Lemma C.2 gives $\eta_E = EX$ and $\eta_{\mathrm{read}} = -XW_{\mathrm{read}}$ for a matrix $X \in \mathbb{R}^{d_{\mathrm{model}} \times d_{\mathrm{model}}}$, the change of basis at h_0. These directions are independent, as in Step 3 of part (i), so $\dim \mathcal{K}(\theta) = d_{\mathrm{model}}^2 + d_{\mathrm{model}}(d_{\mathrm{model}}-1)/2 + 1 + s$. Both values are the counts, and their difference is d_{model}^2.

Part (v)

With $\varepsilon = 0$, $N(h) = \sqrt{d_{\mathrm{model}}}\, h/\|h\|$ ignores the length of each token, so the block reads only rows on the sphere of radius $\sqrt{d_{\mathrm{model}}}$. Let V_θ be the space of directions that move only the weights of the block and leave its output unchanged to first order at every input whose rows lie on this sphere; it contains the symmetries inside the block. We show that the flat directions of the residual network are those of $\mathrm{CO}(d_{\mathrm{model}})$ and those of V_θ, while the skipless network has at least two copies of $\mathrm{CO}(d_{\mathrm{model}})$ and V_θ. The bound then holds whatever V_θ is.

The residual network. Write $h_1 = xE + g(x)$, with $g(x) = f(N(xE)W_{\mathrm{read}})W_{\mathrm{write}}$, and $y = N(h_1)D$, and let η be flat. Step 1 of part (ii) runs as before, except that with $\varepsilon = 0$ the derivative $\mathrm{D}N(h)$ projects onto the vectors orthogonal to h and scales by $\sqrt{d_{\mathrm{model}}}/\|h\|$, so its kernel is spanned by h. Hence $N(h_1)Y$ is orthogonal to h_1; as h_1 takes every value in an open set, $Y_s = 0$, and $\mathrm{D}_\theta h_1\, \eta = h_1 Y + \mu(x)\, h_1$ with a number $\mu(x)$, for attention one per token. Subtracting the rotation η^Y of Step 3 there leaves a flat η' with $\eta'_D = 0$ and $\mathrm{D}_\theta h_1\, \eta' = \mu'(x)\, h_1$.

Take the inputs $x = tw$ with $t > 0$. Since N is homogeneous of degree 0, the block output $g(tw) = g(w)$ and its derivative $\gamma(w)$ along η' do not depend on t, so

$$t\, w\eta'_E + \gamma(w) \;=\; \mu'(tw)\, \big(t\, wE + g(w)\big) \qquad \text{for all } t > 0.$$

For generic parameters wE and $g(w)$ are linearly independent for all w in an open set, since otherwise the block would map every point of the sphere to a multiple of itself. Comparing the parts along wE, along $g(w)$ and outside their span gives $w\eta'_E = c_w\, wE$ and $\gamma(w) = c_w\, g(w)$ with a number c_w. Then $w\eta'_E$ is parallel to wE for all w, so η'_E vanishes on the kernel of E, $\eta'_E = ET$ with vT parallel to v for every v, and $\eta'_E = cE$ with one number c. The scaling $E \mapsto (1+\tau)E$, $W_{\mathrm{write}} \mapsto (1+\tau)W_{\mathrm{write}}$ multiplies h_1 by $1 + \tau$ and is the scale in $\mathrm{CO}(d_{\mathrm{model}})$. Subtracting c times its direction leaves a flat η'' that moves only the weights of the block, with $\mathrm{D}_\theta g\, \eta'' = 0$ at the inputs tw. Their normalized rows cover an open set of the sphere, and the derivative of the block along η'' is analytic on the sphere, so $\eta'' \in V_\theta$. The directions of $\mathrm{CO}(d_{\mathrm{model}})$ move E by $E(Y + cI)$, which vanishes only for

$Y = 0$ and $c = 0$. So $\mathcal{K}(\theta)$ is the direct sum of these directions and V_θ, and the residual network has functional dimension $P - \dim \mathrm{CO}(d_{\mathrm{model}}) - \dim V_\theta$.

The skipless network. Its block also reads only rows on the sphere, so the directions of V_θ are flat, and so are the changes of basis in $\mathrm{CO}(d_{\mathrm{model}})$ at h_0 and at h_1 (Proposition 3.1). These directions are independent. A combination moves E by $E(Y_0 + c_0 I)$ and D by $-Y_1 D$, so $Y_0 = 0$, $c_0 = 0$ and $Y_1 = 0$. What is left is a multiple of the scaling $W_{\mathrm{write}} \mapsto (1 + \tau) W_{\mathrm{write}}$ plus a direction of V_θ, and the scaling lies in V_θ only if the block vanishes on the sphere. So the skipless network has functional dimension at most $P - 2 \dim \mathrm{CO}(d_{\mathrm{model}}) - \dim V_\theta$.

Generic parameters. The dimension of V_θ depends only on the weights of the block and takes its smallest value v for generic weights. The generic functional dimensions are therefore $P - \dim \mathrm{CO}(d_{\mathrm{model}}) - v$ with the skip and at most $P - 2 \dim \mathrm{CO}(d_{\mathrm{model}}) - v$ without it, and the skip adds at least $\dim \mathrm{CO}(d_{\mathrm{model}}) = d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 + 1$.

D The symmetry group of a transformer is not a product over layers

Wang and Wang (2025) characterize the symmetry group of a transformer block. For a full transformer they state that the group is the product of the groups of its blocks, since residual connections and LayerNorm prevent any coupling across layers (Wang and Wang, 2025, Theorem 7). They do not define the group of the full model. We take the model to be the map from tokens to outputs, so that the embedding and the readout are among its parameters, as in the models they name (GPT-2, BERT, LLaMA and Qwen) and in their end-to-end checks, which compare the logits and generated tokens of GPT-2 models. In this case two transformations of the whole stream, applied at every hidden state from the embedding to the readout, are counterexamples.

Table 5: Relative change of the output under the two transformations of the whole stream, worst case over three random transformers ($d_{\mathrm{model}} = 8$, two heads, three layers, GELU MLPs of width 16, generic gains and biases). The notebook also checks the assumptions of Wang and Wang (2025) before and after each transformation, and that both transformations change the function of the bare stack of blocks. Notebook, Section 7 (Appendix F).

Model	Global permutation	Global rotation
pre-norm LayerNorm, untied readout	6×10^{-16}	2×10^{-15}
pre-norm RMSNorm, untied readout	5×10^{-16}	9×10^{-16}
pre-norm LayerNorm, tied readout	5×10^{-16}	fails, at least 0.16
post-norm LayerNorm, tied readout	9×10^{-16}	not applicable

1. A *global permutation* of the stream coordinates, with every read, write, normalization gain and bias permuted along. It preserves the function at every parameter value, for every placement of the normalization and with a tied or an untied readout. Verma and Elbayad (2024, Section 3.3) apply it to merge BERT models, embeddings included.
2. A *global rotation* Q, orthogonal and, for LayerNorm, fixing the all-ones vector. In a pre-norm model with an untied readout it preserves the function when every read W after a normalization with gain g becomes $\mathrm{diag}(g)^{-1} Q^\top \mathrm{diag}(g)\, W$ and every LayerNorm bias b becomes $b\, \mathrm{diag}(g)^{-1} Q\, \mathrm{diag}(g)$; the gains are unchanged. This is the construction behind Theorem 1 of Ashkboos et al. (2024a).

Both transformations change the token embedding: its columns, the coordinates of the stream, are permuted or rotated. No element of the product of per-block groups changes the embedding, so both lie outside that product. The LayerNorm transformers of Table 5 satisfy the assumptions of Wang and Wang (2025), including pairwise distinct gains and generic biases, and so do the transformed parameters.

Their proof transforms the output of one block and argues that the residual term prevents the transformation (Wang and Wang, 2025, Theorem 30); applied from the embedding onward, the transformation acts on the residual term as well. Their lemma on LayerNorm holds the gain and the

bias fixed (Wang and Wang, 2025, Lemma 26); the permutation moves both, and in a pre-norm block the normalized vector is only read, so the rotation does not need LayerNorm to commute with it. For a stack of blocks that acts on a fixed input and is read in fixed coordinates, both transformations change the function, and the counterexample does not apply.

The two transformations are instances of the one change of basis that the skips leave (Theorem A.3(iii)), restricted to what the normalization lets through: the rotation needs a normalization that reads the whole vector, and the permutation passes through per-channel gains as well. With a tied readout the rotation fails: the readout after the final gain would have to become $\mathrm{diag}(g)^{-1}Q^\top \mathrm{diag}(g)\,E^\top$, while tying fixes it to $Q^\top E^\top$.

E Counts at model scale

Table 6 applies the exact reduction of Proposition 5.1 to published models. Without skips, every hidden state keeps its own change of basis, and each turns one block into a triangular matrix; with skips and an untied readout, the whole network keeps one (Section 5.2). Under RMSNorm each change of basis is orthogonal and frees $d_{\mathrm{model}}(d_{\mathrm{model}}-1)/2$ parameters, and under LayerNorm the same holds with $d_{\mathrm{model}}-1$ in place of d_{model} (Proposition A.23). Llama-3.1 uses RMSNorm (Grattafiori et al., 2024), and its parameter counts are from the released weights; for the two ViT-B, with $d_{\mathrm{model}}=768$ and $L=24$ sub-blocks, we assume standard pre-LayerNorm blocks.

Table 6: Parameters that can be removed from trained networks without changing their function. Notebook, Section 8 (Appendix F).

Model	Skips	Parameters	Removable exactly
ViT-B of Ji et al. (2026)	no	86.6M	7.3M (8.5 %)
ViT-B	yes	86.6M	0.29M (0.3 %)
Llama-3.1-8B	yes	8.0B	8.4M (0.10 %)
Llama-3.1-70B	yes	70.6B	34M (0.05 %)
Llama-3.1-405B	yes	405.9B	134M (0.03 %)

The gap between the two ViT-B, 24 copies of $\mathrm{O}(767)$ or 7.1M dimensions of symmetry, 8.1 percent of the parameters, is what the skips remove (Theorem 3.3), and under the equality condition what the residual network gains in functional dimension. In general, a transformer layer with an MLP of width $4d_{\mathrm{model}}$ has about $12d_{\mathrm{model}}^2$ parameters, and under normalization the skips of its two sub-blocks remove about d_{model}^2 dimensions of symmetry, a twelfth of the parameters of the blocks.

Inside the attention blocks, with or without skips, each head also has a change of basis between its queries and keys and one between its values and outputs: $12 \cdot 2Hd_h^2 = 1.2$M dimensions for ViT-B. With the one copy of $\mathrm{O}(767)$ that the residual ViT-B keeps, this gives 1.5M, the count of van Nierop (2024, Eq. 17).

F Numerical checks and reproducibility

All computations are in double precision, in a notebook that will be released in a public Git repository, coming soon; its sections are listed at the end of this appendix. Unless stated otherwise, weights have independent $\mathcal{N}(0, 1/\text{fan-in})$ entries, inputs have independent $\mathcal{N}(0,1)$ entries, and networks compared with and without skips share their weights; the notebook gives the few exceptions.

Exactness checks. For Theorems A.3 and A.24 and Proposition 5.1 the transformed weights are constructed as in the proofs, and both networks are evaluated on random inputs; the reported number is $\|f' - f\|/\|f\|$ over the batch, and values near 10^{-15} are floating-point rounding. At depth six, with biases, the sweep of Theorem A.3(i) turns identity units with trainable U and V into identity skips, with U and V equal to I up to 2×10^{-15} and the function unchanged up to 2×10^{-15}, and its forward version absorbs learned affine skips with skip biases up to 10^{-15}. In Theorem A.24(ii) a

scaled orthogonal skip is absorbed to 10^{-9}, which is the RMSNorm constant $\varepsilon = 10^{-8}$ acting on the overall scale (Lemma A.1), and a generic skip fails at 0.43. For Lemma A.2, on skipless ReLU, ReLU^2 and ReGLU networks of the size of Figure 4 with $\varepsilon = 10^{-1}$, 10^{-3} and 10^{-6}, a change of basis $e^{0.3S}$ at h_0, h_1 or h_2, or the scaling $e^{0.3}$ at h_3, multiplies each output by the factor Λ of the proof up to 5×10^{-12}, and with $\varepsilon = 0$ leaves the function unchanged up to 10^{-13}; the derivatives match part (ii) up to 4×10^{-12} of the output (Section 17 of the notebook). The checks of Section 5 use a pre-RMSNorm network with $d_{\text{model}} = 8$, $L = 5$ tanh blocks of width 16 and learned skips near the identity, with $\varepsilon = 10^{-2}$. Making every skip triangular changes the function by 5×10^{-16}, and making it symmetric positive definite by 4×10^{-15}. With a readout tied to the embedding through the gain of the final normalization, making the first four skips triangular and leaving the last one full changes it by 7×10^{-16} and removes 112 parameters; making all five triangular and inserting one 8×8 matrix before the gain also changes it by 7×10^{-16} and removes $140 - 64 = 76$, and spending also the rotation that this matrix frees at h_0, on a triangular block of the embedding, and the gain, which it absorbs, changes it by 10^{-15} and removes 112, no more than without the matrix. Without normalization and with a tied readout, setting the first four skips and 8 rows of the embedding to the identity changes the function by 10^{-13} and removes 320 parameters. For identity skips, a global rotation changes the tied network by 0.22, and one inserted matrix restores it to 8×10^{-16}. Setting $r = 2$ rows of each low-rank factor U to the identity changes the function by 3×10^{-15}, and without normalization the sweep of Theorem A.3 turns the low-rank skips into identity skips with a change of 6×10^{-16}. In a separate residual network of two blocks without normalization ($d_{\text{model}} = 8$, width 16), a change of basis G with $\|I - G\| = 0.41$ at the two hidden states after the first block, so that only the skip of that block joins two bases, changes the output by 0.28, 0.25, 0.050 and 0.0046 when the branch of the first block is scaled by 0.1, 1, 10 and 100, which makes it about 0.05, 0.5, 5 and 50 times the size of the skip, of the order of $\|h\|\|I - G\|/\|B(h)\|$ when the branch dominates (Section 3.2).

Functional dimension. The parameter-to-output map is evaluated on a fixed batch of inputs, and its Jacobian with respect to all parameters is formed exactly by automatic differentiation, or by central finite differences with step 10^{-6} for the small tanh networks of Table 7. For exact Jacobians the rank is read at the largest gap of the spectrum, the largest ratio s_i/s_{i+1} of consecutive singular values, 10^5 to 10^{12} for the rows of Table 7; for finite differences at the relative threshold 10^{-7}, which lies inside the gap between the genuine singular values, above 10^{-6} of the largest, and the noise of the differences, below 10^{-8}. The rank on a batch is a lower bound on the functional dimension, and where P minus it equals the symmetry count, it is the functional dimension (Theorem 3.3(ii)). The values of Theorem C.1(i) hold exactly for all five activations at $d_{\text{model}} = 4$, $m = 8$, in three seeds, with and without biases, with gaps of 10^{11} to 10^{14}; with biases the inputs must cross the kink of every unit, so these checks use inputs of scale 5. For one attention block ($d_{\text{model}} = 4$, input and output dimension 6, sequences of three tokens, $(H, d_h) = (2, 2)$, $(4, 1)$ and $(1, 4)$, causal and bidirectional, with and without biases, three seeds each), all 72 networks have the values of Theorem C.1(i), with gaps of at least 4×10^{11}. Under pre-RMSNorm with $\varepsilon = 10^{-2}$ ($d_{\text{model}} = 4$, three seeds each), the residual networks with one block have the value of Theorem C.1(ii) for all five activations and for attention, and the skipless ones have $P - d_{\text{model}}(d_{\text{model}} - 1) - s$, so the skip adds exactly $d_{\text{model}}(d_{\text{model}} - 1)/2$ in these checks.

The networks of Figure 4. Every network of Figure 4 is drawn 20 times, and so are the same networks with GeGLU and GLU blocks, which have the counts of SwiGLU. Except for the skipless ReGLU networks under RMSNorm, the rank equals the count in every draw, at gaps of at least 6×10^4 without normalization and 10^7 under RMSNorm with $\varepsilon = 10^{-5}$. In the skipless ReGLU networks under RMSNorm, at gaps between 40 and 2×10^4, the rank read at the largest gap is 263 in 14 draws, the count 264 in 5 and 253 in one, and with hidden units added 283 in 19 draws and the count 284 in one. With $\varepsilon = 10^{-12}$ the skipless ReGLU network has rank 233 in every draw, at gaps of at least 10^4: this is $P - Ld_{\text{model}}^2 - (d_{\text{model}}(d_{\text{model}} - 1)/2 + 1) - s$, the count of the network in which every normalization but the last is removed. The residual ReGLU network then has rank $281 = P - (d_{\text{model}}(d_{\text{model}} - 1)/2 + 1) - s$ in every draw (Section 6.2). With $\varepsilon = 10^{-2}$ the rank equals the count in every draw for all six activations, ReGLU included, at gaps of at least 2.8×10^9; the notebook also plots these spectra as in Figure 4.

The size of ε. RMSNorm divides a row $h^{(t)}$ by $\sqrt{r^2+\varepsilon}$, where $r^2 = \|h^{(t)}\|^2/d_{\text{model}}$ is its mean square, so ε acts relative to r^2. With inputs of unit variance, the median of r^2 over the inputs, and then over the draws, is 0.74 at h_0 and between 0.15 and 0.33 after the blocks of the skipless networks, and it grows along the stream of the residual networks to between 1.6 and 1.9 at h_3; at these medians ε/r^2 lies between 5×10^{-6} and 7×10^{-5} for $\varepsilon=10^{-5}$, and is about a thousand times larger for $\varepsilon=10^{-2}$. In a skipless network every block reads a normalized state, so the scale of the inputs enters only at h_0; the stream of a residual network carries it to every hidden state. At $\varepsilon=10^{-5}$ the singular values with indices 234 to 264 of the skipless ReGLU network, those of the 31 directions of Section 6.2, lie between $10^{-4.5}$ and $10^{-12.6}$ of the largest in the median over the draws, except the last, which lies at the floor. The factors $\varepsilon/(r^2+\varepsilon)$ that break these directions depend on the input, and they are large for the inputs whose hidden states have a small mean square: in every draw, between 11 and 592 of the 4000 inputs have a hidden state after a block with $r^2<10^{-3}$, where $\varepsilon/r^2>10^{-2}$, and in 13 of the 20 draws up to 378 inputs have one that is exactly zero, with every gate closed.

Trained networks. We also train the networks of Figure 4, with $\tanh$, GELU, ReGLU and SwiGLU blocks, on four classification datasets with at least four classes: two generated with scikit-learn, and satimage and segmentation from the Penn Machine Learning Benchmarks, with the features as inputs and one output per class. Training uses AdamW in single precision, a learning rate chosen on a validation set, early stopping and five seeds. A network counts as trained when it beats logistic regression with a tuned penalty on the test set, as 436 of 480 do. On inputs of the dataset and on Gaussian inputs alike, the rank equals the count in 398 of the 436, and the skips add 48 and 18 (Figure 5). Of the 38 exceptions, 36 are skipless ReGLU networks under RMSNorm; one residual GELU network is 8 below the count and one residual $\tanh$ network 1 below. Two further notebooks train these networks, store them and compute their Jacobians.

Learned skips. The $\tanh$ networks of Table 7 with learned skips or with trainable identity units have the functional dimension of the residual network, 224, as Theorem 3.2(ii) requires. Under RMSNorm with $\varepsilon=10^{-2}$, learned skips have $264-234=30=3\,d_{\text{model}}(d_{\text{model}}+1)/2$ more functional dimension than identity skips, as Theorem 4.1(iii) predicts. On residual networks without biases or normalization ($d_{\text{model}}=4$, $L=3$, width 8), spending the last change of basis on d_{model} columns of the first read leaves exactly the redundancy s of the symmetries inside the blocks, the last row of Table 4.

Low-rank skips. Networks with skips $I+UV^\top$ ($L=3$ $\tanh$ blocks of width $2d_{\text{model}}$, $d_{\text{model}}=4$ with $r=1,2$ and $d_{\text{model}}=6$ with $r=2,3$, exact Jacobians) have redundancy $d_{\text{model}}^2+2Ld_{\text{model}}r$ without normalization, all of U and V besides the global change of basis, and $d_{\text{model}}(d_{\text{model}}-1)/2+Lr(2r-1)$ under RMSNorm with $\varepsilon=10^{-2}$. Besides the global rotation and the r^2 dimensions of the $\mathrm{GL}(r)$ inside each product, each skip adds $r(r-1)$. To first order, rotations $I+\Omega_{\ell-1}$ and $I+\Omega_\ell$ of the two hidden states that it joins change it by $(\Omega_\ell-\Omega_{\ell-1})-\Omega_{\ell-1}UV^\top+UV^\top\Omega_\ell$, which keeps it within rank r of the identity when $\Omega_\ell-\Omega_{\ell-1}$ lies in the tangent space $\{XV^\top+UY^\top\}$ of the matrices of rank r at $UV^\top$; the antisymmetric matrices in this space have dimension $r(r-1)$ when $2r\le d_{\text{model}}$ and U, V are generic, which the notebook confirms in every case.

Skipless networks as limits. For Proposition 6.3 we take skipless networks with an embedding $\mathbb{R}^6\to\mathbb{R}^4$, $L=3$ blocks of width 8 and a readout $\mathbb{R}^4\to\mathbb{R}^6$, without biases or normalization, on 400 inputs; the attention network has two causal heads of width two on sequences of five tokens. For the residual networks F_t, the largest error relative to the largest output of S is at most 0.98, 0.094 and 0.0095 at $t=10$, 100 and 1000 over $\tanh$, GELU, ReLU, SiLU, ReLU^2, GLU, ReGLU, SwiGLU, GeGLU and attention blocks, and t times it changes by less than 3 percent between $t=100$ and $t=1000$, as an error of order $1/t$ requires.

Where the redundancy exceeds the count. For ReLU networks the exact Jacobian is taken with the activation pattern held fixed. On skipless ReLU networks without biases ($d_{\text{model}}=4$, $L=3$, width 8), 7 of 8 random draws attain the bound of Theorem 3.3(ii), redundancy 88 including the rescalings of hidden units, and all 8 residual draws attain their bound, 40. In the eighth skipless draw, five units of the last block are active on exactly the same inputs, those on which one unit of the block before is active, so there they act as one linear map of $\mathbb{R}^4$: their 40 parameters carry $40-16=24$ dimensions of redundancy instead of the 5 rescalings of the count, and the redundancy is $88+19=107$. Such

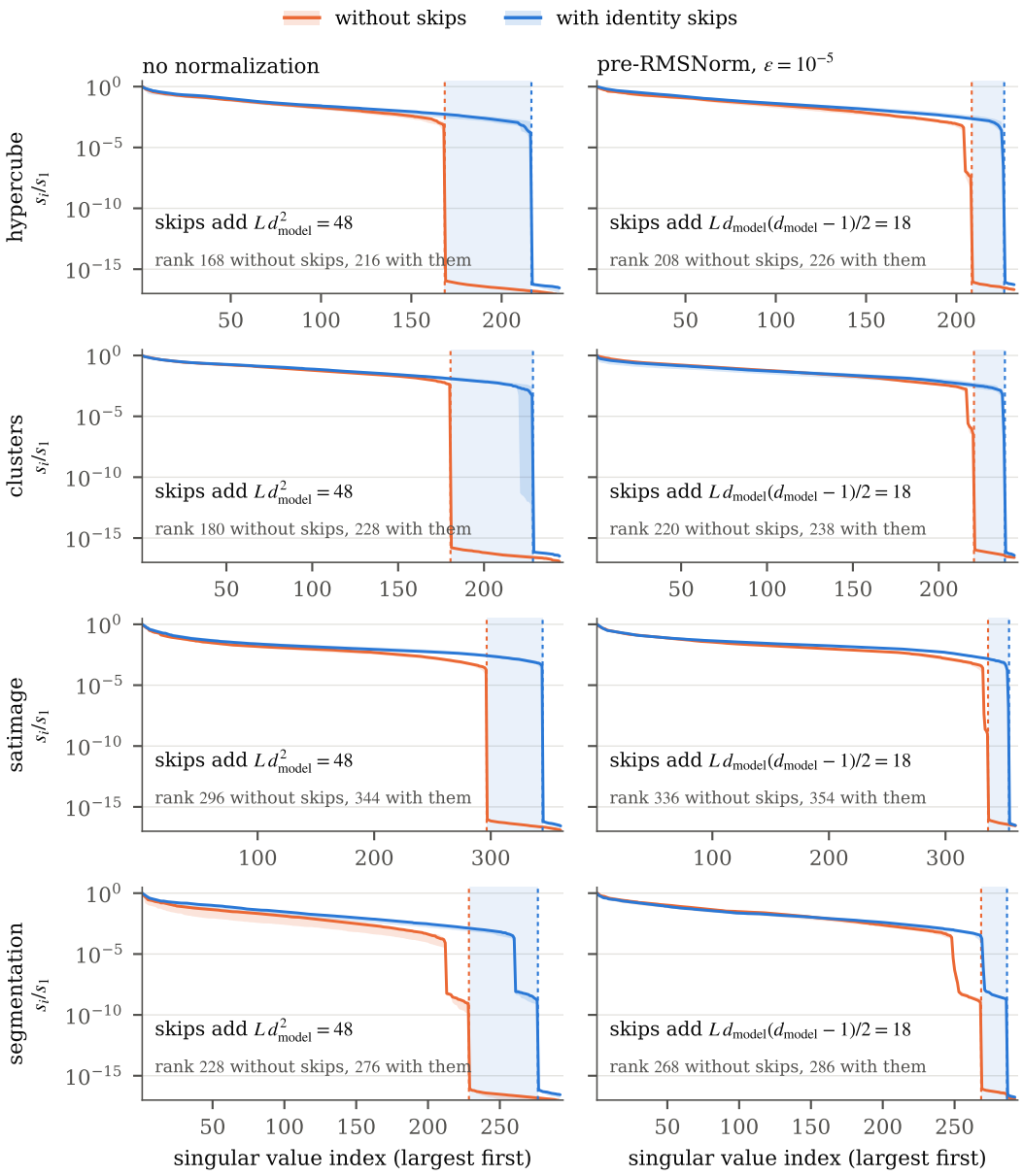


Figure 5: Singular values of the Jacobian at trained weights, on 1000 inputs of each dataset: the GELU networks of Figure 4, trained on four datasets (rows); median and range over five seeds. The skips add Ld^2_{model}, and $Ld_{\text{model}}(d_{\text{model}} - 1)/2$ under RMSNorm, as at random weights.

coincidences need a block that reads the output of another block, so they do not occur for one block, where Theorem C.1 applies; this is consistent with Grigsby et al. (2023). On the 3000 inputs of Section 1 of the notebook, a second draw also seems to exceed the bound, by 6, because one of its units is active on only 0.08 percent of the inputs; with 30,000 inputs or more it attains the bound, and the eighth draw stays at 107 (Section 15 of the notebook). Skipless networks whose first hidden state is an embedding table read through RMSNorm are also more redundant than their symmetries (Table 7).

Certificates at depth. The rows of Table 7 at $L = 8$ are certified as those at $L = 3$, with the activation pattern from ball arithmetic and the Jacobian from reverse mode (Section 16 of the notebook). Without skips these networks start from odd pairs of units, plus noise: ReLU^2 units (u, w) and $(-u, -w)$, and ReGLU units with gates u and $-u$ that share their value and write. With identity

Table 7: Functional-dimension checks. *Func. dim* is the rank of the Jacobian on the batch; where the redundancy, P minus it, equals the prediction, the rank meets the upper bound of Theorem 3.3(ii) and is the functional dimension. Every network has an embedding and a separate readout, and no biases unless marked; normalization is pre-norm. A *mixed* skip is a pair of trainable identity units U, V per block. The MLP networks have $d_{\text{model}} = 4$, width 8, $L = 3$ unless marked, and input and output dimension 6, except the rows marked *large*, which have $d_{\text{model}} = 16$, $L = 6$, width 16 and input and output dimension 20. The transformers have $d_{\text{model}} = 8$, two heads of width four, MLP width 16, a vocabulary of 16 tokens and two layers; each attention block has $2Hd_h^2$ symmetries inside. †The exact rank modulo a prime certifies the value at two parameter points; without skips at $L = 3$, the rank in floating point sits at the double-precision floor. ‡The symmetry count is 268; an embedding table read through RMSNorm adds $d_{\text{model}}(d_{\text{model}} + 1)/2 = 36$, because N maps $\mathbb{R}^{d_{\text{model}}}$ onto an open ball, so the rows of the table can absorb any invertible map near the identity, and with continuous inputs the measured value is 268.

Network	Skip	Normalization	P	Func. dim	Redundant	Predicted
tanh	none	none	240	176	64	$(L{+}1)d_{\text{model}}^2 = 64$
tanh	identity	none	240	224	16	$d_{\text{model}}^2 = 16$
tanh	learned	none	288	224	64	$(L{+}1)d_{\text{model}}^2 = 64$
tanh	mixed	none	336	224	112	$(2L{+}1)d_{\text{model}}^2 = 112$
tanh, biases	none	none	286	206	80	$(L{+}1)(d_{\text{model}}^2{+}d_{\text{model}}) = 80$
tanh, biases	identity	none	286	254	32	$d_{\text{model}}^2 + (L{+}1)d_{\text{model}} = 32$
tanh	none	RMSNorm, $\varepsilon = 10^{-9}$	240	212	28	$(L{+}1)\dim \mathrm{CO}(4) = 28$
tanh	identity	RMSNorm, $\varepsilon = 10^{-9}$	240	233	7	$\dim \mathrm{CO}(4) = 7$
tanh	learned	RMSNorm, $\varepsilon = 10^{-9}$	288	260	28	$(L{+}1)\dim \mathrm{CO}(4) = 28$
tanh	none	RMSNorm, $\varepsilon = 10^{-2}$	240	216	24	$(L{+}1)\dim \mathrm{O}(4) = 24$
tanh	identity	RMSNorm, $\varepsilon = 10^{-2}$	240	234	6	$\dim \mathrm{O}(4) = 6$
tanh	learned	RMSNorm, $\varepsilon = 10^{-2}$	288	264	24	$(L{+}1)\dim \mathrm{O}(4) = 24$
ReLU^2	none	none	240	152	88†	$(L{+}1)d_{\text{model}}^2 + Lm = 88$
ReLU^2	identity	none	240	200	40†	$d_{\text{model}}^2 + Lm = 40$
ReGLU	none	none	336	224	112†	$(L{+}1)d_{\text{model}}^2 + 2Lm = 112$
ReGLU	identity	none	336	272	64†	$d_{\text{model}}^2 + 2Lm = 64$
ReLU^2, $L = 8$	none	none	560	352	208†	$(L{+}1)d_{\text{model}}^2 + Lm = 208$
ReLU^2, $L = 8$	identity	none	560	480	80†	$d_{\text{model}}^2 + Lm = 80$
ReGLU, $L = 8$	none	none	816	544	272†	$(L{+}1)d_{\text{model}}^2 + 2Lm = 272$
ReGLU, $L = 8$	identity	none	816	672	144†	$d_{\text{model}}^2 + 2Lm = 144$
tanh, large	none	none	3712	1920	1792	$(L{+}1)d_{\text{model}}^2 = 1792$
tanh, large	identity	none	3712	3456	256	$d_{\text{model}}^2 = 256$
tanh, large	none	RMSNorm, $\varepsilon = 10^{-2}$	3712	2872	840	$(L{+}1)\dim \mathrm{O}(16) = 840$
tanh, large	identity	RMSNorm, $\varepsilon = 10^{-2}$	3712	3592	120	$\dim \mathrm{O}(16) = 120$
Transformer	identity	RMSNorm, $\varepsilon = 10^{-2}$	1280	1124	156	$\dim \mathrm{O}(8) + 4Hd_h^2 = 156$
Transformer	none	RMSNorm, $\varepsilon = 10^{-2}$	1280	976	304‡	$5\dim \mathrm{O}(8) + 4Hd_h^2 = 268$

skips the weights are Gaussian, with the writes scaled by $1/L$. From Gaussian weights, the skipless ReGLU networks at $L = 8$ attain the count on 12,000 inputs, and the ReLU^2 ones do not. In each ReLU^2 draw, 1 to 5 units of the last three blocks stay off on 2×10^6 fresh inputs, and a unit that is never on leaves its 8 parameters redundant, 7 more than its rescaling. With inputs added on which the rarely active units fire, the rank stays below the count by 35, 7 and 29: 7 for each unit that stays off, and one more in the third draw.

DyT and nanoGPT. DyT networks ($d_{\text{model}} = 4$, $L = 3$ blocks of width 8, exact Jacobians, three draws) lose $L = 3$ dimensions to the skips, the scale of each hidden state, which the learnable α of the next DyT absorbs. The nanoGPT models of Table 8 follow the `model.py` of nanoGPT (Karpathy, 2023), with token and position embeddings, pre-LayerNorm blocks with $\varepsilon = 10^{-5}$, causal attention with four heads of size two, GELU MLPs of width $4d_{\text{model}}$, a final LayerNorm and a readout, and no biases; without normalization every LayerNorm is the identity. The Jacobians are exact, over all logits of 64 random token sequences per layer. With skips the redundancy is, per layer, 144 with LayerNorm: 32 from the changes of basis inside the heads, 16 from the LayerNorm gains against the next matrix, 56 from the zero mean of each LayerNorm output, seen by every column of the next matrix, and 40 from the mean of each write, which every later LayerNorm erases. Once per model come 60 more: the global rotation, 21; the means of the 16 embedding rows, 16; the final LayerNorm

against the readout, 16; and a vector moved from the position to the token embedding, 7. Without normalization it is 32 per layer and $72 = 64 + 8$ per model. For one and two layers we construct all of these symmetries, and in every draw they span the null space of the Jacobian.

Table 8: Redundancy of nanoGPT ($d_{\text{model}} = 8$, four heads of size two, a vocabulary and a context of 8, no biases, untied readout): P minus the rank of the exact Jacobian, the same in all four draws of every network and equal to the prediction. N layers make $L = 2N$ blocks. With skips, each layer is redundant by 144 with LayerNorm and by 32 without normalization, and the model once more by 60 and 72 (Appendix F).

	$N = 1$	2	3	4	Prediction
LayerNorm					
P	984	1768	2552	3336	
without skips	246	432	618	804	$144N + 60 + 21L$
with skips	204	348	492	636	$144N + 60$
No normalization					
P	960	1728	2496	3264	
without skips	232	392	552	712	$32N + 72 + 64L$
with skips	104	136	168	200	$32N + 72$

Notebook. The notebook runs on a CPU in an environment pinned with uv, with Python 3.11. Its sections follow the note and are listed below, with their times on a CPU with two cores. The two long experiments save what Figure 4 and Table 8 need, from which the notebook draws the figure and the table in seconds unless it is asked to rerun them; the times below are those of the full computation.

Section of the notebook	What it reproduces	Time
1. The gauge fix, the counts and RMSNorm	exactness checks of Theorems A.3 and A.24 and Proposition 5.1; finite-difference rows of Table 7; exact ReLU and degree-two Jacobians	1 min
2. The last row of Table 4	the last row of Table 4	23 s
3. A dominant block, and learned, tied and low-rank skips	the checks of Sections 3.2, 4 and 5	under 1 s
4. The redundancy of low-rank skips	the low-rank skips of Section 4	36 s
5. DyT reads	the DyT networks	9 s
6. Exact reductions after training	the reductions of Section 5.1 and Proposition A.23	under 1 s
7. The symmetry group of a transformer is not a product	Table 5	under 1 s
8. Counts at model scale	Table 6	under 1 s
9. One block	Theorem C.1, for MLP and attention blocks	44 s
10. Figure 4	Figures 1 and 4 and the checks of their networks	28 min
11. Exact Jacobians	the rows at $d_{\text{model}} = 16$, the transformers and the embedding table of Table 7	4 min
12. Certificates modulo a prime	the certificates at $L = 3$ in Table 7	19 s
13. nanoGPT	Table 8 and the constructed symmetries	38 min
14. Skipless networks as limits of residual networks	Proposition 6.3	under 1 s
15. ReLU draws above the count	the ReLU draws above the count in Appendix F	2 min
16. Certificates at depth	the certificates at $L = 8$ in Table 7	5 min
17. Homogeneous blocks under RMSNorm	the checks of Lemma A.2	2 s